LeanForControl

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Chapter 1

Analysis toolbox

Facts about real functions, limits and integrals with no differential-equation content of their own. Each section says which later results consume it.

1.1 Continuity toolbox

Continuity gives the last root before a sign change and the first exit of a curve from an open region. First sphere crossing is the open-ball specialization of the general exit lemma.

Lemma 1 (Greatest zero of a sign-changing continuous function)**.** *Let g be continuous on $[t_0, t_1]$ with $g(t_0) \le 0 < g(t_1)$. Then g has a* last *root: there is $a \in [t_0, t_1)$ with $g(a) = 0$ and $g(t) > 0$ for every $t \in (a, t_1]$.*

Proof. □

Lemma 2 (Boundary crossing by a continuous curve)**.** *A continuous curve on $[t_0, t_1]$, with $t_0 \le t_1$, starting in an open set D and ending outside D meets ∂D on that interval.*

Proof. The image of the interval is connected. If it avoided the boundary, meeting D would force the whole image to lie in D. □

Lemma 3 (First exit from an open set)**.** *If a continuous curve on $[t_0, T]$ starts in an open set D and is outside D at $t_1 \in [t_0, T]$, some $\tau \in [t_0, t_1]$ satisfies $\varphi(\tau) \in \partial D$ and $\varphi(t) \in D$ for $t \in [t_0, \tau)$.*

Proof. The nonempty compact set of exit times has a minimum. A boundary crossing by that time must occur at the minimum itself. □

Theorem 4 (First crossing of a sphere)**.** *Let φ be continuous on $[t_0, T]$ with $\|\varphi(t_0) - x_{\mathrm{eq}}\| < \rho$, and suppose some $t_1 \in [t_0, T]$ satisfies $\rho \le \|\varphi(t_1) - x_{\mathrm{eq}}\|$. Then there is a* first *hitting time $\tau \in [t_0, T]$ of the sphere of radius ρ:*

$$\|\varphi(\tau) - x_{\mathrm{eq}}\| = \rho, \qquad \|\varphi(s) - x_{\mathrm{eq}}\| \le \rho \quad \forall\, s \in [t_0, \tau].$$

Proof. Apply lemma 3 to the open ball. Its boundary lies in the sphere; before the first exit the curve remains in the ball. □

A field cut off outside a set stays Lipschitz.

1.2 Dini derivatives

The upper right Dini derivative replaces the ordinary derivative where a function is only known to be continuous, which is the situation of the comparison lemma. As in the literature it takes values in $[-\infty, +\infty]$: the forward difference quotients may be unbounded, and the extended-real lim sup always exists, so no boundedness hypotheses are needed. Bounds against a real number are characterized by the difference quotients themselves.

Definition 5 (Upper right Dini derivative)**.** The *upper right Dini derivative* of $f : \mathbb{R} \to \mathbb{R}$ at t is

$$D^+ f(t) \;=\; \limsup_{h \to 0^+} \frac{f(t+h) - f(t)}{h}.$$

When f is differentiable at t this equals $f'(t)$.

1.2.1 Relationship to the classical derivative

Lemma 6 (Dini derivative of a differentiable function)**.** *If f has right derivative L at t, then $D^+ f(t) = L$: the Dini derivative agrees with the classical one wherever the latter exists.*

Proof. □

1.2.2 Upper bounds

Lemma 7 (Upper bound on the Dini derivative)**.** *Let the forward difference quotients of f at t be bounded above and cobounded below along $h \to 0^+$. Then $D^+ f(t) \le L$ if and only if for every $\varepsilon > 0$ the quotient $\frac{f(t+h)-f(t)}{h}$ is eventually at most $L + \varepsilon$ as $h \to 0^+$.*

Proof. □

1.2.3 Grönwall's inequality in Dini form

1.3 Interval integrals

Estimates and integrability side conditions for the interval integral, used wherever a differential equation is handled in integral form. The last three concern an integral whose endpoints move: normalizing by the window length makes it a mean value, after which continuity, antitonicity and decay each need only a hypothesis on the endpoints themselves. They support the class $\mathcal{KL}$ construction of the chapter on comparison functions.

Lemma 8 (Difference of integrals over adjacent intervals)**.** *For μ continuous on $[a, t]$ and $s \in [a, t]$,*

$$\int_a^t \mu - \int_a^s \mu = \int_s^t \mu.$$

Proof. □

Lemma 9 (Integral bound from a pointwise multiplicative bound)**.** *Let u take values in a normed space and v be real, both integrable on $[a,b]$ with $a \le b$. If $\|u(s)\| \le L\, v(s)$ for all $s \in [a,b]$, then*

$$\Big\| \int_a^b u \Big\| \le L \int_a^b v.$$

Proof. □

Lemma 10 (Integral bound from a constant pointwise bound)**.** *If $\|u(s)\| \le C$ for all $s \in [a,b]$ with $a \le b$, then*

$$\Big\| \int_a^b u \Big\| \le C\,(b-a).$$

Proof. □

Lemma 11 (Integral of a constant)**.** $\int_a^b C \,\mathrm{d}s = (b-a)\,C$ *for a real constant C.*

Proof. □

Lemma 12 (Interval integrability of a continuous composition)**.** *If f is jointly continuous and z is continuous on the segment between t_0 and t_1, then $s \mapsto f(s, z(s))$ is interval-integrable between them. This is the integrability side condition every integral-form solution argument needs.*

Proof. □

Lemma 13 (Differentiating an integral in its upper limit)**.** *For μ continuous on $[a,b]$ and $t \in (a,b)$,*

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big|_{x=t} \int_a^x \mu(\tau)\,\mathrm{d}\tau \;=\; \mu(t).$$

Proof. □

Lemma 14 (Continuity of an integral in its upper limit)**.** *If f is integrable on $[a,t]$ with $a \le t$, then $s \mapsto \int_a^s f(\tau)\,\mathrm{d}\tau$ is continuous on $[a,t]$.*

Proof. □

Lemma 15 (Continuity of an integral in both endpoints)**.** *Let f be locally integrable on $(0,\infty)$, and let a, b be continuous on a set S with $a(\eta), b(\eta) > 0$ in S. Then*

$$\eta \;\longmapsto\; \int_{a(\eta)}^{b(\eta)} f(x)\,\mathrm{d}x$$

is continuous on S.

Proof. □

Lemma 16 (The sliding average of an antitone function is antitone)**.** *Let f be antitone and locally integrable on $(0,\infty)$, and let a, b be monotone on a set S with $0 < a(\eta) < b(\eta)$. Then the mean value*

$$\eta \;\longmapsto\; \frac{1}{b(\eta)-a(\eta)} \int_{a(\eta)}^{b(\eta)} f(x)\,\mathrm{d}x$$

is antitone on S.

Proof. □

Lemma 17 (Sliding average of an eventually vanishing function)**.** *Let f be nonneg, antitone and locally integrable on $(0,\infty)$ with $f(s)\to 0$ as $s\to+\infty$, and let $a(\eta)<b(\eta)$ with $a(\eta)\to+\infty$ as $\eta\to+\infty$. Then*

$$\frac{1}{b(\eta)-a(\eta)}\int_{a(\eta)}^{b(\eta)} f(x)\,\mathrm{d}x \;\longrightarrow\; 0 \qquad (\eta\to+\infty).$$

The mean is squeezed between 0 and $f(a(\eta))$; no hypothesis relating a to b is needed.

Proof. □

1.4 Monotonicity above a level

Lemma 18 (Nonincreasing above a level)**.** *Let v be continuous on $[a,b]$ with right derivative $v'(t)$ at every $t\in[a,b)$, and suppose $v'(t)\le 0$ whenever $v(t)>\eta$. Then $v(t)\le\max\{v(a),\eta\}$ for all $t\in[a,b]$.*

Proof. Let $m=\max\{v(a),\eta\}$ and $\varepsilon>0$. The barrier $B(t)=m+\varepsilon\,(1+t-a)$ starts above v, and wherever v touches it, $v>\eta$, so $v'\le 0<\varepsilon=B'$. The barrier comparison theorem gives $v\le B$ on $[a,b]$; let $\varepsilon\to 0$. □

1.5 Norm inequalities

1.6 Positive definite functions

Positive definiteness in the sense of functions, not matrices: a function that vanishes at a point and is positive elsewhere. It is the shape every Lyapunov function and every comparison bound takes.

Definition 19 (Positive definite function)**.** A function W is *positive definite* on S about x_0 when it is continuous on S, $W(x_0)=0$, and $W(x)>0$ for every $x\in S\ \ \{x_0\}$.

Chapter 2

Ordinary differential equations

2.1 Gronwall–Bellman inequality

A function obeying an integral inequality is bounded by the solution of the corresponding integral equation. The two specializations are what the rest of the development actually calls: constant Λ gives exponential growth, and constant Λ and μ gives the $Ce^{\mu(t-a)}$ bound behind every continuous-dependence estimate.

Theorem 20 (Gronwall–Bellman inequality)**.** *Khalil, Lemma A.1. Let $\Lambda, \mu : [a,b] \to \mathbb{R}$ be continuous with $\mu \geq 0$, and let $y : [a,b] \to \mathbb{R}$ be continuous satisfying*

$$y(t) \;\leq\; \Lambda(t) + \int_a^t \mu(s)\, y(s)\, \mathrm{d}s \qquad \forall\, t \in [a,b].$$

Then

$$y(t) \;\leq\; \Lambda(t) + \int_a^t \Lambda(s)\, \mu(s)\, e^{\int_s^t \mu(\tau)\,\mathrm{d}\tau}\, \mathrm{d}s \qquad \forall\, t \in [a,b].$$

Proof. An integrating-factor argument. Write $z(t) = \int_a^t \mu y$, $M(t) = \int_a^t \mu$, and $v = z + \Lambda - y$, so that $v \geq 0$ is exactly the hypothesis. The transform $w = e^{-M} z$ has, by the product rule and lemma 13,

$$\dot{w} = e^{-M}\mu\,(y - z) = e^{-M}\mu\,(\Lambda - v) \;\leq\; e^{-M}\mu\Lambda,$$

the inequality because e^{-M}, μ and v are all nonnegative — this is the only place $\mu \geq 0$ is used, and it is essential: a sign change would reverse it. Integrating from a, where $w(a) = 0$, gives $w(t) \leq \int_a^t e^{-M(s)}\mu(s)\Lambda(s)\, ds$. Multiplying by $e^{M(t)}$ and folding $e^{M(t)}e^{-M(s)} = e^{\int_s^t \mu}$ turns this into the stated bound on $z(t)$, and $y \leq \Lambda + z$ finishes. □

Lemma 21 (Gronwall–Bellman with a constant rate)**.** *Khalil, Lemma A.1 (the case $\Lambda(t) \equiv \Lambda$ constant). Let μ be continuous and nonnegative on $[a,b]$, and let y be continuous with*

$$y(t) \;\leq\; C + \int_a^t \mu(s)\, y(s)\, \mathrm{d}s \qquad \forall\, t \in [a,b].$$

Then $y(t) \leq C\, e^{\int_a^t \mu(\tau)\,\mathrm{d}\tau}$ on $[a,b]$.

Proof. theorem 20 at $\Lambda \equiv C$ gives $y(t) \le C + \int_a^t C\mu(s)e^{\int_s^t \mu}\,ds$, and the integral evaluates in closed form: $s \mapsto -C\,e^{M(t)-M(s)}$ is an antiderivative of the integrand, so the integral is $Ce^{M(t)} - C$. The two C's cancel. □

Lemma 22 (Gronwall–Bellman with constant data)**.** *Khalil, Lemma A.1 (the case Λ and $\mu \ge 0$ both constant). Let $\mu \ge 0$ be constant and let y be continuous with*

$$y(t) \;\le\; C + \int_a^t \mu\, y(s)\,\mathrm{d}s \qquad \forall\, t \in [a,b].$$

Then $y(t) \le C\,e^{\mu(t-a)}$ on $[a,b]$. This is the form the continuous-dependence estimates use, and the reason their bounds carry a factor $e^{L(t-t_0)}$.

Proof. lemma 21 with $\int_a^t \mu = \mu(t-a)$. □

2.2 Existence, uniqueness and continuous dependence

Definition 23 (Integral solution)**.** A function x, defined on the segment between t_0 and t_1, is an *integral solution* of $\dot{x} = F(t,x)$ with initial value x_0 when

$$x(t) \;=\; x_0 + \int_{t_0}^t F(s,\,x(s))\,\mathrm{d}s$$

for every t between t_0 and t_1.

Lemma 24 (Re-anchoring an integral solution)**.** *Let x be a continuous integral solution of $\dot{x} = F(t,x)$ with initial value x_0 on the segment between t_0 and t_1, with F jointly continuous. Then for any s, r on that segment, x is an integral solution on the segment between s and r, with initial value $x(s)$:*

$$x(t) \;=\; x(s) + \int_s^t F(w, x(w))\,\mathrm{d}w \qquad \textit{for } t \textit{ between } s \textit{ and } r.$$

Leaving both endpoints free means the domain need not shrink — taking $s = t_1$ and $r = t_0$ re-anchors at one end of the original segment while retaining all of it.

Proof. Subtract the defining equations at t and at s and split the integral at s, which is legitimate because $w \mapsto F(w, x(w))$ is interval-integrable on each piece. □

Lemma 25 (Time reflection of an integral solution)**.** *Let x be an integral solution of $\dot{x} = F(t,x)$ with initial value x_0 on the segment between t_0 and t_1. Then $\sigma \mapsto x(t_0 + t_1 - \sigma)$ is an integral solution, with the same initial value x_0, of the reflected field*

$$(r, y) \;\longmapsto\; -F(t_0 + t_1 - r,\; y)$$

on the segment between t_1 and t_0.

The map $\sigma \mapsto t_0 + t_1 - \sigma$ is an involution exchanging the two endpoints, so reflection turns a statement anchored at t_0 into one anchored at t_1 and nothing more. Its purpose is to let an argument that is inherently forward-marching — a Grönwall bootstrap, say — be applied unchanged in backward time.

Proof. Substituting $r \mapsto t_0 + t_1 - r$ in the defining integral reverses the orientation of the interval, contributing one sign, and the negation of the field contributes another; the two cancel. □

2.2.1 The integral and differential formulations

An ODE can be written as a differential equation, $\dot{x} = F(t, x)$, or as the equivalent integral equation. The continuous-dependence and Gronwall arguments want the integral form, which presupposes no differentiability and carries its initial condition; the Lyapunov arguments want the differential form, so that $V \circ \varphi$ can be differentiated. The results below are the two directions of the fundamental theorem of calculus relating them.

They are provision rather than current practice: nothing in the stability chapters has yet been routed through this equivalence, so it is the intended path from a trajectory to the results of this chapter rather than an exercised one.

Lemma 26 (Integral curves are integral solutions)**.** *Let x be an integral curve of F on the segment between t_0 and t_1, with $s \mapsto F(s, x(s))$ continuous there. Then x is an integral solution on that segment, anchored at its own value $x(t_0)$:*

$$x(t) \;=\; x(t_0) + \int_{t_0}^{t} F(s, x(s))\,\mathrm{d}s.$$

Continuity of x itself is not assumed — it follows from differentiability.

Proof. The fundamental theorem of calculus. The derivative hypothesis holds on an open neighbourhood of each interior point, so it yields the right-derivative form the theorem needs, and continuity of F along x gives the integrability side condition. □

Lemma 27 (Integral solutions are integral curves)**.** *Let x be an integral solution of $\dot{x} = F(t, x)$ on the segment between t_0 and t_1, with $s \mapsto F(s, x(s))$ continuous there. Then x is an integral curve of F on that segment: it is differentiable, with $\dot{x}(t) = F(t, x(t))$.*

Proof. Differentiating the primitive: the integral term has derivative $F(t, x(t))$ by the fundamental theorem of calculus, and the constant x_0 contributes nothing. Transport that along the defining equation, which says x agrees with the primitive. □

Lemma 28 (Integral and differential forms agree)**.** *Given continuity of $s \mapsto F(s, x(s))$ on the segment between t_0 and t_1, the two formulations agree: x is an integral solution anchored at $x(t_0)$ if and only if it is an integral curve of F there. Continuity of F along x is exactly the price of the equivalence — without it the integral form is the weaker notion.*

Proof. The two implications are lemma 27 and lemma 26. □

Lemma 29 (Integral and differential forms agree, forward time)**.** *lemma 28 in forward time: for $t_0 \le t_1$, the equivalence holds over $[t_0, t_1]$. This is the interval the finite-forward stability predicates are phrased over, so this is the form the stability track would use to reach the results of this chapter.*

Proof. □

Lemma 30 (Integral form for an autonomous field)**.** *Let g be a continuous* autonomous *field and let x be an integral curve of g on $[t_0, t_1]$, $t_0 \le t_1$. Then x is an integral solution there, anchored at $x(t_0)$.*

This is the autonomous, forward-time specialization of lemma 29, and so the shape a trajectory of an autonomous system has: continuity of the field is the only hypothesis, the continuity of x along which F must be continuous being supplied by the curve itself.

Proof. □

2.2.2 Time invariance and local existence

A time-invariant field has no preferred time origin: a solution can be shifted to start at time 0.

Lemma 31 (Time invariance)**.** *If x is an integral curve of an autonomous field g on a set s, then $t \mapsto x(t + \Delta t)$ is an integral curve of the same g on $s - \Delta t$, for every Δt.*

An autonomous field has no preferred time origin, so translating a solution translates its interval of definition and changes nothing else. For a time-varying field the translated solution would solve the translated *equation instead.*

Proof. □

Lemma 32 (Shifting a solution segment to the origin)**.** *lemma 31 in the form the finite-segment predicates need: a solution segment of an autonomous field on $[t_0, t_1]$ becomes one on $[0, t_1 - t_0]$ under $s \mapsto x(s + t_0)$.*

This is why the autonomous stability predicates may fix the initial time at 0 without loss: any segment can be shifted there.

Proof. □

Lemma 33 (Local existence for a C^1 field)**.** *Let g be an autonomous field that is C^1 at x_0. Then there is a $T > 0$ and a curve φ with $\varphi(0) = x_0$ that is an integral curve of g on $[0, T]$: a nontrivial solution segment exists from every point at which the field is C^1.*

The anchor 0 is a construction choice rather than a restriction — by lemma 31 a segment from any other anchor follows.

Reference: the Picard–Lindelöf local existence theorem.

Proof. Picard–Lindelöf local existence: being C^1 at x_0 gives a closed ball on which g is Lipschitz, and hence a solution on some $(-\varepsilon, \varepsilon)$; take $T := \varepsilon/2$. □

Theorem 34 (Picard–Lindelöf on a compact interval)**.** *Let $f : \mathbb{R} \to E \to E$ be jointly continuous and K-Lipschitz in the state variable, uniformly in time, on a Banach space E. Then for every $t_0 \le t_1$ and every $x_0 \in E$ there is an α with $\alpha(t_0) = x_0$ solving $\dot{\alpha} = f(t, \alpha)$ on all of $[t_0, t_1]$.*

Global — rather than merely local — Lipschitz continuity is what makes the solution reach t_1: it forces linear growth in the state, ruling out finite-time blowup.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.2 (Global Existence and Uniqueness), proved in his Appendix C.1. Difference: Khalil assumes f piecewise continuous in t; here it is jointly continuous.*

Proof. Mathlib's Picard–Lindelöf theorem supplies a solution on a time interval of length Δt only when $\|f\| \le L$ on a ball of radius a about x_0 with $L\,\Delta t \le a$. For a K-Lipschitz field the sharpest available bound on that ball is $L = M + Ka$, where M bounds $\|f(t, x_0)\|$, so the condition becomes $M\Delta t \le a(1 - K\Delta t)$ — solvable in a exactly when $K\Delta t < 1$. One application therefore never covers a long interval.

Take instead a step h with $Kh \le 1/2$; then $a = 2Mh$ and $L = 2M$ satisfy the hypotheses, since $\|f(t, x)\| \le M + Ka = M(1 + 2Kh) \le 2M$ and $Lh = 2Mh = a$. This gives a solution on $[s, s + h]$ from any initial state. Chain n such steps, splicing consecutive segments at their shared endpoint: away from a junction the glued curve agrees with one piece on a neighbourhood, and at a junction the two one-sided derivatives combine. Choosing n with $nh \ge t_1 - t_0$ covers $[t_0, t_1]$. □

Lemma 35 (Right derivatives of an integral curve)**.** *If α is an integral curve of f on $[t_0, t_1]$, then for every $s \in [t_0, t_1)$ it has right derivative $f(s, \alpha(s))$ at s, i.e. α has derivative $f(s, \alpha(s))$ within $[s, \infty)$ at s.*

Proof. □

Lemma 36 (Gronwall separation of two solutions)**.** *Let f be K-Lipschitz in the state variable, uniformly in time, and let α, β be integral curves of f on $[t_0, t_1]$. Then*

$$d(\alpha(t), \beta(t)) \le d(\alpha(t_0), \beta(t_0))\, e^{K(t-t_0)} \qquad \forall\, t \in [t_0, t_1].$$

Uniqueness (lemma 37) is the case where the initial distance is zero; continuous dependence on the initial state is the statement read as a bound on the whole interval.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.4, in the unperturbed case: two solutions of the* same *field, so the perturbation bound is zero.*

Proof. □

Lemma 37 (Uniqueness of solutions)**.** *Let f be K-Lipschitz in the state variable, uniformly in time, and let α, β be integral curves of f on $[t_0, t_1]$ with $\alpha(t_0) = \beta(t_0)$. Then $\alpha = \beta$ on $[t_0, t_1]$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), the uniqueness half of Theorem 3.2.*

Proof. The initial distance in lemma 36 is zero, so the bound forces the distance to vanish throughout. □

Theorem 38 (Picard–Lindelöf on the forward ray)**.** *Let f be jointly continuous and K-Lipschitz in the state variable, uniformly in time. Then for every t_0 and x_0 there is an α with $\alpha(t_0) = x_0$ solving $\dot\alpha = f(t, \alpha)$ on all of $[t_0, \infty)$, and any other such solution agrees with it on $[t_0, \infty)$.*

Uniqueness is agreement on $[t_0, \infty)$, not equality of functions: the defining condition says nothing about $t < t_0$, so solutions may differ there.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.2, whose t_1 is arbitrary, extended to the forward ray.*

Proof. For each n take a solution α_n on $[t_0, t_0 + n]$ (theorem 34). By uniqueness (lemma 37) these agree wherever two of them are both defined, so $\Phi(t) := \alpha_{\lceil t - t_0 \rceil}(t)$ is well defined and agrees with α_n on all of $[t_0, t_0 + n]$. Near any $t \ge t_0$ choose $n > t - t_0$; then $[t_0, t_0 + n]$ is a neighbourhood of t within $[t_0, \infty)$ on which $\Phi = \alpha_n$, so Φ inherits the derivative there. Uniqueness on the ray follows by applying the compact-interval uniqueness at each n. □

Theorem 39 (Picard–Lindelöf, scalar integral form)**.** *Let $g : \mathbb{R} \to \mathbb{R} \to \mathbb{R}$ be jointly continuous and L-Lipschitz in its state argument, uniformly in time, and let $t_0 \le t_1$. Then for every x_0 there is a z, continuous on $[t_0, t_1]$, that is an integral solution of $\dot z = g(t, z)$ with $z(t_0) = x_0$ and has right derivative $g(s, z(s))$ at each $s \in [t_0, t_1)$.*

This is the scalar specialization of theorem 34; it discharges the perturbed-solution hypothesis that theorem 43 takes for generality.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.2, scalar case.*

Proof. Apply theorem 34 with $E = \mathbb{R}$. Continuity on $[t_0, t_1]$ is automatic for an integral curve, which converts the differential form into the integral one (lemma 29); the right derivatives are lemma 35. □

2.2.3 Continuous dependence

Two solutions of the same equation, started at different points or subject to different perturbations, separate no faster than exponentially — at a rate fixed by the Lipschitz constant of the field. Everything downstream that needs uniqueness gets it from here, by taking the two solutions to agree initially and the perturbation to vanish, which collapses the bound to zero.

Theorem 40 (Continuous dependence on initial states and parameters)**.** *Khalil, Theorem 3.4. Let f be jointly continuous and L-Lipschitz in its state argument on the segment between t_0 and t_1, let y be an integral solution of $\dot{y} = f(t, y)$ with $y(t_0) = y_0$, and let z be an integral solution of the perturbed equation $\dot{z} = f(t, z) + g(t, z)$ with $z(t_0) = z_0$, where $\|g(t, x)\| \le \mu$. Then for every t on that segment,*

$$\|y(t) - z(t)\| \;\le\; \|y_0 - z_0\|\, e^{L|t-t_0|} \;+\; \frac{\mu}{L}(e^{L|t-t_0|} - 1).$$

Two solutions may separate, but no faster than exponentially, at a rate set by the Lipschitz constant alone; the two error sources — a different starting point and a perturbed field — contribute additively. The bound is stated on the unordered segment, so it runs in either time direction.

The perturbation g is assumed only bounded and integrable along z, never continuous, so z need not be differentiable anywhere. That is why the theorem is phrased over the integral formulation rather than the differential one.

Proof. The forward case is a Grönwall bootstrap. Subtracting the two integral equations and applying the triangle inequality gives

$$\|y(\tau) - z(\tau)\| \;\le\; \|y_0 - z_0\| + \mu(\tau - t_0) + L \int_{t_0}^{\tau} \|y(s) - z(s)\|\, \mathrm{d}s,$$

the Lipschitz hypothesis bounding the f-difference and the uniform bound on g the rest. Adding μ/L to both sides absorbs the linear term $\mu(\tau - t_0)$ into the integral, putting the inequality in the shape lemma 22 expects, and Grönwall delivers the exponential.

That bootstrap is inherently forward-marching, so rather than repeat it for the backward case, reflect: lemma 25 sends $\sigma \mapsto t_0 + t_1 - \sigma$ and negates the field, turning the backward instance into a forward one on $[t_1, t_0]$ to which the same argument applies unchanged. □

The same mechanism bounds a single solution from below: an equilibrium cannot be approached faster than exponentially.

Theorem 41 (Continuous dependence on parameters)**.** *Khalil, Theorem 3.5. If y solves $\dot{y} = f(t, y)$ and z solves $\dot{z} = f(t, z) + g(t, z)$ with $\|g(t, x)\| \le \alpha$ and $\|z_0 - y_0\| \le \alpha$, and $\alpha(1 + 1/L)e^{L(t_1 - t_0)} \le \varepsilon$, then $\|y(t) - z(t)\| \le \varepsilon$ for all $t \in [t_0, t_1]$.*

Proof. Specialise theorem 40 with $\mu = \alpha$ and $\|y_0 - z_0\| \le \alpha$: for $t \in [t_0, t_1]$ it gives $\|y(t) - z(t)\| \le \alpha e^{L(t-t_0)} + \frac{\alpha}{L}(e^{L(t-t_0)} - 1) = \alpha(1 + 1/L)e^{L(t-t_0)} - \alpha/L$. Drop the $-\alpha/L$ and bound $e^{L(t-t_0)}$ by $e^{L(t_1-t_0)}$ to get at most ε. □

2.3 Comparison lemma

The comparison lemma bounds a function satisfying a differential *inequality* by the solution of the corresponding differential *equation*. The inequality is stated with the Dini derivative, so v need only

be continuous — which is what makes the lemma applicable to $\|x(t)\|$, not differentiable where a trajectory crosses the origin.

Lemma 42 (Integral form from a pointwise derivative)**.** *Let u be continuous on $[t_0, t_1]$ with $u(t_0) = u_0$ and $\dot{u}(t) = f(t, u(t))$ at every interior point, and let f be jointly continuous. Then u is an integral solution:*

$$u(s) = u_0 + \int_{t_0}^{s} f(\tau, u(\tau))\,\mathrm{d}\tau.$$

.

Proof. The fundamental theorem of calculus on $[t_0, s]$; the integrand is continuous there, being a composition of continuous maps. □

Theorem 43 (Comparison lemma)**.** *Khalil, Lemma 3.4. Let f be jointly continuous and L-Lipschitz in its state argument on $[t_0, t_1]$, and suppose*

1. *u is continuous on $[t_0, t_1]$ and solves $\dot{u} = f(t, u)$ with $u(t_0) = u_0$;*
2. *v is continuous on $[t_0, t_1]$ with $v(t_0) \le u_0$, satisfies the Dini subsolution inequality $D^+v(t) \le f(t, v(t))$, and has forward difference quotients bounded above at each t;*
3. *for every $\lambda > 0$ the perturbed equation $\dot{z} = f(t, z) + \lambda$, $z(t_0) = u_0$, has a solution on $[t_0, t_1]$.*

Then $v(t) \le u(t)$ for all $t \in [t_0, t_1]$.

The perturbed solutions in (3) are taken as a hypothesis for generality, so that callers with solutions already in hand need not re-derive them. Under the hypotheses already listed they always exist: $f + \lambda$ is Lipschitz exactly when f is, so theorem 39 discharges (3).

Proof. Fix $\lambda > 0$ and let z_λ be as in hypothesis (3).

Claim: $v \le z_\lambda$ on $[t_0, t_1]$. Otherwise $v(t_{\mathrm{bad}}) > z_\lambda(t_{\mathrm{bad}})$ for some t_{bad}, and since $v(t_0) \le z_\lambda(t_0)$ the continuous difference $v - z_\lambda$ has a last zero $a < t_{\mathrm{bad}}$, with $v > z_\lambda$ on $(a, t_{\mathrm{bad}}]$. Being below v immediately to the right of a forces $\dot{z}_\lambda(a) \le D^+v(a)$, so

$$f(a, z_\lambda(a)) + \lambda \;\le\; D^+v(a) \;\le\; f(a, v(a)) = f(a, z_\lambda(a)),$$

contradicting $\lambda > 0$.

Now let $\varepsilon > 0$ and choose λ small enough that theorem 41 — which applies because f is Lipschitz in its state argument, and which treats λ as a perturbation of size λ — gives $\|u - z_\lambda\| \le \varepsilon/2$ on $[t_0, t_1]$. Then $v \le z_\lambda \le u + \varepsilon$. As ε was arbitrary, $v \le u$. □

2.4 Lie derivatives

The rate of change of a function $V(t, x)$ along the integral curves of a time-dependent vector field, with no differential-equation content beyond the chain rule: what every Lyapunov, Chetaev and invariance argument differentiates.

Definition 44 (Lie derivative along a vector field)**.** For $V : \mathbb{R} \times E \to \mathbb{R}$ and a time-dependent vector field $v : \mathbb{R} \times E \to E$, the *Lie derivative of V along v* is

$$L_v V(t, x) = \frac{\partial V}{\partial t}(t, x) + \frac{\partial V}{\partial x}(t, x)\, v(t, x),$$

the derivative of V at (t, x) in the direction $(1, v(t, x))$: the Lie derivative of V, as a function on $\mathbb{R} \times E$, along the autonomous field $(t, x) \mapsto (1, v(t, x))$.

Lemma 45 (Lie derivative of a time-invariant function)**.** *If V does not depend on t and is differentiable at x, then $L_v V(t, x) = DV(x)\, v(t, x)$.*

Proof. □

Lemma 46 (Lie derivative of $V(x) + ct$)**.** *If V does not depend on t and is differentiable at x, then the Lie derivative of $(t, x) \mapsto V(x) + ct$ along v is $DV(x)\, v(t, x) + c$.*

Proof. □

Lemma 47 (Chain rule along an integral curve)**.** *Let γ be an integral curve of v on I, and V differentiable at $(t, \gamma(t))$ for $t \in I$. Then for $t \in I$, within I,*

$$\frac{d}{dt} V(t, \gamma(t)) = L_v V(t, \gamma(t))$$

(definition 44).

Proof. □

Lemma 48 (The Lie derivative along a solution)**.** *Let γ be an integral curve of v on $[t, \infty)$, V differentiable at $(\tau, \gamma(\tau))$ for $\tau \geq t$, and let $\tau \mapsto V(\tau, \gamma(\tau))$ have right derivative w at t. Then $L_v V(t, \gamma(t)) = w$.*

Proof. By lemma 47 the right derivative is also $L_v V(t, \gamma(t))$, and right derivatives are unique. □

Lemma 49 (Nonpositive Lie derivative along an integral curve)**.** *Let γ be an integral curve of v on an interval s, V differentiable at $(t, \gamma(t))$ for $t \in s$, and $L_v V(t, \gamma(t)) \leq 0$ at every interior point t of s. Then $t \mapsto V(t, \gamma(t))$ is nonincreasing on s.*

Proof. The mean value theorem, with the derivative given by lemma 47. □

Lemma 50 (Exponential bound from a linear Lie-derivative bound)**.** *Let γ be an integral curve of v on $[t_0, t_1]$, V differentiable at $(t, \gamma(t))$ for $t \in [t_0, t_1]$, and $c \in \mathbb{R}$ with $L_v V(t, \gamma(t)) \leq -c\, V(t, \gamma(t))$ for $t \in (t_0, t_1)$. Then $V(t, \gamma(t)) \leq e^{-c(t-t_0)}\, V(t_0, \gamma(t_0))$ on $[t_0, t_1]$.*

Proof. The derivative of $e^{ct} V(t, \gamma(t))$ is $e^{ct}(cV + L_v V) \leq 0$ (lemma 47), so it does not increase. □

Lemma 51 (Continuity of the Lie derivative)**.** *If V is C^1 and v is continuous, then $(t, x) \mapsto L_v V(t, x)$ is continuous.*

Proof. □

2.5 ω-limit sets

The points a curve or sequence returns arbitrarily close to, arbitrarily late: its cluster points at infinity, which is the ω-limit of topological dynamics for a single orbit. These are the topological facts behind LaSalle's invariance principle, in continuous and in discrete time alike: a continuous function converging along the orbit is constant on its ω-limit set, an orbit trapped in a closed set has its ω-limit set there, and an orbit trapped in a compact set converges to its ω-limit set.

Definition 52 (ω-limit set)**.** The *ω-limit set* of $\varphi : \tau \to X$, for τ a preordered index set such as $\mathbb{R}$ or $\mathbb{N}$, is the set of cluster points of φ as $t \to \infty$:

$$\omega(\varphi) = \bigcap_T \overline{\{\varphi(t) : t \geq T\}},$$

the points y such that $\varphi(t)$ is frequently in every neighbourhood of y as $t \to \infty$.

Lemma 53 (Membership in the ω-limit set)**.** *$y \in \omega(\varphi)$ (definition 52) if and only if for every neighbourhood U of y there are arbitrarily large t with $\varphi(t) \in U$.*

Proof. □

Lemma 54 (The ω-limit set as Mathlib's ω-limit)**.** *definition 52 is the ω-limit, in the sense of topological dynamics, of the one-member family $\{\varphi\}$.*

Proof. □

Lemma 55 (A limit of $g \circ \varphi$ is the value of g on $\omega(\varphi)$)**.** *Let $g : X \to Y$ be continuous, Y Hausdorff, and $g(\varphi(t)) \to L$ as $t \to \infty$. Then $g(y) = L$ for every $y \in \omega(\varphi)$ (definition 52).*

Proof. $g(y)$ is a cluster point of $g \circ \varphi$, which converges to L; in a Hausdorff space the two coincide. □

Lemma 56 (The ω-limit set of a curve trapped in a closed set)**.** *If Ω is closed and $\varphi(t) \in \Omega$ for all large t, then $\omega(\varphi) \subseteq \Omega$.*

Proof. A point outside Ω has the open neighbourhood $X \setminus \Omega$, which φ eventually avoids. □

Lemma 57 (Convergence to a set containing the ω-limit set)**.** *If $\varphi(t)$ lies in a compact set for all large t and $\omega(\varphi) \subseteq M$, then $\varphi(t) \to M$: for every open $W \supseteq M$, $\varphi(t) \in W$ for all large t.*

Proof. Mathlib's `eventually_mapsTo_of_isCompact_absorbing_of_isOpen_of_omegaLimit_subset` (lemma 54): a family eventually absorbed by a compact set is eventually inside every open set containing its ω-limit set. □

Chapter 3

Systems

The most general systems the library considers are $\dot{x} = F(t, x, u)$ in continuous time and $x^+ = F(k, x, u)$ in discrete time, each with an output $y = h(t, x, u)$. As for the linear systems of the next chapter, there is one object per time domain: the data has the same shape, but a continuous trajectory is an integral curve and a discrete one is a recursion.

Special cases are constructors on the general object, each with a predicate recognising it: *time invariance* drops t, and *autonomy* drops u. Autonomy is used here in the control-theory sense, *without input*; much of the dynamical-systems literature uses the word for time invariance instead.

3.1 Continuous time

Definition 58 (Continuous-time system)**.** A *continuous-time system* with state space E, input space U and output space Y is the data of two maps

$$F : \mathbb{R} \times E \times U \to E, \qquad h : \mathbb{R} \times E \times U \to Y,$$

to be read as $\dot{x} = F(t, x, u)$, $y = h(t, x, u)$. The structure is only the data.

Definition 59 (Time-invariant continuous-time system)**.** The continuous-time system $\dot{x} = f(x, u)$, $y = g(x, u)$, whose equations do not depend on time.

Definition 60 (Time invariance, continuous-time system)**.** A continuous-time system is *time-invariant* when $F(t, \cdot, \cdot)$ and $h(t, \cdot, \cdot)$ are the same for every t.

Lemma 61 (Time-invariant continuous-time systems are time-invariant)**.** *The system of definition 59 is time-invariant (definition 60).*

Proof. □

Lemma 62 (Time-invariant continuous-time systems come from time-free data)**.** *A time-invariant continuous-time system (definition 60) is* timeInvariant *f g for some f and g. The converse of lemma 61.*

Proof. Read f and g off at $t = 0$; time invariance says every other time agrees with it. □

Definition 63 (Autonomous continuous-time system)**.** The continuous-time system $\dot{x} = f(t, x)$, $y = g(t, x)$, which has no input: its input space is the one-point type.

Definition 64 (Autonomy, continuous-time system)**.** A continuous-time system is *autonomous* when $F(t, x, u)$ and $h(t, x, u)$ do not depend on u.

Lemma 65 (Autonomous continuous-time systems are autonomous)**.** *The system of definition 63 is autonomous (definition 64).*

Proof. □

Lemma 66 (Autonomous continuous-time systems come from input-free data)**.** *If a continuous-time system is autonomous (definition 64) and its input space is nonempty, there are f and g with $F(t, x, u) = f(t, x)$ and $h(t, x, u) = g(t, x)$ for every input u.*

Proof. Fix any input u_0 and read f and g off at it. □

Definition 67 (The unforced system)**.** The autonomous system (definition 63) $\dot{x} = F(t, x, 0)$, $y = h(t, x, 0)$ obtained by holding the input at 0. Stability of a system with input means stability of this system.

Lemma 68 (Trajectories of the unforced system)**.** *A map is a trajectory of the unforced system (definition 67) on I exactly when it is a trajectory of s under the zero input on I.*

Proof. □

Well-posedness is kept apart from stability. The stability notions quantify over every solution, so they need no uniqueness; forward completeness enters only where a limit as $t \to \infty$ is taken. Theorems that need either property assume it.

Definition 69 (Forward completeness from a set)**.** A continuous-time system is *forward complete* from a set S at time t_0 under the input u when every trajectory on a segment $[t_0, t_1]$ (definition 76) with $x(t_0) \in S$ agrees on that segment with a trajectory on $[t_0, \infty)$.

Definition 70 (Forward completeness)**.** A continuous-time system is *forward complete* under the input u when it is forward complete (definition 69) from every state at every initial time.

Definition 71 (Uniqueness of solutions)**.** A continuous-time system has *unique solutions* under the input u when any two trajectories on a segment $[t_0, t_1]$ (definition 76) that agree at t_0 agree on all of $[t_0, t_1]$.

Theorem 72 (Lipschitz systems have unique solutions)**.** *If $x \mapsto F(t, x, u(t))$ is K-Lipschitz for every t, the system has unique solutions under u (definition 71).*

Proof. Grönwall's inequality, in the form of Mathlib's `ODE_solution_unique_of_mem_Icc_right`: a derivative within $[t_0, t_1]$ at a point of $[t_0, t_1)$ is a right derivative there. □

Theorem 73 (Globally Lipschitz systems are forward complete)**.** *If $(t, x) \mapsto F(t, x, u(t))$ is continuous and K-Lipschitz in x uniformly in t, the system is forward complete under u (definition 70).*

Proof. From the endpoint $x(t_0)$ of a segment, Picard–Lindelöf gives a solution on $[t_0, \infty)$ (theorem 38), which agrees with the segment by uniqueness (theorem 72). □

3.1.1 The state transition map

The state transition map $\varphi(t; t_0, x_0)$ is the solution from x_0 at time t_0, evaluated at time t, on its maximal interval of existence. The state transition matrix of a linear system is its linear case.

Definition 74 (State equation of a continuous-time system)**.** Under an input signal $u : \mathbb{R} \to U$, the right-hand side of the state equation is the time-dependent vector field $(t, x) \mapsto F(t, x, u(t))$.

Definition 75 (Output equation of a continuous-time system)**.** Under an input signal u, the output equation is $(t, x) \mapsto h(t, x, u(t))$. An instantaneous relation: no derivative appears.

Definition 76 (Trajectory of a continuous-time system)**.** A map $x : \mathbb{R} \to E$ is a *trajectory* of s under the input u on a set $I \subseteq \mathbb{R}$ when

$$\dot{x}(t) = F(t, x(t), u(t)), \qquad \forall\, t \in I,$$

the derivative being taken within I. Stated on a set: a map can satisfy the equation on an interval without being defined for all time.

Definition 77 (Trajectory for all time of a continuous-time system)**.** A trajectory (definition 76) defined on all of $\mathbb{R}$.

Lemma 78 (Trajectories of a continuous-time system on the whole line)**.** *definition 76 at $I = \mathbb{R}$ is definition 77.*

Proof. □

Lemma 79 (Restriction of a trajectory of a continuous-time system)**.** *A trajectory for all time is a trajectory on every set.*

Proof. □

Inputs that switch — steps, sample-and-hold, bang-bang — produce solutions with corners at the switching times. With the input taken right-continuous, such a solution still has a right derivative everywhere, given by the new value of the input; these right trajectories are the solution concept for input-to-state stability, and every trajectory is one.

A set is forward invariant when trajectories that start in it stay in it.

Definition 80 (Forward invariant set)**.** A set $S \subseteq E$ is *forward invariant* (or positively invariant) for s under the input u when every trajectory on a segment $[t_0, t_1]$ (definition 76) that starts in S stays in S:

$$x(t_0) \in S \;\Rightarrow\; x(t) \in S \quad \forall\, t \in [t_0, t_1].$$

The derivative of a function along the system is the rate at which it changes along trajectories. It takes the input value as an argument: stability uses the input-free system, input-to-state notions a general input.

Definition 81 (Derivative along the system)**.** For $V : \mathbb{R} \times E \to \mathbb{R}$ and an input value $u \in U$, the *derivative of V along the system $\dot{x} = F(t, x, u)$* is

$$\dot{V}(t, x, u) = \frac{\partial V}{\partial t}(t, x) + \frac{\partial V}{\partial x}(t, x)\, F(t, x, u),$$

the Lie derivative of V along the field $F(\cdot, \cdot, u)$ (definition 44).

The derivative form above excludes the solutions produced by discontinuous inputs, which have corners where the input jumps. The integral form admits them, and is the one input-driven notions such as input-to-state stability are stated with; for continuous right-hand sides the two agree.

Definition 82 (Carathéodory trajectory of a continuous-time system)**.** A map $x : \mathbb{R} \to E$, with E a Banach space, is a *Carathéodory trajectory* of s under the input u on a set $I \subseteq \mathbb{R}$ when, for all $a \leq b$ in I, the map $\tau \mapsto F(\tau, x(\tau), u(\tau))$ is integrable on $[a, b]$ and

$$x(b) - x(a) = \int_a^b F(\tau, x(\tau), u(\tau))\, \mathrm{d}\tau.$$

Unlike definition 76 it does not require x to be differentiable, so it admits the solutions produced by discontinuous or merely measurable inputs, which input-to-state stability quantifies over.

Definition 83 (Carathéodory trajectory for all time)**.** A Carathéodory trajectory (definition 82) on all of $\mathbb{R}$.

Lemma 84 (Restriction of a Carathéodory trajectory)**.** *A Carathéodory trajectory on I is one on every subset of I; in particular one for all time is one on every set.*

Proof. □

Theorem 85 (Carathéodory and classical trajectories agree on a segment)**.** *Let $t_0 \leq t_1$ and let $\tau \mapsto F(\tau, x(\tau), u(\tau))$ be continuous on $[t_0, t_1]$. Then x is a Carathéodory trajectory on $[t_0, t_1]$ (definition 82) if and only if it is a trajectory there (definition 76).*

Proof. Both are equivalent to the integral equation anchored at t_0 (definition 23), by lemma 28. The anchored equation gives the equation between any $a \leq b$ by subtracting its values at b and a. □

Theorem 86 (Carathéodory and classical trajectories agree on an interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval (order-connected) and let $\tau \mapsto F(\tau, x(\tau), u(\tau))$ be continuous on I. Then x is a Carathéodory trajectory on I (definition 82) if and only if it is a trajectory there (definition 76).*

Order-connectedness cannot be dropped: on $\{0\} \cup [1, 2]$ the integral form relates $x(1)$ to $x(0)$, the derivative form does not.

Proof. Both notions reduce to closed segments inside I, where they agree by theorem 85. For the integral form, the equation between $a \leq b$ only involves $[a, b] \subseteq I$. For the derivative form, at $t \in I$ choose $a \leq t \leq b$ in I, taking $a < t$ if I has a point left of t and $a = t$ otherwise, and likewise for b. Then $[a, b] \subseteq I$ agrees with I near t, so a derivative within $[a, b]$ at t is one within I. □

3.2 Discrete time

Definition 87 (Discrete-time system)**.** A *discrete-time system* with state space E, input space U and output space Y is the data of two maps

$$F : \mathbb{N} \times E \times U \to E, \qquad h : \mathbb{N} \times E \times U \to Y,$$

to be read as $x^+ = F(k, x, u)$, $y = h(k, x, u)$. The same data as definition 58 with time in $\mathbb{N}$; what differs is the reading, a shift rather than a derivative.

Definition 88 (Time-invariant discrete-time system)**.** The discrete-time system $x^+ = f(x,u)$, $y = g(x,u)$, whose equations do not depend on time.

Definition 89 (Time invariance, discrete-time system)**.** A discrete-time system is *time-invariant* when $F(k,\cdot,\cdot)$ and $h(k,\cdot,\cdot)$ are the same for every k.

Lemma 90 (Time-invariant discrete-time systems are time-invariant)**.** *The system of definition 88 is time-invariant (definition 89).*

Proof. □

Lemma 91 (Time-invariant discrete-time systems come from time-free data)**.** *A time-invariant discrete-time system (definition 89) is* timeInvariant *f g for some f and g. The converse of lemma 90.*

Proof. Read f and g off at $k = 0$; time invariance says every other time agrees with it. □

Definition 92 (Autonomous discrete-time system)**.** The discrete-time system $x^+ = f(k,x)$, $y = g(k,x)$, which has no input: its input space is the one-point type.

Definition 93 (Autonomy, discrete-time system)**.** A discrete-time system is *autonomous* when $F(k,x,u)$ and $h(k,x,u)$ do not depend on u.

Lemma 94 (Autonomous discrete-time systems are autonomous)**.** *The system of definition 92 is autonomous (definition 93).*

Proof. □

Lemma 95 (Autonomous discrete-time systems come from input-free data)**.** *If a discrete-time system is autonomous (definition 93) and its input space is nonempty, there are f and g with $F(k,x,u) = f(k,x)$ and $h(k,x,u) = g(k,x)$ for every input u.*

Proof. Fix any input u_0 and read f and g off at it. □

Definition 96 (State equation of a discrete-time system)**.** Under an input sequence $u : \mathbb{N} \to U$, the right-hand side of the state equation is the time-dependent next-state map $(k,x) \mapsto F(k,x,u(k))$. The same expression as definition 74; what differs is what a trajectory does with it, a shift rather than a derivative.

Definition 97 (Output equation of a discrete-time system)**.** Under an input sequence u, the output equation is $(k,x) \mapsto h(k,x,u(k))$.

Definition 98 (Trajectory of a discrete-time system)**.** A sequence $x : \mathbb{N} \to E$ is a *trajectory* of s under the input u on a set $I \subseteq \mathbb{N}$ when

$$x(k+1) = F(k,x(k),u(k)), \qquad \forall\, k \in I.$$

Membership constrains the step *out of* k, so $I = [k_0,\infty)$ is the forward recursion from k_0, and the horizon $[k_0,k_1]$ corresponds to $I = [k_0,k_1)$.

Definition 99 (Trajectory for all time of a discrete-time system)**.** A trajectory (definition 98) satisfying the recursion at every $k \in \mathbb{N}$.

Lemma 100 (Trajectories of a discrete-time system on the whole domain)**.** *definition 98 at $I = \mathbb{N}$ is definition 99.*

Proof. □

Lemma 101 (Restriction of a trajectory of a discrete-time system)**.** *A trajectory at every time is a trajectory on every set.*

Proof. □

Unlike continuous time, a trajectory needs no hypothesis to exist or to be unique: the recursion defines it from any initial time and state for all later times. So no discrete-time solution escapes in finite time, and nothing like forward completeness is ever needed.

Lemma 102 (Trajectories of a discrete-time system exist)**.** *For every input u, initial time k_0 and state x_0 there is a trajectory (definition 98) on $[k_0, \infty)$ with $x(k_0) = x_0$.*

Proof. Define it by the recursion itself. □

Lemma 103 (Trajectories of a discrete-time system are unique)**.** *Two trajectories (definition 98) on $[k_0, \infty)$ under the same input with $x(k_0) = y(k_0)$ satisfy $x(k) = y(k)$ for every $k \geq k_0$.*

Proof. Induction on k. □

Lemma 104 (Trajectories of a time-invariant discrete-time system are iterates)**.** *If $F(k, x, u(k)) = f(x)$ for all k and x, every trajectory (definition 98) on $[k_0, \infty)$ satisfies $x(k_0 + n) = f^n(x(k_0))$ for every $n \in \mathbb{N}$.*

Proof. Induction on n. □

Definition 105 (The unforced discrete-time system)**.** The autonomous system (definition 92) $x^+ = F(k, x, 0)$, $y = h(k, x, 0)$ obtained by holding the input at 0. Stability of a system with input means stability of this system.

Lemma 106 (Trajectories of the unforced discrete-time system)**.** *A sequence is a trajectory of the unforced system (definition 105) on I exactly when it is a trajectory of s under the zero input on I.*

Proof. □

Chapter 4

Contraction theory

A *pairing* replaces the inner product in contraction analysis for norms that come from no inner product. Regular pairings are the pairings for which the norm of a curve can be differentiated through the pairing, which is what contraction arguments use.

4.1 Regular pairings

Every normed space carries a regular pairing, built from the norm alone; on an inner product space the inner product is another.

Chapter 5

Linear systems

5.1 The linear system objects

Linear systems are a special case of the general systems of the Systems chapter: the continuous-time system with $F(t,x,u) = A(t)x + B(t)u$ and $h(t,x,u) = C(t)x + D(t)u$, and its discrete-time counterpart. The objects below carry only the coefficient data. Each is the input of a constructor into the general system, and the semantics — state equation, output equation, trajectories — are defined through that constructor, so every result about general systems applies to linear ones unchanged.

There are two objects per time domain: a *linear time-varying* (LTV) system with coefficient maps $A(t), B(t), C(t), D(t)$, and a *linear time-invariant* (LTI) system with constant matrices. An LTI system is not a different kind of system: it builds the LTV system with constant coefficients, and the general system it describes is the time-invariant system of the Systems chapter, by definition.

5.1.1 Continuous-time LTV systems

Definition 107 (Continuous-time linear time-varying system)**.** A *continuous-time linear time-varying (LTV) system* with states indexed by X, inputs by U, outputs by Y and coefficients in $\mathbb{K}$ is the data of four matrix-valued maps on $\mathbb{R}$,

$$A : \mathbb{R} \to \mathbb{K}^{X\times X}, \quad B : \mathbb{R} \to \mathbb{K}^{X\times U}, \quad C : \mathbb{R} \to \mathbb{K}^{Y\times X}, \quad D : \mathbb{R} \to \mathbb{K}^{Y\times U},$$

describing the continuous-time system (definition 58) $\dot{x} = A(t)x + B(t)u$, $y = C(t)x + D(t)u$ (definition 108).

Definition 108 (The system of an LTV system)**.** The continuous-time system (definition 58) with $F(t,x,u) = A(t)x + B(t)u$ and $h(t,x,u) = C(t)x + D(t)u$.

Definition 109 (State equation of an LTV system)**.** The right-hand side of the state equation, $(t,z) \mapsto A(t)\,z + B(t)\,u(t)$: the vector field (definition 74) of definition 108.

Lemma 110 (The unforced state equation of an LTV system)**.** *At zero input the state equation (definition 109) reduces to* $(t,z) \mapsto A(t)\,z$.

Proof. □

Lemma 111 (The state equation of an LTV system at the origin)**.** *At $z = 0$ the state equation (definition 109) contributes only the input term, $A(t)\,0 + B(t)u(t) = B(t)u(t)$. With the input switched off the origin is therefore an equilibrium of the system.*

Proof. □

Definition 112 (Output equation of an LTV system)**.** The output equation, $(t, z) \mapsto C(t)\,z + D(t)\,u(t)$: the output map (definition 75) of definition 108.

Definition 113 (Trajectory of an LTV system)**.** A map x is a *trajectory* of s under the input u on a set $I \subseteq \mathbb{R}$ when it is a trajectory (definition 76) of definition 108:

$$\dot{x}(t) = A(t)\,x(t) + B(t)\,u(t), \qquad \forall\, t \in I.$$

Definition 114 (Trajectory for all time of an LTV system)**.** A trajectory (definition 77) for all time of definition 108.

Definition 115 (Carathéodory trajectory of an LTV system)**.** A Carathéodory trajectory (definition 82) of definition 108.

Definition 116 (Time invariance of an LTV system)**.** A continuous-time LTV system is *time-invariant* when each of its four coefficients takes the same value at every pair of times.

Lemma 117 (Time-invariant LTV systems are time-invariant systems)**.** *If an LTV system is time-invariant (definition 116), so is the system it describes (definition 60).*

Proof. □

5.1.2 Continuous-time LTI systems

Definition 118 (Continuous-time linear time-invariant system)**.** A *continuous-time linear time-invariant (LTI) system* is the data of four matrices $A \in \mathbb{K}^{X\times X}$, $B \in \mathbb{K}^{X\times U}$, $C \in \mathbb{K}^{Y\times X}$, $D \in \mathbb{K}^{Y\times U}$, describing the LTV system (definition 107) with these constant coefficients, $\dot{x} = Ax + Bu$, $y = Cx + Du$.

Definition 119 (The LTV system of an LTI system)**.** The LTV system (definition 107) whose coefficients are the constant maps $t \mapsto A$, $t \mapsto B$, $t \mapsto C$, $t \mapsto D$.

Lemma 120 (LTI systems are time-invariant LTV systems)**.** *The LTV system of an LTI system (definition 119) is time-invariant (definition 116).*

Proof. □

Definition 121 (The system of an LTI system)**.** The continuous-time system (definition 58) of the LTV system of s (definition 108, definition 119).

Lemma 122 (An LTI system is a linear time-invariant system)**.** *The system of an LTI system (definition 121) is the time-invariant system (definition 59) with $f(x, u) = Ax + Bu$ and $g(x, u) = Cx + Du$: the two routes from the LTI data to a general system agree.*

Proof. □

5.1.3 Discrete-time LTV systems

Definition 123 (Discrete-time linear time-varying system)**.** A *discrete-time linear time-varying (LTV) system* is the same data as definition 107 with the coefficients indexed by $\mathbb{N}$ instead of $\mathbb{R}$, describing the discrete-time system (definition 87) $x(k+1) = A(k)x(k) + B(k)u(k)$, $y(k) = C(k)x(k) + D(k)u(k)$ (definition 124).

The forward time domain $\mathbb{N}$ rather than $\mathbb{Z}$ matches the discrete solution theory, whose state transition matrix is an ordered product over $[t_0, t)$.

Definition 124 (The system of a discrete-time LTV system)**.** The discrete-time system (definition 87) with $F(k, x, u) = A(k)x + B(k)u$ and $h(k, x, u) = C(k)x + D(k)u$.

Definition 125 (State equation of a discrete-time LTV system)**.** The right-hand side of the state equation, $(k, z) \mapsto A(k)\, z + B(k)\, u(k)$: the next-state map (definition 96) of definition 124.

Lemma 126 (The unforced state equation of a discrete-time LTV system)**.** *At zero input the state equation (definition 125) reduces to* $(k, z) \mapsto A(k)\, z$.

Proof. □

Lemma 127 (The state equation of a discrete-time LTV system at the origin)**.** *At* $z = 0$ *the state equation (definition 125) contributes only the input term,* $A(k)\, 0 + B(k)u(k) = B(k)u(k)$. *With the input switched off the origin is therefore an equilibrium of the system.*

Proof. □

Definition 128 (Output equation of a discrete-time LTV system)**.** The output equation, $(k, z) \mapsto C(k)\, z + D(k)\, u(k)$: the output map (definition 97) of definition 124.

Definition 129 (Trajectory of a discrete-time LTV system)**.** A sequence x is a *trajectory* of s under the input u on a set $I \subseteq \mathbb{N}$ when it is a trajectory (definition 98) of definition 124:

$$x(k+1) = A(k)\, x(k) + B(k)\, u(k), \qquad \forall\, k \in I.$$

Definition 130 (Trajectory for all time of a discrete-time LTV system)**.** A trajectory (definition 99) for all time of definition 124.

Definition 131 (Time invariance of a discrete-time LTV system)**.** A discrete-time LTV system is *time-invariant* when each of its four coefficients takes the same value at every pair of times.

Lemma 132 (Time-invariant discrete-time LTV systems are time-invariant systems)**.** *If a discrete-time LTV system is time-invariant (definition 131), so is the system it describes (definition 89).*

Proof. □

5.1.4 Discrete-time LTI systems

Definition 133 (Discrete-time linear time-invariant system)**.** A *discrete-time linear time-invariant (LTI) system* is the data of four matrices A, B, C, D, describing the discrete-time LTV system (definition 123) with these constant coefficients, $x(k+1) = Ax(k) + Bu(k)$, $y(k) = Cx(k) + Du(k)$.

Definition 134 (The LTV system of a discrete-time LTI system)**.** The discrete-time LTV system (definition 123) whose coefficients are the constant maps $k \mapsto A$, $k \mapsto B$, $k \mapsto C$, $k \mapsto D$.

Lemma 135 (Discrete-time LTI systems are time-invariant LTV systems)**.** *The LTV system of a discrete-time LTI system (definition 134) is time-invariant (definition 131).*

Proof. □

Lemma 136 (Time-invariant discrete-time LTV systems come from LTI systems)**.** *A time-invariant discrete-time LTV system (definition 131) is the LTV system of some LTI system. The converse of lemma 135.*

Proof. Read the coefficients off at $k = 0$. □

Definition 137 (The system of a discrete-time LTI system)**.** The discrete-time system (definition 87) of the LTV system of s (definition 124, definition 134).

Lemma 138 (A discrete-time LTI system is a linear time-invariant system)**.** *The system of a discrete-time LTI system (definition 137) is the time-invariant system (definition 88) with $f(x, u) = Ax + Bu$ and $g(x, u) = Cx + Du$: the two routes from the LTI data to a general system agree.*

Proof. □

5.2 Peano–Baker series and the state transition matrix

For a genuinely time-varying coefficient $A(t)$ in a real Banach algebra — matrices under any submultiplicative norm, or the bounded operators on a Banach space — this section constructs the state transition matrix $\Phi(t, t_0)$ as the Peano–Baker series and proves Theorem 5.1: Φ solves $\dot{\Phi}(t, t_0) = A(t)\Phi(t, t_0)$ with $\Phi(t_0, t_0) = 1$, and consequently $x(t) := \Phi(t, t_0)x_0$ solves $\dot{x} = A(t)x$, $x(t_0) = x_0$.

Standing hypothesis. Throughout this section A is continuous. The term-wise estimates additionally assume a bound $\|A(s)\| \le M$ on the segment between the two times named; the results about Φ itself need none, since A is bounded on every compact interval.

Definition 139 (Peano–Baker term)**.** The k-th term of the Peano–Baker series for a time-varying state matrix $A(t)$:

$$P_0(t, t_0) := I, \qquad P_{k+1}(t, t_0) := \int_{t_0}^{t} A(s)\, P_k(s, t_0)\, \mathrm{d}s.$$

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5 (Peano-Baker series).

Definition 140 (State transition matrix)**.** The *state transition matrix* $\Phi(t, t_0)$, given by the Peano–Baker series

$$\Phi(t, t_0) := \sum_{k=0}^{\infty} P_k(t, t_0).$$

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, Theorem 5.1 (Peano-Baker series).

Lemma 141 (Continuity of the Peano–Baker terms)**.** *Each term $P_k(t, t_0)$ of the Peano–Baker series is continuous in t, for A continuous.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Induction on k: P_0 is constant, and P_{k+1} is the primitive of a continuous integrand (definition 139), hence continuous by the fundamental theorem of calculus. □

Lemma 142 (Factorial bound on the Peano–Baker terms)**.** *If $\|A(s)\| \le M$ for all s between t_0 and t_1, then*

$$\|P_k(t, t_0)\| \;\le\; \|I\| \cdot \frac{(M|t - t_0|)^k}{k!} \qquad \text{for every } t \text{ between } t_0 \text{ and } t_1.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Induction on k: bound the integrand $A(s)P_k(s, t_0)$ pointwise on the segment between t_0 and t, using submultiplicativity of the matrix norm and the induction hypothesis. Split on whether $t_0 \le t$ or $t \le t_0$ and integrate the majorant via the power rule, flipping the integral's orientation (`intervalIntegral.integral_symm`) in the backward case. □

Lemma 143 (Summability of the Peano–Baker series)**.** *The Peano–Baker series $\sum_{k=0}^{\infty} P_k(t, t_0)$ converges absolutely, for A bounded between t_0 and t (in either order).*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Compare against the exponential series $\sum_k (M|t-t_0|)^k/k!$ via lemma 142 and the comparison test. □

Theorem 144 (The state transition matrix solves the matrix ODE)**.** *The state transition matrix solves the matrix ODE*

$$\dot{\Phi}(t, t_0) = A(t)\, \Phi(t, t_0)$$

for t strictly between t_0 and t_1.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Differentiate the Peano–Baker series term by term, using lemma 142 as the uniform bound needed for term-by-term differentiation of a series. The $k = 0$ term is constant, so peel it off and match the shifted series' derivative against $A(t)\Phi(t, t_0)$ using linearity of the sum under left multiplication by $A(t)$. □

Theorem 145 (The state transition matrix at the initial time)**.** $\Phi(t_0, t_0) = I$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Every term of the Peano–Baker series beyond $k = 0$ integrates over the degenerate interval $[t_0, t_0]$ and vanishes. □

5.2.1 Uniqueness

5.2.2 Semigroup, invertibility and composition

Uniqueness is what makes these available: two expressions that solve the same matrix initial value problem must agree, and each identity below is an instance of that.

Theorem 146 (Semigroup property for a globally bounded system)**.** *If A is continuous and bounded on $\mathbb{R}$, then $\Phi(t,s)\,\Phi(s,\tau) = \Phi(t,\tau)$ for all t, s, τ.*

Proof. One of the three times lies between the other two. If it is s, this is **??**. Otherwise compose through the middle time and cancel $\Phi(\tau,s)\Phi(s,\tau) = I$ (**??**). □

Theorem 147 (Derivative in the initial time)**.** *With the first argument held fixed, the state transition matrix is differentiable in its second argument, with*

$$\frac{\partial}{\partial t}\Phi(\tau,t) = -\Phi(\tau,t)\,A(t)$$

for t strictly between τ and t_1 — the mirror of theorem 144, which differentiates in the first argument.

Reference: Khalil, Nonlinear Systems *(3rd ed.), proof of Theorem 4.12.*

Proof. By **??**, $\Phi(\tau,t) = \Phi(t,\tau)^{-1}$. Differentiate $t \mapsto \Phi(t,\tau)$ by theorem 144 and the inverse by $d(M^{-1}) = -M^{-1}\,dM\,M^{-1}$: $-\Phi(\tau,t)\,A(t)\,\Phi(t,\tau)\,\Phi(\tau,t) = -\Phi(\tau,t)\,A(t)$. □

Lemma 148 (Exponential growth bound)**.** *If $\|A(s)\| \le M$ on the segment between t_0 and t, then*

$$\|\Phi(t,t_0)\| \le \|I\|\,e^{M|t-t_0|}.$$

Proof. Sum the term bounds of lemma 142: they are $\|I\|$ times the terms of the exponential series. □

The matrix-level forms below, with the times ordered, predate the general results above and are being retired.

5.2.3 Variation of constants

With a forcing term $B(t)u(t)$, the solution acquires a convolution against Φ. The difficulty is that the elapsed time appears both as the upper limit of integration and inside the integrand; the proof factors the two apart rather than differentiating Φ in its second argument.

5.2.4 The forced response as a trajectory

The solution theory above is stated on the coefficient maps A and B, which is where the analysis lives. The results below restate it for the system object of definition 107: the variation-of-constants formula is a *trajectory* in the sense of definition 114, which the stability theory quantifies over.

Definition 149 (Forced response)**.** The *forced response* of $\dot{x} = A(t)x + B(t)u(t)$ from the state x_0 at time t_0, given by the variation-of-constants formula

$$x(t) = \Phi(t,t_0)\,x_0 + \int_{t_0}^{t} \Phi(t,\tau)\,B(\tau)\,u(\tau)\,\mathrm{d}\tau,$$

with Φ the state transition matrix (definition 140).

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, Theorem 5.2.

Theorem 150 (The forced response is a trajectory)**.** *Let s be a continuous-time linear system with continuous coefficients A and B, and let u be a continuous input. Then the forced response (definition 149) from any state x_0 at any time t_0 is a trajectory of s (definition 114) — it satisfies the state equation at every $t \in \mathbb{R}$, with a two-sided derivative.*

Restriction to an interval, in particular to the ray $[t_0, \infty)$ that the stability predicates quantify over, is lemma 79.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Theorem 5.2.*

Proof. For $t \neq t_0$, apply **??** on a window having t in its interior; since the window is unordered this covers $t < t_0$ as well as $t > t_0$, and continuity of A supplies the bound on the compact segment.

At $t = t_0$ no such window exists, because the formula is anchored there and t_0 is an endpoint of every window. Instead take the integral form (**??**) on $[t_0, t_0 + 1]$ and on $[t_0 - 1, t_0]$; each yields a one-sided derivative at t_0 (lemma 27), and the two glue to a two-sided one. □

Theorem 151 (The homogeneous response is a trajectory)**.** *For a continuous-time linear system with continuous coefficients, the homogeneous response (**??**) from any state x_0 at any time t_0 is a trajectory (definition 114) under zero input.*

Definition 8.1 is stated about this map, so this is the form in which the solution theory reaches the stability theory.

Proof. theorem 150 at a system with no inputs, where the forcing integral vanishes (**??**). The dummy system is an artefact of the proof: only A appears in the statement, and no hypothesis on B is needed. □

Theorem 152 (The homogeneous response is a trajectory on the forward ray)**.** *The homogeneous response (**??**) is a trajectory of the unforced system (definition 113) on $[t_0, \infty)$.*

Proof. theorem 151, restricted to the ray by lemma 79. □

Theorem 153 (Trajectories of the unforced system are homogeneous responses)**.** *Let s have continuous A and let φ be a trajectory of the unforced system on $[t_0, t_1]$ (definition 113). Then $\varphi(t) = \Phi(t, t_0)\,\varphi(t_0)$ for every $t \in [t_0, t_1]$.*

Proof. **??**, after reducing the unforced state equation by lemma 110. □

Theorem 154 (Linear systems are forward complete)**.** *If A is continuous, the unforced system (definition 67) of definition 108 is forward complete (definition 70).*

Proof. A trajectory on $[t_0, t_1]$ agrees there with the homogeneous response from its initial state (theorem 153), which is a trajectory on $[t_0, \infty)$ (theorem 152). □

5.2.5 The time-invariant case: the matrix exponential

For a constant state matrix the Peano–Baker series sums to the matrix exponential, which is Hespanha's (6.1)–(6.2). The point of proving it is economy: every property of Φ established above for a time-varying A then holds for $e^{A(t-t_0)}$ by specialization, with no separate LTI theory. The two exponential facts that need no solution theory at all — the semigroup law (P6.3) and the

finite-polynomial form via Cayley–Hamilton (P6.5) — are proved as statements about exp and A alone.

Lemma 155 (Peano–Baker terms for a constant matrix)**.** *For a constant state matrix A, the k-th Peano–Baker term collapses to*

$$\Phi_k(t, t_0) = \frac{(t - t_0)^k}{k!} A^k.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, equation (6.1).*

Proof. Induction on k. The $k = 0$ term is the identity. For the step, $\Phi_{k+1}(t, t_0) = \int_{t_0}^{t} A\,\Phi_k(s, t_0)\,ds$; substituting the inductive hypothesis pulls A^{k+1} out of the integral and leaves $\int_{t_0}^{t}(s - t_0)^k ds = (t - t_0)^{k+1}/(k + 1)$, which combines with $1/k!$ to give $1/(k + 1)!$. □

Lemma 156 (The LTI state transition matrix as a series)**.** *For a constant state matrix A, the state transition matrix is the power series*

$$\Phi(t, t_0) = \sum_{k=0}^{\infty} \frac{(t - t_0)^k}{k!} A^k.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, equation (6.1).*

Proof. Sum lemma 155 termwise; the Peano–Baker series converges for any continuous state matrix, a constant one included. □

Theorem 157 (The LTI state transition matrix is $e^{A(t-t_0)}$)**.** *For a constant state matrix A, the state transition matrix is the matrix exponential:*

$$\Phi(t, t_0) = e^{A(t-t_0)}.$$

This is the bridge that lets every property already proved for Φ in the time-varying setting — existence, uniqueness, the semigroup law, invertibility, variation of constants — be reused for LTI systems by specialization rather than reproved.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, equation (6.2).*

Proof. Both sides are power series in A: lemma 156 for the left, the defining series of exp for the right. Match them termwise. □

5.2.6 The forced response of a time-invariant system

For a time-invariant system the state transition matrix is the matrix exponential, so the variation-of-constants formula collapses to the closed form every textbook writes.

Lemma 158 (Forced response of a time-invariant system)**.** *For constant A and B the forced response (definition 149) is*

$$x(t) = e^{A(t-t_0)} x_0 + \int_{t_0}^{t} e^{A(t-\tau)}\, B\, u(\tau)\, \mathrm{d}\tau.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6.*

Proof. theorem 157 identifies Φ with the matrix exponential, applied once to the homogeneous term and once under the integral sign. □

Theorem 159 (The LTI response is a trajectory)**.** *For an LTI system (definition 118) with continuous input u, the closed form*

$$t \mapsto e^{A(t-t_0)}x_0 + \int_{t_0}^{t} e^{A(t-\tau)}\, B\, u(\tau)\, \mathrm{d}\tau$$

is a trajectory (definition 114) of that system.

Proof. theorem 150 at the LTV system of the LTI system (definition 119), whose coefficient maps are constant and so continuous, rewritten by lemma 158. □

5.3 The matrix exponential and complexification

5.3.1 Complexification

Spectral arguments must happen over $\mathbb{C}$, because $\mathbb{R}$ is not algebraically closed, while the systems of interest have real coefficients. Entrywise complexification is the bridge, and these record what it commutes with. The one with content is the transfer of positive definiteness: a real positive-definite matrix remains positive definite against *complex* vectors, which is what lets a Lyapunov certificate be evaluated against an eigenvector.

Definition 160 (Complexification of a real matrix)**.** For a real matrix A, its *complexification* $A_{\mathbb{C}}$ is obtained by applying $\mathbb{R} \hookrightarrow \mathbb{C}$ to each entry.

Lemma 161 (Positive definiteness survives complexification)**.** *If a real matrix P is positive definite, so is its complexification $P_{\mathbb{C}}$, as a Hermitian matrix over $\mathbb{C}$.*

Proof. Write a complex vector as $v = a + ib$ with a, b real. Then $v^* P_{\mathbb{C}} v = a^{\mathsf{T}} P a + b^{\mathsf{T}} P b$: the imaginary part cancels because P is symmetric, and the real part is positive because $v \neq 0$ forces a or b to be nonzero. □

Over a field the spectrum of a matrix is its set of eigenvalues, which is how a condition on eigenpairs meets the spectral radius.

Lemma 162 (The spectrum of a matrix is its set of eigenvalues)**.** *For a square matrix M over a field K, $\mu \in \sigma(M)$ if and only if $Mv = \mu v$ for some $v \neq 0$.*

Proof. □

5.3.2 Spectral facts about the exponential

These say nothing about trajectories, initial times or state transition matrices — they are facts about exp and A alone, and are proved as such. The complexification bridge is what lets spectral arguments, which must happen over $\mathbb{C}$, yield bounds on a real exponential; P6.5 collapses the exponential series to a finite polynomial in A. The eigenpair lemma is the half of spectral mapping that is pure power-series algebra, and is what turns an eigenvalue condition on A into a statement about the growth of e^{At}.

Lemma 163 (Complexification commutes with the matrix exponential)**.** *Entrywise complexification commutes with the matrix exponential: for a real matrix A,*

$$\overline{\exp(A)} = \exp(\overline{A}),$$

where $\overline{\ }$ denotes applying $\mathbb{R} \hookrightarrow \mathbb{C}$ entrywise. This is the bridge used to transfer spectral facts, which live over $\mathbb{C}$, to norm bounds on a real exponential.

Proof. Entrywise complexification is a continuous ring homomorphism, and exp commutes with any such. □

Lemma 164 (The exponential acts on an eigenvector by exponentiating its eigenvalue)**.** *Let M be a square matrix and let (μ, v) be an eigenpair of M, so that $Mv = \mu v$. Then*

$$e^M v = e^\mu\, v.$$

Reference: standard power-series functional calculus for the exponential.

Proof. Induction gives $M^k v = \mu^k v$ for every k. The map $N \mapsto Nv$ is linear and continuous, so it commutes with the sum defining $e^M = \sum_k (k!)^{-1} M^k$, giving $e^M v = \sum_k (k!)^{-1} \mu^k v = e^\mu v$. □

Lemma 165 (A scalar multiple scales the eigenvalues)**.** *If $Av = \mu v$ then $(cA)v = (c\mu)v$.*

Proof. □

Lemma 166 (Powers power the eigenvalues)**.** *If $Av = \mu v$ then $A^k v = \mu^k v$ for every k.*

Proof. □

Lemma 167 (A scalar shift shifts the eigenvalues)**.** *For a square matrix A over a commutative ring and scalars α, μ, a vector v satisfies $(A + \alpha I)v = (\mu + \alpha)v$ if and only if $Av = \mu v$.*

Proof. □

Lemma 168 (The kernel form of an eigenpair)**.** *$(\mu I - A)v = 0$ if and only if $Av = \mu v$.*

Proof. □

Lemma 169 (Negative spectrum gives a contractive exponential)**.** *If every eigenpair (μ, v) of the complexification of a real matrix A, with $v \neq 0$, has $\mathrm{Re}(\mu) < 0$, then $r(e^{A_{\mathbb{C}}}) < 1$.*

Reference: standard spectral mapping for the matrix exponential.

Proof. Reverse spectral mapping: every spectral value of $e^{A_{\mathbb{C}}}$ is e^μ for an eigenvalue μ of $A_{\mathbb{C}}$, and $|e^\mu| = e^{\mathrm{Re}\,\mu} < 1$. □

Theorem 170 (Semigroup property of the matrix exponential)**.** *Hespanha, P6.3. For every $t, \tau \in \mathbb{R}$,*

$$e^{At} e^{A\tau} = e^{A(t+\tau)}.$$

Proof. tA and τA commute, being scalar multiples of the same matrix, so the exponential of the sum factors. The general $e^X e^Y = e^{X+Y}$ needs commutativity and is false without it; here it is free. □

Theorem 171 (The matrix exponential as a finite polynomial)**.** *Hespanha, P6.5. There are scalar functions* $\alpha_0, \dots, \alpha_{n-1} : \mathbb{R} \to \mathbb{R}$ *with*

$$e^{At} = \sum_{i=0}^{n-1} \alpha_i(t)\, A^i \qquad \text{for every } t.$$

The matrix exponential — an infinite series in A *— is a* finite *polynomial in* A *of degree less than* n*, at the cost of coefficients that are transcendental in* t*.*

Proof. Cayley–Hamilton reduces every power A^k to a combination of $A^0, \dots, A^{n-1}$, say with coefficients $\bar{a}_i(k)$; substituting into the exponential series and exchanging the finite sum over i with the infinite sum over k exhibits $\alpha_i(t) = \sum_k t^k \bar{a}_i(k)/k!$ as a witness.

The exchange needs those series to converge, which holds because the reduced coefficients grow at most geometrically, $|\bar{a}_i(k)| \le CM^k$, dominating each α_i by a scalar exponential series. That growth bound is the crux of the proof and is established by iteration, not by spectral theory: "multiply by X, reduce modulo the characteristic polynomial" is a fixed linear operation on the n-dimensional space of polynomials of degree $< n$, so one step scales the ℓ^1 size of the coefficient vector by at most a constant factor. No eigenvalues and no Jordan form appear anywhere. □

5.4 The discrete-time state transition matrix

For $x(t+1) = A(t)x(t)$ the state transition matrix is an ordered product rather than a series, and the whole theory becomes finite: every proof below is an induction, with no convergence, no differentiation and no Gronwall estimate. Two differences from the continuous case are substantive rather than merely technical. There is no standing bound on A to carry, since a finite product always converges; and Φ need not be invertible, so the semigroup property genuinely requires $\tau \le s \le t$ instead of holding in any order.

Definition 172 (Discrete state transition matrix)**.** The discrete-time *state transition matrix* $\Phi(t, t_0)$:

$$\Phi(t, t_0) := \begin{cases} A(t-1)A(t-2)\cdots A(t_0+1)A(t_0) & t > t_0, \\ I & t \le t_0. \end{cases}$$

Formally, the ordered product of A over $[t_0, t)$, which for $t \le t_0$ is the empty product I. The system runs forward only, so the $t < t_0$ branch carries no meaning; it is there to make the definition total.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, equation (5.11).

Lemma 173 (The discrete state transition matrix at the initial time)**.** $\Phi(t_0, t_0) = I$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Property P5.5.*

Proof. The defining product is over the empty range $[t_0, t_0)$. □

Lemma 174 (One-step recursion)**.** *For* $t \ge t_0$*,* $\Phi(t+1, t_0) = A(t)\,\Phi(t, t_0)$*. With lemma 173 this is the existence half of Hespanha's P5.5: the discrete state transition matrix solves the matrix recursion.*

Proof. Peel the leading factor $A(t)$ off the defining product. □

Theorem 175 (Discrete state transition matrix: uniqueness)**.** *Hespanha, P5.5. If $Z(t_0) = I$ and $Z(t+1) = A(t)Z(t)$ for all $t \geq t_0$, then $Z(t) = \Phi(t, t_0)$ for all $t \geq t_0$.*

Proof. Induction on t from t_0: the base case is lemma 173 and the step chains the recursion against lemma 174. No analytic content at all, unlike the continuous-time counterpart — the recursion pins each value outright. □

Theorem 176 (Uniqueness of the discrete state response)**.** *If $z(t_0) = x_0$ and $z(t+1) = A(t)z(t)$ for all $t \geq t_0$, then $z(t) = \Phi(t, t_0)x_0$ for all $t \geq t_0$: the vector form of theorem 175.*

Proof. □

Theorem 177 (Columns of the discrete state transition matrix)**.** *Hespanha, P5.6. For each i, the i-th column of $\Phi(t, t_0)$ is the unique solution of $z(t+1) = A(t)z(t)$ with $z(t_0) = e_i$, for $t \geq t_0$.*

Proof. theorem 176 at $x_0 = e_i$, using $\Phi(t, t_0)e_i = (\Phi(t, t_0))_{\cdot,i}$. □

Theorem 178 (Semigroup property, discrete)**.** *Hespanha, P5.7. For $\tau \leq s \leq t$,*

$$\Phi(t, s)\,\Phi(s, \tau) = \Phi(t, \tau).$$

Unlike the continuous-time case there is no inverse: Φ is a product of matrices that need not be nonsingular, so the ordering $\tau \leq s \leq t$ is a genuine restriction here rather than an artifact.

Proof. Induction on t from s, chaining lemma 174 at both s and τ. □

5.4.1 Discrete variation of constants

Lemma 179 (Discrete variation of constants at the initial time)**.** *The discrete variation-of-constants formula*

$$x(t) := \Phi(t, t_0)x_0 + \sum_{\tau=t_0}^{t-1} \Phi(t, \tau+1)B(\tau)u(\tau)$$

takes the value x_0 at $t = t_0$: the forcing sum is over the empty range and $\Phi(t_0, t_0) = I$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Section 5.3 (discrete variation of constants).*

Proof. □

Theorem 180 (Discrete variation of constants: existence)**.** *For $t \geq t_0$, the formula of lemma 179 satisfies the forced recursion*

$$x(t+1) = A(t)\,x(t) + B(t)\,u(t).$$

Proof. Peel the top term off the sum defining $x(t+1)$ and fold $A(t)\Phi(t, \tau+1)$ into $\Phi(t+1, \tau+1)$ termwise by lemma 174, applied once more to the leading $\Phi(t, t_0)$ term. The peeled term is exactly $B(t)u(t)$, since $\Phi(t+1, t+1) = I$. □

Theorem 181 (Discrete variation of constants: uniqueness)**.** *Any two solutions of $z(t+1) = A(t)z(t) + B(t)u(t)$ with the same value at t_0 agree for all $t \geq t_0$.*

Proof. Induction on t from t_0. Unlike **??**, no Lipschitz condition or Gronwall estimate is needed: the recursion determines each successive value outright, so uniqueness is immediate. □

5.4.2 The discrete responses as trajectories

The variation-of-constants formula and its unforced part, named, and read on the system object (definition 123): they are exactly the trajectories of the system from a given initial state, which is the form in which the solution theory reaches the stability theory.

Definition 182 (Forced response, discrete time)**.** The *forced response* of $x(t+1) = A(t)x(t) + B(t)u(t)$ from the state x_0 at time t_0,

$$x(t) = \Phi(t, t_0)\, x_0 + \sum_{\tau=t_0}^{t-1} \Phi(t, \tau+1)\, B(\tau)\, u(\tau),$$

with Φ the discrete state transition matrix (definition 172). It solves the recursion forward from t_0 only.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, Section 5.3.

Definition 183 (Homogeneous response, discrete time)**.** The *homogeneous response* of $x(t+1) = A(t)x(t)$ from the state x_0 at time t_0, $x(t) = \Phi(t, t_0)\, x_0$: the forced response (definition 182) with the input switched off.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, Section 5.3.

Lemma 184 (The discrete forced response at the initial time)**.** *The forced response (definition 182) takes the value x_0 at $t = t_0$.*

Proof. lemma 179. □

Lemma 185 (The discrete forced response solves the recursion)**.** *For $t \geq t_0$ the forced response (definition 182) satisfies $x(t+1) = A(t)x(t) + B(t)u(t)$.*

Proof. theorem 180. □

Lemma 186 (The discrete homogeneous response at the initial time)**.** *The homogeneous response (definition 183) takes the value x_0 at $t = t_0$.*

Proof. $\Phi(t_0, t_0) = I$ (lemma 173). □

Lemma 187 (The discrete homogeneous response solves the unforced recursion)**.** *For $t \geq t_0$ the homogeneous response (definition 183) satisfies $x(t+1) = A(t)x(t)$.*

Proof. $\Phi(t+1, t_0) = A(t)\Phi(t, t_0)$ (lemma 174). □

Theorem 188 (The discrete state transition matrix of an LTI system)**.** *For a constant state matrix A, the discrete state transition matrix (definition 172) is $\Phi(t, t_0) = A^{t-t_0}$.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Section 5.3.*

Proof. $\Phi(t, t_0)$ is the ordered product of $t - t_0$ factors, all equal to A. □

Theorem 189 (The discrete forced response is a trajectory)**.** *For a discrete-time LTV system s and input u, the forced response (definition 182) from x_0 at t_0 is a trajectory of s under u on $[t_0, \infty)$ (definition 129).*

Proof. lemma 185, which is the trajectory condition on the nose. □

Theorem 190 (Discrete trajectories are forced responses)**.** *If x is a trajectory of s under u on $[t_0, \infty)$ (definition 129), then $x(t)$ is the forced response (definition 182) from $x(t_0)$ at t_0, for every $t \geq t_0$.*

Proof. Both satisfy the forced recursion from the same initial value, so they agree by theorem 181. □

Theorem 191 (The discrete homogeneous response is a trajectory)**.** *The homogeneous response (definition 183) from x_0 at t_0 is a trajectory of the unforced system on $[t_0, \infty)$ (definition 129).*

Proof. lemma 187, after reducing the unforced state equation by lemma 126. □

Theorem 192 (Discrete unforced trajectories are homogeneous responses)**.** *If x is a trajectory of the unforced system on $[t_0, \infty)$ (definition 129), then $x(t) = \Phi(t, t_0)\, x(t_0)$ for every $t \geq t_0$.*

Proof. theorem 176, after reducing the unforced state equation by lemma 126. □

Chapter 6

Comparison functions

Comparison functions are the vocabulary of stability estimates: a class $\mathcal{K}$ function bounds how far a trajectory can go, a class $\mathcal{L}$ function how fast it decays, and a class $\mathcal{KL}$ function both at once. This chapter builds them, and the tools that turn a cruder bound into one of them.

6.1 Class $\mathcal{K}$ and $\mathcal{K}_\infty$ functions

Definition 193 (Class $\mathcal{K}$ function)**.** A *class* $\mathcal{K}$ function on $[0, a)$ is a continuous strictly increasing map $\alpha : [0, a) \to [0, b)$ with $\alpha(0) = 0$. All class K functions have an inverse $\alpha^{-1} : [0, b) \to [0, a)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.2 (class $\mathcal{K}$). Differences:

- Khalil writes the codomain as $[0, \infty)$; here the bound b is tracked in the type.
- The inverse is bundled as data rather than derived.

Definition 194 (Class $\mathcal{K}_\infty$ function)**.** A *class* $\mathcal{K}_\infty$ function is a continuous strictly increasing map $\alpha : [0, \infty) \to [0, \infty)$ with $\alpha(0) = 0$ and $\alpha(r) \to \infty$ as $r \to \infty$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.2 (class $\mathcal{K}_\infty$). The inverse is bundled as data rather than derived.

6.1.1 Building class $\mathcal{K}$ functions

A recognition criterion and the closure properties. The restatements of the definition itself (strict monotonicity as an iff, the two inverse equations, and so on) are left to the Lean API; they say nothing the definition above does not.

Definition 195 (Class $\mathcal{K}$ from a strictly increasing function)**.** Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(0) = 0$ and $f(a) = b$ and be continuous and strictly increasing on $[0, a]$, with $a, b > 0$. Then f is a class $\mathcal{K}$.

Definition 196 (Restriction of a class $\mathcal{K}$ function)**.** If α is class $\mathcal{K}$ on $[0, a) \to [0, b)$ and $0 < c < a$, then α restricted to $[0, c)$ is class $\mathcal{K}$ on $[0, c) \to [0, \alpha(c))$.

Definition 197 (Restriction to a prescribed codomain)**.** definition 196 with the codomain bound named: if $\alpha(c) = e$, the restriction of α to $[0, c)$ is class $\mathcal{K}$ on $[0, c) \to [0, e)$.

Definition 198 (Inverse of a class $\mathcal{K}$ function)**.** Class $\mathcal{K}$ is closed under inversion. The inverse of a class K function is also class K.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, first bullet. Khalil's domain $[0, \alpha(a))$ is the codomain bound b carried by the type.

Definition 199 (Composition of class $\mathcal{K}$ functions)**.** Class $\mathcal{K}$ is closed under composition: if α is class $\mathcal{K}$ on $[0, a) \to [0, b)$ and β is class $\mathcal{K}$ on $[0, b) \to [0, c)$, then $\beta \circ \alpha$ is class $\mathcal{K}$ on $[0, a) \to [0, c)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, third bullet.

6.1.2 Building class $\mathcal{K}_\infty$ functions

The same recognition criterion and closure properties on the unbounded domain, together with the closure properties that need the global domain (sum, positive scaling, minimum), the power functions as the standard examples, and the restriction back to class $\mathcal{K}$.

Definition 200 (Class $\mathcal{K}_\infty$ from a strictly increasing function)**.** Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(0) = 0$, be continuous and strictly increasing on $[0, \infty)$, and have $f(r) \to \infty$ as $r \to \infty$. Then f is a class $\mathcal{K}_\infty$ function, its inverse being the inverse of f on $[0, \infty)$.

Definition 201 (Inverse of a class $\mathcal{K}_\infty$ function)**.** Class $\mathcal{K}_\infty$ is closed under inversion: if α is class $\mathcal{K}_\infty$, so is α^{-1}.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, second bullet.

Definition 202 (Composition of class $\mathcal{K}_\infty$ functions)**.** Class $\mathcal{K}_\infty$ is closed under composition: if α and β are class $\mathcal{K}_\infty$, so is $\beta \circ \alpha$, with inverse $\alpha^{-1} \circ \beta^{-1}$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fourth bullet.

Definition 203 (Sum of class $\mathcal{K}_\infty$ functions)**.** Class $\mathcal{K}_\infty$ is closed under pointwise addition: if α and β are class $\mathcal{K}_\infty$, so is $\alpha + \beta$.

Definition 204 (Positive multiple of a class $\mathcal{K}_\infty$ function)**.** Class $\mathcal{K}_\infty$ is closed under multiplication by a positive constant: if α is class $\mathcal{K}_\infty$ and $c > 0$, so is $c\alpha$.

Definition 205 (Minimum of class $\mathcal{K}_\infty$ functions)**.** Class $\mathcal{K}_\infty$ is closed under pointwise minimum: if α and β are class $\mathcal{K}_\infty$, so is $\min(\alpha, \beta)$.

Definition 206 (Powers of a class $\mathcal{K}_\infty$ function)**.** For every $p > 0$, the map $r \mapsto r^p$ is class $\mathcal{K}_\infty$.

Definition 207 (A class $\mathcal{K}_\infty$ function restricts to class $\mathcal{K}$)**.** Every class $\mathcal{K}_\infty$ function α restricts, for each $a > 0$, to a class $\mathcal{K}$ function on $[0, a) \to [0, \alpha(a))$.

Lemma 208 (Class $\mathcal{K}_\infty$ majorant of a monotone function)**.** *Let $\omega : \mathbb{R} \to \mathbb{R}$ satisfy $\omega(0) = 0$, be monotone on $[0, \infty)$, and be continuous at 0 from the right. Then there is a class $\mathcal{K}_\infty$ function α with $\omega(r) \le \alpha(r)$ for all $r \ge 0$.*

Proof. lemma 235 supplies a continuous, strictly increasing f with $f(0) = 0$, $f(r) \to \infty$ and $\omega \le f$ on $[0, \infty)$; by definition 200 that f is class $\mathcal{K}_\infty$. □

6.2 Inverses of antitone functions

A function W that is continuous and strictly antitone on $(0,\infty)$, blows up at 0^+ and vanishes at $+\infty$ surjects onto $(0,\infty)$, and its right inverse has the same shape. This is the mechanism behind the class $\mathcal{KL}$ construction of the next section: W is a mean-value bound and its inverse supplies the decay rate.

Lemma 209 (Surjectivity onto $(0,\infty)$ from the two boundary limits)**.** *Let $W : \mathbb{R} \to \mathbb{R}$ be continuous on $(0,\infty)$ with $W(\eta) \to +\infty$ as $\eta \to 0^+$ and $W(\eta) \to 0$ as $\eta \to +\infty$. Then for every $s > 0$ there exists $c > 0$ with $W(c) = s$.*

Proof. □

Lemma 210 (The right inverse inverts)**.** *Under the surjectivity conditions of lemma 209, $W\,(\mathrm{invFunOn}\,W\,(0,\infty)\,s) = s$ for all $s > 0$.*

Proof. □

Lemma 211 (Positivity of the right inverse)**.** *Under the surjectivity conditions of lemma 209,* $\mathrm{invFunOn}\,W\,(0,\infty)\,s > 0$ *for all $s > 0$.*

Proof. □

Lemma 212 (The inverse of a strictly antitone function is strictly antitone)**.** *If W is strictly antitone on $(0,\infty)$, then* $\mathrm{invFunOn}\,W\,(0,\infty)$ *is strictly antitone on $(0,\infty)$.*

Proof. □

Lemma 213 (The inverse vanishes at $+\infty$)**.** *Under the conditions of lemma 209 and with W strictly antitone,* $\mathrm{invFunOn}\,W\,(0,\infty)\,s \to 0$ *as $s \to +\infty$.*

Proof. □

Lemma 214 (Continuity of the inverse)**.** *Under the conditions of lemma 209 and with W strictly antitone,* $\mathrm{invFunOn}\,W\,(0,\infty)$ *is continuous on $(0,\infty)$.*

Proof. □

Lemma 215 (The inverse blows up at 0^+)**.** *Under the conditions of lemma 209 and with W strictly antitone,* $\mathrm{invFunOn}\,W\,(0,\infty)\,s \to +\infty$ *as $s \to 0^+$.*

Proof. □

6.3 Class $\mathcal{KL}$ functions and the Osgood construction

Definition 216 (Class $\mathcal{L}$ function)**.** A *class $\mathcal{L}$* function on $[0,\infty)$ is continuous, strictly positive, and strictly decreasing, and satisfies $\sigma(s) \to 0$ as $s \to \infty$. It carries the time decay of an asymptotic stability estimate: the product of a class $\mathcal{K}$ function of the initial deviation with a class $\mathcal{L}$ function of elapsed time is a class $\mathcal{KL}$ bound.

Definition 217 (Singular class $\mathcal{L}$ function)**.** A *singular class* $\mathcal{L}$ function on $(0,\infty)$ is continuous, strictly positive and antitone, with $U(s)\to 0$ as $s\to\infty$ and $U(s)\to\infty$ as $s\to 0^+$. It differs from class $\mathcal{L}$ in two ways: the domain excludes 0, where the function is instead required to blow up, and the decrease is only required to be weak. It arises as the inverse of the sliding-window function in the class $\mathcal{KL}$ characterization of asymptotic stability.

Definition 218 (Class $\mathcal{KL}$ function)**.** A *class* $\mathcal{KL}$ function on $[0,a)\times[0,\infty)$ is continuous, class $\mathcal{K}$ in the first argument, and for each fixed $r>0$ is strictly decreasing and tends to 0 as $s\to\infty$. It arises as the bound $\|x(t)\|\le\beta(\|x_0\|,t)$ in asymptotic stability estimates.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.3. Difference: Khalil says $\beta(\cdot,s)$ is class $\mathcal{K}$; here that is spelled out as separate fields, so no class $\mathcal{K}$ instance is required for each s.

Definition 219 (Global class $\mathcal{KL}$ function)**.** A *global class* $\mathcal{KL}$ function is a map $\beta:[0,\infty)\times[0,\infty)\to\mathbb{R}$ that is class $\mathcal{K}_\infty$ in the first argument (continuous, strictly increasing, zero at zero, radially unbounded) and, for each fixed $r\ge 0$, is antitone and tends to 0 as $s\to\infty$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.3 with $a=\infty$.

6.3.1 Building class $\mathcal{KL}$ functions

Two constructions and the two composition rules, in the bounded and the global setting. The product $\alpha(r)\gamma(s)$ of a class $\mathcal{K}$ and a class $\mathcal{L}$ function is the easy one; the Sontag-style cap is the one that does real work.

Definition 220 (Class $\mathcal{KL}$ from a class $\mathcal{K}$ times a class $\mathcal{L}$)**.** If α is class $\mathcal{K}$ on $[0,a)$ and γ is class $\mathcal{L}$, then $\beta(r,s):=\alpha(r)\gamma(s)$ is class $\mathcal{KL}$ on $[0,a)$.

The cap is built in the form $\min(\alpha(r),\sqrt{\alpha(r)\,U(s)})$, so the decay of that particular expression is recorded first.

Lemma 221 (Decay of a geometric-mean cap)**.** *Let $c\ge 0$ and let $U:\mathbb{R}\to\mathbb{R}$ satisfy $U(s)\to 0$ as $s\to+\infty$. Then $\min(c,\sqrt{c\,U(s)})\to 0$ as $s\to+\infty$.*

Proof. □

Theorem 222 (Class $\mathcal{KL}$ from a singular class $\mathcal{L}$ cap)**.** *Let α be class $\mathcal{K}$ on $[0,a)$ and let U be a singular class $\mathcal{L}$ function (positive, antitone, $U(s)\to 0$ as $s\to\infty$ and $U(s)\to\infty$ as $s\to 0^+$). Then*

$$\beta(r,s):=\begin{cases}\alpha(r) & s=0,\\ \min(\alpha(r),\sqrt{\alpha(r)U(s)}) & s>0,\end{cases}$$

is class $\mathcal{KL}$ on $[0,a)$.

Proof. Every field but continuity is immediate from the corresponding property of α and U, the decay in s being lemma 221. Continuity is checked in two regimes. For $s>0$ the min branch is a composition of continuous maps. At $(r,0)$, pick a buffer radius $r_1\in(r,a)$; since $U(s)\to\infty$ as $s\to 0^+$, on a neighbourhood we have $\alpha(r_1)<U(s)$, hence $\alpha(r')\le U(s)$ and so $\alpha(r')\le\sqrt{\alpha(r')U(s)}$ — the minimum is locally just α, which is continuous. □

Definition 223 (Post-composition with a class $\mathcal{K}$ function)**.** Let β be class $\mathcal{KL}$ on $[0,a)$ and α be class $\mathcal{K}$ on $[0,b)$, and suppose $\beta(r,s)<b$ for every $r\in[0,a)$ and $s\geq 0$. Then $(r,s)\mapsto\alpha(\beta(r,s))$ is class $\mathcal{KL}$ on $[0,a)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the outer α_1). The range hypothesis is additional: α_1 is defined only on $[0,b)$, so β's values must land there, which Khalil leaves implicit.

Definition 224 (Post-composition with a class $\mathcal{K}_\infty$ function)**.** If β is class $\mathcal{KL}$ on $[0,a)$ and α is class $\mathcal{K}_\infty$, then $(r,s)\mapsto\alpha(\beta(r,s))$ is class $\mathcal{KL}$ on $[0,a)$. This is definition 223 with the range hypothesis discharged by α being defined on all of $[0,\infty)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet, with α_1 of class $\mathcal{K}_\infty$; no range hypothesis is then needed.

Definition 225 (Pre-composition in the radius argument)**.** If β is class $\mathcal{KL}$ on $[0,b)$ and α is class $\mathcal{K}$ on $[0,a)\to[0,b)$, then $(r,s)\mapsto\beta(\alpha(r),s)$ is class $\mathcal{KL}$ on $[0,a)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the inner α_2).

Definition 226 (Restriction of a class $\mathcal{KL}$ function)**.** If β is class $\mathcal{KL}$ on $[0,a)\times[0,\infty)$ and $0<a'\leq a$, then β is class $\mathcal{KL}$ on $[0,a')\times[0,\infty)$.

Definition 227 (Global class $\mathcal{KL}$ from a product)**.** If α is class $\mathcal{K}_\infty$ and γ is class $\mathcal{L}$, then $\beta(r,s):=\alpha(r)\gamma(s)$ is global class $\mathcal{KL}$.

Definition 228 (Post-composition, global case)**.** If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r,s)\mapsto\alpha(\beta(r,s))$ is global class $\mathcal{KL}$. No range hypothesis is needed, since α is defined on all of $[0,\infty)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the outer α_1), in the global case $a=\infty$.

Definition 229 (Pre-composition in the radius argument, global case)**.** If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r,s)\mapsto\beta(\alpha(r),s)$ is global class $\mathcal{KL}$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the inner α_2), in the global case $a=\infty$.

Theorem 230 (Capping a class $\mathcal{KL}$ function by a class $\mathcal{K}$ function)**.** *If β is class $\mathcal{KL}$ on $[0,a)$ and α is class $\mathcal{K}$ on $[0,a)$, then $(r,s)\mapsto\min(\beta(r,s),\alpha(r))$ is class $\mathcal{KL}$ on $[0,a)$.*

Capping a class $\mathcal{KL}$ bound by a function of the initial condition alone preserves the class. Its use is to recover a range condition: a bound obtained only as a majorant — from a smoothing construction, say — carries no upper control, and capping restores it without disturbing the bound, since the minimum still dominates whatever β dominated.

Proof. Each field is inherited. The minimum of two strictly increasing functions of r is strictly increasing; α does not depend on s, so antitonicity in s survives the minimum; and $\min(\beta(r,s),\alpha(r))\to\min(0,\alpha(r))=0$ as $s\to\infty$ because $\alpha\geq 0$. □

Definition 231 (Capping a global class $\mathcal{KL}$ function)**.** If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r,s)\mapsto\min(\beta(r,s),\alpha(r))$ is global class $\mathcal{KL}$ — capping a $\mathcal{KL}$ bound by a function of the initial condition alone preserves the class.

6.3.2 Smoothing a bound into a class $\mathcal{KL}$ function

A candidate bound that has the right qualitative behaviour — monotone in the initial deviation, decaying in time — need not itself be class $\mathcal{KL}$: it may fail continuity or strict monotonicity. These two results replace such a candidate by a genuine class $\mathcal{KL}$ function dominating it, which is how the stability characterizations produce their bounds. The first lemma below is the continuity criterion the construction rests on.

Lemma 232 (Joint continuity from separate continuity and monotonicity)**.** *Let $f : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be nondecreasing in its first argument, nonincreasing in its second, and continuous in each argument separately. Then f is jointly continuous.*

Proof. Near (r_0, s_0), monotonicity gives $f(r_-, s_+) \le f(r, s) \le f(r_+, s_-)$ whenever $r_- < r < r_+$ and $s_- < s < s_+$. Given $u > f(r_0, s_0)$, continuity of $f(r_0, \cdot)$ gives $s_- < s_0$ with $f(r_0, s_-) < u$, and then continuity of $f(\cdot, s_-)$ gives $r_+ > r_0$ with $f(r_+, s_-) < u$; so $f < u$ on the neighbourhood $(-\infty, r_+) \times (s_-, \infty)$. The lower bound is symmetric. □

Lemma 233 (Class $\mathcal{KL}$ majorant)**.** *Let $\psi : \mathbb{R} \to \mathbb{R} \to \mathbb{R}$ satisfy, for all $r, s \ge 0$:*

1. *$\psi(r, s) \ge 0$;*
2. *$\psi(0, s) = 0$;*
3. *$r \mapsto \psi(r, s)$ is monotone on $[0, \infty)$;*
4. *$s \mapsto \psi(r, s)$ is antitone on $[0, \infty)$;*
5. *$\psi(r, s) \to 0$ as $s \to \infty$;*
6. *$r \mapsto \psi(r, 0)$ is continuous at 0 within $[0, \infty)$.*

Then there is a global class $\mathcal{KL}$ function β with $\psi(r, s) \le \beta(r, s)$ for all $r, s \ge 0$.

Proof. Clamp both arguments of ψ at 0 and average it over a window in each variable: first $\psi_1(r, s) = \int_0^1 \psi(r, s - v)\, dv$, over $[s-1, s]$, then $\psi_2(r, s) = \int_1^2 \psi_1(ru, s)\, du$, over $[r, 2r]$. Each window sits on the side that makes the average dominate, so $\psi \le \psi_1 \le \psi_2$, and ψ_2 is still nondecreasing in r and nonincreasing in s.

- $\psi_1(r, \cdot)$ is a difference of primitives of $\psi(r, \cdot)$, so continuous; $\psi_2(r, \cdot)$ is then continuous by dominated convergence, the integrands being bounded by $\psi(2r, 0)$.
- For $r > 0$, $\psi_2(r, s) = r^{-1} \int_r^{2r} \psi_1(\cdot, s)$ is a quotient of primitives, so continuous; at $r \le 0$, $0 \le \psi_2(r, s) \le \psi(2 \max(r, 0), 0) \to \psi(0, 0) = 0$ by continuity of $\psi(\cdot, 0)$ at 0.

By lemma 232, ψ_2 is jointly continuous. Also $\psi_2(r, s) \le \psi(2r, s - 1) \to 0$. Finally $\beta(r, s) = \psi_2(r, s) + re^{-s}$ is class $\mathcal{KL}$, the second term making it strictly increasing in r. □

6.4 Class $\mathcal{K}$ bounds for positive definite functions

A Lyapunov function is positive definite: a pointwise condition, awkward to combine with estimates. The theorem below converts it into monotone bounds in the single variable $\|x\|$, after which every Lyapunov argument can be run in the comparison-function calculus. It rests on two smoothing results, which replace a monotone bound by a continuous strictly increasing one.

Lemma 234 (Class $\mathcal{K}$ minorant of a positive monotone function)**.** *Let $r > 0$ and let $\psi : \mathbb{R} \to \mathbb{R}$ satisfy $\psi(0) = 0$, $\psi(s) > 0$ for $s \in (0, r]$, and ψ nondecreasing on $[0, r]$. Then there are $b > 0$ and a function f with $f(0) = 0$, $f(r) = b$, f continuous and strictly increasing on $[0, r]$, and $f(s) \le \psi(s)$ for all $s \in [0, r]$.*

That is: a positive nondecreasing function admits a class $\mathcal{K}$ minorant on any bounded interval. This is what turns a positive-definite V into the lower comparison bound of theorem 237.

Proof. Take $f(s) = r^{-1} \int_0^s \psi$ (with the argument of ψ clamped to $[0, r]$, so that it is monotone, hence integrable, on all of $\mathbb{R}$). It is continuous as a primitive, and $f(0) = 0$. For $s < s'$ in $[0, r]$, $f(s') - f(s) = r^{-1} \int_s^{s'} \psi > 0$ since $\psi > 0$ on (s, s'), so f is strictly increasing and $b = f(r) > 0$. Finally, ψ is nondecreasing, so $\int_0^s \psi \le s\,\psi(s) \le r\,\psi(s)$, i.e. $f(s) \le \psi(s)$. □

Lemma 235 (Class $\mathcal{K}_\infty$ majorant of a monotone function)**.** *Let $\varphi : \mathbb{R} \to \mathbb{R}$ satisfy $\varphi(0) = 0$, be monotone on $[0, \infty)$, and be continuous at 0 from the right. Then there is an f with $f(0) = 0$, continuous and strictly increasing on $[0, \infty)$, $f(s) \to \infty$ as $s \to \infty$, and $\varphi(s) \le f(s)$ for all $s \ge 0$.*

That is: a nondecreasing function admits a class $\mathcal{K}_\infty$ majorant. The unbounded domain is what distinguishes this from lemma 236, which it implies.

Continuity at 0 is necessary: the unit step has no continuous majorant vanishing at 0.

Proof. Take $f(s) = s + g(s)$ with $g(s) = \int_1^2 \varphi(su)\,du$, the average of φ over $[s, 2s]$ (with φ clamped to $\varphi(\max(t, 0))$ so that it is monotone, hence integrable, on all of $\mathbb{R}$). Monotonicity of φ gives $\varphi(s) \le g(s) \le \varphi(2s)$ and that g is nondecreasing, so f is strictly increasing, $f \ge \varphi$, and $f(s) \ge s \to \infty$. For $s > 0$, $g(s) = s^{-1}(\Phi(2s) - \Phi(s))$ with Φ the (continuous) primitive of φ, so g is continuous there; at 0, $0 \le g(s) \le \varphi(2s) \to 0$ by continuity of φ at 0. □

Lemma 236 (Class $\mathcal{K}$ majorant on a bounded interval)**.** *Let $r > 0$ and let φ satisfy $\varphi(0) = 0$, be monotone on $[0, r]$, and be continuous at 0 from the right. Then there are $b > 0$ and an f with $f(0) = 0$, $f(r) = b$, f continuous and strictly increasing on $[0, r]$, and $\varphi(s) \le f(s)$ for all $s \in [0, r]$.*

Proof. Clamp φ to $\varphi_{\text{ext}}(s) = \varphi(\min(s, r))$, which is monotone on all of $[0, \infty)$, still vanishes at 0 and agrees with φ near 0 (so is still continuous there), and apply lemma 235. Restricting the resulting majorant to $[0, r]$ gives f; $b = f(r) > 0$ because f is strictly increasing from $f(0) = 0$. □

Theorem 237 (Class $\mathcal{K}$ sandwich bounds)**.** *Let $W : \mathbb{R}^n \to \mathbb{R}$ be positive definite (definition 19) about x_{eq} on a set containing $\overline{B}(x_{\text{eq}}, r)$. Then there are class $\mathcal{K}$ functions α_1, α_2 on $[0, r)$ with*

$$\alpha_1(\|x - x_{\text{eq}}\|) \;\le\; W(x) \;\le\; \alpha_2(\|x - x_{\text{eq}}\|) \qquad \forall\, \|x - x_{\text{eq}}\| < r.$$

The conclusion holds on the open *ball: α_1 and α_2 are class $\mathcal{K}$ on $[0, r)$, so neither is defined at $\|x - x_{\text{eq}}\| = r$.*

This is what lets a Lyapunov argument be run entirely in terms of comparison functions: positive definiteness is a pointwise condition, and this converts it into monotone bounds in the single variable $\|x - x_{\mathrm{eq}}\|$.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.3 (proved in his Appendix C.4). Differences:*

- *The bound holds on the open ball, with* α_1, α_2 *class* $\mathcal{K}$ *on* $[0, r)$*; Khalil states it on* $[0, r]$.
- *The radially unbounded* $\mathcal{K}_\infty$ *case is not derived here.*

Proof. Translate to the origin, $V(y) = W(y + x_{\mathrm{eq}})$. For the lower bound take $\psi(s) = \inf\{V(y) : s \le \|y\| \le r\}$, the infimum of V over the annulus of inner radius s: it vanishes at 0, is positive for $s > 0$ by compactness and positive definiteness, and is nondecreasing. For the upper bound take $\phi(s) = \sup\{V(y) : \|y\| \le s\}$, similarly nondecreasing and vanishing at 0. Neither is continuous or strictly monotone, so smooth them: lemma 234 gives a class $\mathcal{K}$ minorant of ψ and lemma 236 a class $\mathcal{K}$ majorant of ϕ. Since $\psi(\|y\|) \le V(y) \le \phi(\|y\|)$ pointwise, these sandwich V. □

6.5 The decay system and the comparison bound

A class $\mathcal{K}$ decay rate produces a class $\mathcal{KL}$ bound. This is the step that turns a Lyapunov estimate — which controls only the *rate* of decrease — into an asymptotic stability statement, and it is where the comparison system $\dot{y} = -\alpha(y)$ enters: a Lyapunov function along a trajectory is a subsolution of it.

Definition 238 (The decay system of a class $\mathcal{K}$ function)**.** For α class $\mathcal{K}$ on $[0, a)$ (definition 193), the *decay system* is the scalar time-invariant system

$$\dot{y} = -\alpha(y),$$

as a continuous-time system (definition 58) with no input. Its meaningful states are $y \in [0, a)$.

Theorem 239 (The decay system has a class $\mathcal{KL}$ solution operator)**.** *Let* α *be class* $\mathcal{K}$ *on* $[0, a)$ *and at most linear near the origin,* $\alpha(x) \le Lx$ *on* $(0, \mathrm{base}]$*. Then there is a class* $\mathcal{KL}$ *function* σ *(definition 218) with* $\sigma(y_0, 0) = y_0$ *such that for every* $y_0 \in [0, a)$ *and every* t_0*,* $t \mapsto \sigma(y_0, t - t_0)$ *is a trajectory on* $[t_0, \infty)$ *(definition 76) of the decay system* $\dot{y} = -\alpha(y)$ *(definition 238).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.4. Differences:*

- *Khalil assumes* α *locally Lipschitz; here the hypothesis is the linear bound* $\alpha(x) \le Lx$ *near the origin, which is what the construction uses.*
- *Uniqueness of the solution is not asserted.*

Proof. **Osgood's construction.** Let $\eta(y) = -\int_{\mathrm{base}}^{y} \mathrm{d}x/\alpha(x)$ be the time for the solution started at base to reach y. Since $\alpha > 0$ on $(0, a)$, η is continuous and strictly decreasing there; and since $\alpha(x) \le Lx$ near 0, $\eta(y) \ge (\log \mathrm{base} - \log y)/L \to +\infty$ as $y \to 0^+$, so the origin is not reached in finite time. Setting $\sigma(r, s) := \eta^{-1}(\eta(r) + s)$, extended by $\sigma(0, s) = 0$, the class $\mathcal{KL}$ conditions follow from monotonicity and continuity of η^{-1}, and differentiating $\eta(\sigma) = \eta(r) + s$ gives $\dot{\sigma} = -\alpha(\sigma)$ for $s > 0$. At $s = 0$ the one-sided derivative is the limit of these, $-\alpha(r)$, by continuity of σ and α. □

Lemma 240 (Lipschitz class $\mathcal{K}$ minorant)**.** *Let α be a class $\mathcal{K}$ function on $[0,a)$. Then there are $b'>0$, $L>0$ and a class $\mathcal{K}$ function $\beta:[0,a)\to[0,b')$ with $\beta(x)\le\alpha(x)$ for $x\in[0,a)$, $\beta(x)\le Lx$ for $x\ge 0$, and β L-Lipschitz on all of $\mathbb{R}$.*

The range of β cannot in general be the range $[0,b)$ of α: $\alpha(x)=1-\sqrt{1-x}$ on $[0,1)$ approaches 1 more slowly than any Lipschitz function can.

Proof. Extend α to a nondecreasing $g:\mathbb{R}\to[0,b]$ by 0 on the left and b from a on, and set $\beta(x)=a^{-1}\int_0^x g$, so $b'=\beta(a)$. Since $0\le g\le b$, β is (b/a)-Lipschitz, and with $\beta(0)=0$ this gives $\beta(x)\le(b/a)\,x$. Since $g>0$ on $(0,\infty)$, β is strictly increasing, hence class $\mathcal{K}$ by definition 195. Finally g is nondecreasing, so for $x\in[0,a)$, $\int_0^x g\le x\,\alpha(x)\le a\,\alpha(x)$, i.e. $\beta(x)\le\alpha(x)$. □

Theorem 241 (Subsolutions of the decay system decay)**.** *Let α be class $\mathcal{K}$ on $[0,a)$. Then there is a class $\mathcal{KL}$ function σ with $\sigma(r,0)\le r$ such that: whenever v is continuous on $[t_0,t]$, takes values in $[0,a)$ there, has bounded forward difference quotients, and is a Dini subsolution of the decay system (definition 238),*

$$D^+v(s)\ \le\ -\alpha(v(s)),$$

it obeys $v(t)\le\sigma(v(t_0),\,t-t_0)$.

A differential inequality *with a class $\mathcal{K}$ decay rate therefore yields a class $\mathcal{KL}$ bound — which is what converts a Lyapunov decay estimate into an asymptotic stability statement. No hypothesis is placed on α beyond being class $\mathcal{K}$: the linear-growth condition the construction needs is obtained internally by passing to a Lipschitz minorant.*

Proof. Replace α by a class $\mathcal{K}$ minorant $\beta\le\alpha$ that is Lipschitz near the origin; this only weakens the Dini hypothesis, and supplies the linear bound $\beta(x)\le Lx$ that theorem 239 requires. That theorem gives a class $\mathcal{KL}$ σ whose sections solve $\dot y=-\beta(y)$ with $\sigma(r,0)=r$. Since v is a Dini subsolution of the same equation and starts at $\sigma(v(t_0),0)$, theorem 43 bounds v by it — the perturbed solutions that lemma requires coming from theorem 39, whose hypotheses hold because β is globally Lipschitz (lemma 240). □

Theorem 242 (Subsolutions above a level decay to it)**.** *Let α be class $\mathcal{K}$ on $[0,a)$, and σ the class $\mathcal{KL}$ function of theorem 241. Let v be continuous on $[t_0,t]$ with values in $[0,a)$, with right derivative v' on $[t_0,t)$ satisfying $v'(s)\le-\alpha(v(s))$ whenever $v(s)>\eta$. Then*

$$v(t)\le\max\{\sigma(v(t_0),t-t_0),\ \eta\}.$$

Proof. If $v>\eta$ on all of $[t_0,t)$, v is a subsolution of the decay system there and theorem 241 applies. Otherwise $v(\tau)\le\eta$ for some $\tau\in[t_0,t)$. Wherever $v>\eta$, $v'\le-\alpha(v)\le 0$, so by lemma 18 on $[\tau,t]$, $v(t)\le\max\{v(\tau),\eta\}=\eta$. □

6.6 Massera's lemma

A class $\mathcal{K}$ function with class $\mathcal{K}$ derivative can flatten any decay rate enough to make a Lyapunov integral along the solutions converge, together with its derivative in the initial state.

Lemma 243 (Massera's lemma)**.** *Let $g:[0,\infty)\to\mathbb{R}$ be positive, continuous and strictly decreasing with $g(t)\to 0$, and $h:[0,\infty)\to\mathbb{R}$ positive, continuous and nondecreasing. There is a class $\mathcal{K}_\infty$*

function G (definition 194) whose derivative G' is class $\mathcal{K}_\infty$, and constants $k_1, k_2 > 0$, such that for every continuous u with $0 \le u(t) \le g(t)$ the integrals below converge and

$$\int_0^\infty G(u(t))\,dt \le k_1, \qquad \int_0^\infty G'(u(t))\,h(t)\,dt \le k_2.$$

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma C.1. Differences: G and G' are class $\mathcal{K}_\infty$ rather than class $\mathcal{K}$ on $[0,\infty)$, since only their values on $[0, g(0)]$ matter; convergence of the integrals is stated.*

Proof. Let $\eta(t) = g(t) + 1/t$: continuous and strictly decreasing on $(0,\infty)$, with $\eta \to \infty$ at 0^+, $\eta \to 0$ at ∞, and $\eta > g$. Its inverse $\theta = \eta^{-1}$ (lemma 214) satisfies $\theta(s) \ge t$ whenever $0 < s \le g(t)$. Let

$$G'(s) = \frac{e^{-\theta(s)}}{h(\theta(s))} + \max(s - g(0), 0), \qquad G(r) = \int_0^r G'(s)\,ds,$$

so that $G'(s) \le e^{-t}/h(t)$ for $0 \le s \le g(t)$. Then $G'(u(t))\,h(t) \le e^{-t}$ and $G(u(t)) \le u(t)\,e^{-t}/h(t) \le (g(0)/h(0))\,e^{-t}$, so $k_1 = g(0)/h(0)$ and $k_2 = 1$. Khalil's piecewise-linear η is replaced by $g(t) + 1/t$, which lies above g for every $t > 0$, so no splitting at t_1 is needed. □

Chapter 7

Stability of continuous-time systems

Stability notions for an equilibrium of an autonomous continuous-time system $\dot{x} = f(t, x)$ (definition 58); a system with input is first unforced (definition 67). Each notion is defined once, for time-varying systems, with the initial time $t_0 \geq 0$ explicit; a time-invariant system is a special case, and that its notions may be read with $t_0 = 0$ is a theorem rather than a second definition.

Bounds — stability, uniform convergence times, exponential estimates — quantify over trajectories on segments $[t_0, t_1]$, so that a solution escaping in finite time counts against stability instead of being silently excluded. Limits quantify over trajectories on rays $[t_0, \infty)$.

7.1 Definitions

Definition 244 (Stability)**.** The equilibrium x_{eq} of an autonomous continuous-time system $\dot{x} = f(t, x)$ is *stable* when for every $\varepsilon > 0$ and $t_0 \geq 0$ there is $\delta = \delta(\varepsilon, t_0) > 0$ such that every trajectory on a segment $[t_0, t_1]$ with $\|x(t_0) - x_{\text{eq}}\| < \delta$ satisfies $\|x(t) - x_{\text{eq}}\| < \varepsilon$ for all $t \in [t_0, t_1]$.

Quantifying over segments rather than over solutions on $[t_0, \infty)$ means a solution that escapes in finite time counts against stability instead of being silently excluded.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*stable*).

Definition 245 (Instability)**.** The equilibrium is *unstable* when it is not stable (definition 244).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4.

Definition 246 (Uniform stability)**.** The equilibrium is *uniformly stable* when δ in definition 244 can be chosen independently of t_0: $\delta = \delta(\varepsilon)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*uniformly stable*).

Definition 247 (Global uniform stability)**.** The equilibrium is *globally uniformly stable* when it is uniformly stable (definition 246) with a choice of $\delta(\varepsilon)$ satisfying $\delta(\varepsilon) \to \infty$ as $\varepsilon \to \infty$. In particular every trajectory is bounded, so none escapes in finite time.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4.

Definition 248 (Local asymptotic stability)**.** The equilibrium is *locally asymptotically stable* when it is stable (definition 244) and for every $t_0 \geq 0$ there is $c = c(t_0) > 0$ such that the system is forward complete from the ball of radius c about x_{eq} at t_0 (definition 69) and every trajectory on $[t_0, \infty)$ starting in that ball satisfies $x(t) \to x_{\text{eq}}$. Forward completeness keeps solutions that escape in finite time from being silently excluded.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*asymptotically stable*).

Definition 249 (Global asymptotic stability)**.** The equilibrium is *globally asymptotically stable* when it is stable (definition 244), the system is forward complete from every state at every $t_0 \geq 0$ (definition 69), and every trajectory on $[t_0, \infty)$ satisfies $x(t) \to x_{\text{eq}}$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Section 4.1.

Definition 250 (Uniform convergence time)**.** Trajectories starting within c of x_{eq} have a *uniform convergence time*: for every $\eta > 0$ there is $T = T(\eta) > 0$ such that every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$, with $\|x(t_0) - x_{\text{eq}}\| < c$ satisfies $\|x(t) - x_{\text{eq}}\| < \eta$ for $t \in [t_0 + T, t_1]$.

Definition 251 (Uniform asymptotic stability)**.** The equilibrium is *uniformly asymptotically stable* when it is uniformly stable (definition 246) and there is $c > 0$ from which convergence has a uniform convergence time (definition 250).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*uniformly asymptotically stable*).

Definition 252 (Global uniform asymptotic stability)**.** The equilibrium is *globally uniformly asymptotically stable* when it is globally uniformly stable (definition 247) and every $c > 0$ admits a uniform convergence time (definition 250).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4.

Definition 253 (Local exponential stability)**.** The equilibrium is *locally exponentially stable* when there are $c, k, \gamma > 0$ such that every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$, with $\|x(t_0) - x_{\text{eq}}\| < c$ satisfies

$$\|x(t) - x_{\text{eq}}\| \leq k \, \|x(t_0) - x_{\text{eq}}\| \, e^{-\gamma(t - t_0)} \qquad \forall\, t \in [t_0, t_1].$$

The constants do not depend on t_0: exponential stability is uniform, as in the reference.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.5.

Definition 254 (Global exponential stability)**.** The equilibrium is *globally exponentially stable* when the estimate of definition 253 holds for every initial state, with no radius c.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.5.

Boundedness is a property of the system near a point that need not be an equilibrium: trajectories stay, or end up and stay, within a prescribed distance of it.

Definition 255 (Uniform boundedness)**.** Trajectories are *uniformly bounded* around x_c when there is $c > 0$ such that for every $0 < a < c$ there is $\beta = \beta(a) > 0$ with

$$\|x(t_0) - x_c\| \leq a \;\Rightarrow\; \|x(t) - x_c\| \leq \beta \qquad \forall\, t \in [t_0, t_1]$$

for every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$. The point x_c need not be an equilibrium. The bound applies to every segment a solution exists on, so finite escape violates it.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6.

Definition 256 (Global uniform boundedness)**.** Trajectories are *globally uniformly bounded* around x_c when the bound of definition 255 holds for every $a > 0$, with no radius c.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6.

Definition 257 (Uniform ultimate boundedness)**.** Trajectories are *uniformly ultimately bounded* with ultimate bound b around x_c when there is $c > 0$ such that for every $t_0 \geq 0$ the system is forward complete from the ball of radius c about x_c (definition 69), and for every $0 < a < c$ there is $T = T(a, b) \geq 0$ with

$$\|x(t_0) - x_c\| \leq a \;\Rightarrow\; \|x(t) - x_c\| \leq b \qquad \forall\, t \geq t_0 + T$$

for every trajectory on $[t_0, \infty)$. The point x_c need not be an equilibrium. Khalil leaves forward completeness implicit; without it, a solution escaping before $t_0 + T$ would never be tested.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6.

Definition 258 (Global uniform ultimate boundedness)**.** Trajectories are *globally uniformly ultimately bounded* with ultimate bound b around x_c when the system is forward complete from every state at every $t_0 \geq 0$ (definition 69) and the ultimate bound of definition 257 holds for every $a > 0$, with no radius c.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6.

7.2 Implications

Lemma 259 (Instability from a fixed escape radius)**.** *Fix $\varepsilon > 0$. If for every $\delta > 0$ some trajectory on a segment $[0, T]$ starts within δ of x_{eq} and reaches distance at least ε from it, then x_{eq} is unstable (definition 245).*

Proof. Stability at $t_0 = 0$ with tolerance ε gives a δ; the trajectory for that δ contradicts it. □

Lemma 260 (Uniform stability implies stability)**.** *A uniformly stable equilibrium (definition 246) is stable (definition 244).*

Proof. □

Lemma 261 (Global asymptotic stability implies local asymptotic stability)**.** *A globally asymptotically stable equilibrium (definition 249) is locally asymptotically stable (definition 248).*

Proof. □

Lemma 262 (Uniform asymptotic stability implies local asymptotic stability)**.** *A uniformly asymptotically stable equilibrium (definition 251) of a forward-complete system (definition 70) is locally asymptotically stable (definition 248). Forward completeness is needed: bounds on solution segments do not by themselves make the segments extend.*

Proof. Uniform stability gives stability; a uniform convergence time gives convergence of every trajectory on a ray, with the same radius c for every t_0. □

Lemma 263 (Global uniform asymptotic stability implies global asymptotic stability)**.** *A globally uniformly asymptotically stable equilibrium (definition 252) is globally asymptotically stable (definition 249), given forward completeness (definition 70).*

Proof. Every trajectory starts in some ball, and every ball has a uniform convergence time. □

Lemma 264 (Exponential stability implies uniform stability)**.** *A locally exponentially stable equilibrium (definition 253) is uniformly stable (definition 246).*

Proof. The estimate bounds the distance by k times the initial distance, so $\delta = \min(c, \varepsilon/k)$ works for every t_0. □

Lemma 265 (Exponential stability implies uniform asymptotic stability)**.** *A locally exponentially stable equilibrium (definition 253) is uniformly asymptotically stable (definition 251).*

Proof. Uniform stability is lemma 264. From the ball of radius c the distance after time T is below $kc\,e^{-\gamma T}$, which is below any $\eta > 0$ for T large, independently of t_0. □

Lemma 266 (Global exponential stability implies local exponential stability)**.** *A globally exponentially stable equilibrium (definition 254) is locally exponentially stable (definition 253).*

Proof. □

Lemma 267 (Global exponential stability implies global uniform asymptotic stability)**.** *A globally exponentially stable equilibrium (definition 254) is globally uniformly asymptotically stable (definition 252).*

Proof. The estimate bounds the distance by k times the initial distance, so $\delta(\varepsilon) = \varepsilon/k$ witnesses global uniform stability. For each radius c the estimate is locally exponential from that ball, which gives a uniform convergence time by lemma 265. □

Lemma 268 (Exponential stability under a change of coordinates)**.** *Let $e : E \to E'$ be continuous linear with $\|x\| \le a\|ex\|$ and $\|ex\| \le b\|x\|$, $a, b > 0$, and let $F'(t, x'_{\mathrm{eq}} + e(x - x_{\mathrm{eq}})) = e(F(t, x))$. If x'_{eq} is locally exponentially stable for s' (definition 253), so is x_{eq} for s.*

Proof. If x is a trajectory of s, then $x'_{\mathrm{eq}} + e(x - x_{\mathrm{eq}})$ is one of s'. Its estimate, with radius c/b and constant abk, is one for x. □

7.3 Comparison-function characterizations

The uniform notions are equivalent to comparison-function bounds on the trajectories: uniform stability to a class $\mathcal{K}$ bound, uniform asymptotic stability to a class $\mathcal{KL}$ bound. The hard direction is the decay in elapsed time. A uniform convergence time says that for each tolerance η there is a delay after which every trajectory from a ball is inside the η-ball. Read the other way round it bounds how far a trajectory can still be after a given elapsed time, but only through a function of η that may be discontinuous. The results below regularize it into a decay envelope, a function of elapsed time alone that is continuous, positive and vanishing at $+\infty$: the time-decay half of a class $\mathcal{KL}$ bound, whose other half is a uniform class $\mathcal{K}$ bound.

Definition 269 (Uniform class $\mathcal{K}$ bound)**.** The trajectories of s *have the uniform class $\mathcal{K}$ bound* $\alpha : [0, a) \to [0, b)$ about x_{eq} when

$$\|x(t) - x_{\mathrm{eq}}\| \le \alpha(\|x(t_0) - x_{\mathrm{eq}}\|)$$

for every trajectory on a segment $[t_0, t_1]$ with $t_0 \ge 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < a$, and every $t \in [t_0, t_1]$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), (4.19).

Definition 270 (Uniform class $\mathcal{K}_\infty$ bound)**.** The trajectories of s *have the uniform class* $\mathcal{K}_\infty$ *bound* α about x_{eq} when $\|x(t) - x_{\mathrm{eq}}\| \le \alpha(\|x(t_0) - x_{\mathrm{eq}}\|)$ for every trajectory on a segment $[t_0, t_1]$ with $t_0 \ge 0$ and every $t \in [t_0, t_1]$, with no restriction on the initial state.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), (4.20).

Lemma 271 (Local uniform decay envelope)**.** *Suppose the trajectories of s have a uniform convergence time from the ball of radius c (definition 250) and a uniform class $\mathcal{K}$ bound α defined on $[0, a_\alpha)$ (definition 269). Let $0 < a \le c$ with $a < a_\alpha$. Then there is a singular class $\mathcal{L}$ function U (definition 217) such that*

$$\|x(t) - x_{\mathrm{eq}}\| \le U(t - t_0)$$

for every trajectory on a segment $[t_0, t_1]$ with $t_0 \ge 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < a$, and every $t \in (t_0, t_1]$.

The bound depends on the elapsed time alone: neither on t_0, nor on the trajectory, nor on where in the a-ball it started.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5.*

Proof. Let $\bar{T}(\eta, r)$ be the least delay after which every trajectory starting within r of x_{eq} lies inside the η-ball; it is finite by hypothesis and antitone in η, but possibly discontinuous. Smooth it by the sliding average

$$W(r, \eta) = \frac{2}{\eta} \int_{\eta/2}^{\eta} \left[\bar{T}(s, r) + \frac{r}{\eta} \right] ds,$$

which equals $\frac{2}{\eta} \int_{\eta/2}^{\eta} \bar{T}(s, r)\, ds + r/\eta$. The averaged term is antitone in η and the r/η term is strictly antitone, so $W(r, \cdot)$ is continuous and strictly antitone on $(0, \infty)$; it dominates r/η, hence tends to $+\infty$ as $\eta \to 0^+$, and $\bar{T}(\cdot, r)$ vanishes beyond $\alpha(r)$, hence $W(r, \eta) \to 0$ as $\eta \to \infty$. Inverting it (lemma 211, lemma 212, lemma 213, lemma 214, lemma 215) gives $U = \mathrm{invFunOn}\, W(r, \cdot)\, (0, \infty)$, singular class $\mathcal{L}$. Finally $W(r, U(s)) = s$ and $W \ge \bar{T} + r/\eta$ give $\bar{T}(U(s), r) < s$, so s is itself a valid convergence time to the $U(s)$-ball, which is the claimed bound. □

Lemma 272 (Global uniform decay envelope)**.** *Suppose the trajectories of s have a uniform convergence time from every ball (definition 250) and a uniform class $\mathcal{K}_\infty$ bound (definition 270). Then there is a family $U : (0, \infty) \times (0, \infty) \to \mathbb{R}$ with*

1. *$U(r, s) > 0$ for all $r > 0$ and $s > 0$;*
2. *$U(r, \cdot)$ antitone on $(0, \infty)$ for each $r > 0$;*
3. *$U(r, s) \to 0$ as $s \to \infty$, for each $r > 0$;*
4. *$r \mapsto U(r + 1, s)$ monotone on $[0, \infty)$ for each $s > 0$;*
5. *$\|x(t) - x_{\mathrm{eq}}\| \le U(r, t - t_0)$ for every $r > 0$, every trajectory on a segment $[t_0, t_1]$ with $t_0 \ge 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < r$, and every $t \in (t_0, t_1]$.*

Conditions (1)–(4) are exactly the hypotheses of **??***; (5) is the decay bound. Unlike the local case (lemma 271) no single envelope can serve every initial state, so the radius remains a parameter.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5 (global case).*

Proof. Every radius is finite, so at radius r the global hypotheses restrict to the local ones (a uniform convergence time from the r-ball, and the class $\mathcal{K}_\infty$ bound read as a class $\mathcal{K}$ bound on $[0, r+1)$). Take $U(r,\cdot)$ to be the envelope of lemma 271 at that radius, namely the inverse of $W(r,\cdot)$. Conditions (1)–(3) and (5) are then that lemma's conclusions, and (4) holds because W is monotone in r — a larger ball has larger optimal convergence times — so inverting reverses nothing. □

Lemma 273 (Uniform convergence from a class $\mathcal{KL}$ bound)**.** *Let β be class $\mathcal{KL}$ on $[0,a)\times[0,\infty)$ (definition 218), and suppose every trajectory on a segment $[t_0,t_1]$, $t_0 \geq 0$, with $\|x(t_0)-x_{\text{eq}}\| < a$ satisfies $\|x(t)-x_{\text{eq}}\| \leq \beta(\|x(t_0)-x_{\text{eq}}\|, t-t_0)$ on the segment. Then for every $c<a$, trajectories from the ball of radius c have a uniform convergence time (definition 250).*

Proof. For $c<0$ no trajectory starts in the ball. Otherwise every deviation is at most $\beta(c,t-t_0)$, by monotonicity of β in its first argument, and $\beta(c,\cdot)\to 0$. □

Lemma 274 (Class $\mathcal{K}$ characterization of stability)**.** *The equilibrium x_{eq} is stable (definition 244) if and only if for each $t_0 \geq 0$ there is a class $\mathcal{K}$ function α_{t_0} on $[0,a_{t_0})$ with $\|x(t)-x_{\text{eq}}\| \leq \alpha_{t_0}(\|x(t_0)-x_{\text{eq}}\|)$ for every trajectory on a segment $[t_0,t_1]$ with $\|x(t_0)-x_{\text{eq}}\| < a_{t_0}$, and every $t\in[t_0,t_1]$. A single α for all t_0 is uniform stability (lemma 275).*

Proof. The argument of lemma 275 with the initial time fixed. □

Lemma 275 (Class $\mathcal{K}$ characterization of uniform stability)**.** *The equilibrium x_{eq} is uniformly stable (definition 246) if and only if the trajectories have a uniform class $\mathcal{K}$ bound (definition 269): there is a class $\mathcal{K}$ function α on $[0,a)$ with*

$$\|x(t)-x_{\text{eq}}\| \leq \alpha(\|x(t_0)-x_{\text{eq}}\|)$$

for every trajectory on a segment $[t_0,t_1]$ with $t_0\geq 0$ and $\|x(t_0)-x_{\text{eq}}\| < a$, and every $t\in[t_0,t_1]$.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5, first half.*

Proof. Forward: the worst deviation $\omega(r)$ reachable from the r-ball is nondecreasing, vanishes at 0 and is continuous there by uniform stability; a continuous strictly increasing majorant (lemma 236) is the class $\mathcal{K}$ bound. Backward: α is continuous at 0 with $\alpha(0)=0$, so a small radius δ has $\alpha(\delta)<\varepsilon$. □

Lemma 276 (Class $\mathcal{KL}$ characterization of uniform asymptotic stability)**.** *The equilibrium x_{eq} is uniformly asymptotically stable (definition 251) if and only if there are a class $\mathcal{KL}$ function β on $[0,a)\times[0,\infty)$ (definition 218) with*

$$\|x(t)-x_{\text{eq}}\| \leq \beta(\|x(t_0)-x_{\text{eq}}\|, t-t_0)$$

for every trajectory on a segment $[t_0,t_1]$ with $t_0\geq 0$ and $\|x(t_0)-x_{\text{eq}}\| < a$, and every $t\in[t_0,t_1]$.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5, second half.*

Proof. Forward: uniform stability gives a class $\mathcal{K}$ bound α (lemma 275) and, with the uniform convergence time, a decay envelope U (lemma 271); $\beta=\min(\alpha,\sqrt{\alpha\,U})$ is class $\mathcal{KL}$ (theorem 222) and bounds the trajectories. Backward: $\beta(\cdot,0)$ is continuous at 0, which gives uniform stability, and lemma 273 gives the convergence time. □

Lemma 277 (Class $\mathcal{KL}$ characterization, global case)**.** *The equilibrium x_{eq} is globally uniformly asymptotically stable (definition 252) if and only if there is a $\beta : [0,\infty) \times [0,\infty) \to \mathbb{R}$ that is*

1. *zero at the origin: $\beta(0,s) = 0$ for $s \geq 0$;*
2. *jointly continuous on $[0,\infty) \times [0,\infty)$;*
3. *strictly increasing in r for each fixed $s \geq 0$;*
4. *radially unbounded: $\beta(r,0) \to \infty$ as $r \to \infty$;*
5. *antitone in s for each fixed $r \geq 0$, with $\beta(r,s) \to 0$ as $s \to \infty$;*

such that $\|x(t) - x_{\mathrm{eq}}\| \leq \beta(\|x(t_0) - x_{\mathrm{eq}}\|, t - t_0)$ for every trajectory on a segment $[t_0, t_1]$ with $t_0 \geq 0$ and every $t \in [t_0, t_1]$, with no restriction on the initial state.

Condition (4) is what makes the characterization global*: without it β is class $\mathcal{KL}$ on a bounded radius and the statement collapses to the local one (lemma 276).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5, global case.*

Proof. Forward: global uniform stability gives a class $\mathcal{K}_\infty$ bound, the convergence times from every ball a family of decay envelopes (lemma 272), and **??** combines them. Backward: $\alpha = \beta(\cdot, 0)$ is class $\mathcal{K}_\infty$ and $\delta = \alpha^{-1}$ gives global uniform stability; from the r-ball every deviation is at most $\beta(r,\cdot) \to 0$. □

The same characterizations hold with the bounds on solutions over $[t_0,\infty)$. A bound on every segment restricts to those solutions, so that direction needs nothing more. The converse needs every segment from the ball to extend to such a solution: forward completeness from some ball about x_{eq} locally, and from every state globally.

Lemma 278 (Class $\mathcal{K}$ characterization for forward-complete systems)**.** *Let the system be forward complete from some ball $B(x_{\mathrm{eq}}, c)$ at every $t_0 \geq 0$ (definition 69). Then x_{eq} is uniformly stable if and only if there is a class $\mathcal{K}$ function α on $[0,a)$ with $\|x(t) - x_{\mathrm{eq}}\| \leq \alpha(\|x(t_0) - x_{\mathrm{eq}}\|)$ for every solution x on $[t_0,\infty)$ with $t_0 \geq 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < a$, and every $t \geq t_0$. The forward direction holds without completeness.*

Proof. lemma 275, with the bound moved between segments and solutions on $[t_0,\infty)$. Forward: restrict a solution to $[t_0, t]$. Converse: shrink the radius to $a' = \min(c, a/2)$, restricting α (definition 196); a segment starting within a' extends by completeness to a solution on $[t_0,\infty)$, which the bound covers. □

Lemma 279 (Class $\mathcal{KL}$ characterization for forward-complete systems)**.** *Let the system be forward complete from some ball $B(x_{\mathrm{eq}}, c)$ at every $t_0 \geq 0$ (definition 69). Then x_{eq} is uniformly asymptotically stable if and only if there is a class $\mathcal{KL}$ function β on $[0,a) \times [0,\infty)$ with $\|x(t) - x_{\mathrm{eq}}\| \leq \beta(\|x(t_0) - x_{\mathrm{eq}}\|, t - t_0)$ for every solution x on $[t_0,\infty)$ with $t_0 \geq 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < a$, and every $t \geq t_0$. The forward direction holds without completeness.*

Proof. lemma 276, with the bound moved between segments and solutions on $[t_0,\infty)$ as in lemma 278, shrinking the radius to $\min(c, a)$ (definition 226). □

Lemma 280 (Class $\mathcal{KL}$ characterization for forward-complete systems, global case)**.** *Let the system be forward complete (definition 70). Then x_{eq} is globally uniformly asymptotically stable if and only if there is a β with the properties of lemma 277 and $\|x(t) - x_{\mathrm{eq}}\| \leq \beta(\|x(t_0) - x_{\mathrm{eq}}\|, t - t_0)$ for every solution x on $[t_0, \infty)$ with $t_0 \geq 0$, and every $t \geq t_0$. The forward direction holds without completeness.*

Proof. lemma 277, with the bound moved between segments and solutions on $[t_0, \infty)$ as in lemma 278. □

7.4 Lyapunov functions

A Lyapunov function is a certificate: its existence implies a stability notion. The certificates are time-varying, $V(t, x)$, and named by what they certify. What separates them is which bounds hold, not whether time appears: a positive definite (definition 19) lower bound gives stability, and a positive definite upper bound as well (decrescence) is what makes the conclusion uniform. A time-invariant $V(x)$ is its own lower and upper bound, so it is a uniform certificate as soon as it decreases along the system.

Definition 281 (Lyapunov function)**.** A function $V : \mathbb{R} \times E \to \mathbb{R}$ is a *Lyapunov function* on a neighbourhood D of x_{eq} when, for all $t \geq 0$ and $x \in D$, V is differentiable at (t, x) and

$$V(t, x_{\mathrm{eq}}) = 0, \qquad W_1(x) \leq V(t, x), \qquad \dot{V}(t, x) \leq 0,$$

for some W_1 positive definite on D (definition 19), with $\dot{V}$ the derivative along the system (definition 81). It certifies stability (definition 244).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.8.

Definition 282 (Uniform Lyapunov function)**.** A Lyapunov function (definition 281) that is also *decrescent*: $V(t, x) \leq W_2(x)$ for all $t \geq 0$, $x \in D$, for some W_2 positive definite on D. The upper bound is what makes the conclusion uniform: it certifies uniform stability (definition 246).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.8.

Definition 283 (Strict uniform Lyapunov function)**.** A uniform Lyapunov function (definition 282) with $\dot{V}(t, x) \leq -W_3(x)$ for all $t \geq 0$, $x \in D$, for some W_3 positive definite on D. It certifies uniform asymptotic stability (definition 251).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.9.

Definition 284 (Exponential Lyapunov function)**.** A function $V : \mathbb{R} \times E \to \mathbb{R}$ is an *exponential Lyapunov function* with exponent $a > 0$ on a neighbourhood D of x_{eq} when there are $k_1, k_2, k_3 > 0$ with, for all $t \geq 0$ and $x \in D$, V differentiable at (t, x) and

$$k_1 \|x - x_{\mathrm{eq}}\|^a \leq V(t, x) \leq k_2 \|x - x_{\mathrm{eq}}\|^a, \qquad \dot{V}(t, x) \leq -k_3 \|x - x_{\mathrm{eq}}\|^a.$$

It certifies local exponential stability (definition 253).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.10.

Lemma 285 (Time-invariant Lyapunov functions are uniform)**.** *Let $V : E \to \mathbb{R}$ be positive definite on a neighbourhood D of x_{eq} (definition 19) and differentiable at its points, with $\nabla V(x) \cdot f(t, x) \leq 0$ for all $t \geq 0$, $x \in D$. Then $(t, x) \mapsto V(x)$ is a uniform Lyapunov function (definition 282).*

Proof. Take $W_1 = W_2 = V$. The derivative along the system of a function of x alone is its directional derivative along the vector field. □

Lemma 286 (Strict time-invariant Lyapunov functions are strict uniform)**.** *Let $V : E \to \mathbb{R}$ be positive definite on a neighbourhood D of x_{eq} and differentiable at its points, with $\nabla V(x) \cdot f(t,x) \leq -W_3(x)$ for all $t \geq 0$, $x \in D$, for some W_3 positive definite on D. Then $(t,x) \mapsto V(x)$ is a strict uniform Lyapunov function (definition 283).*

Proof. As for lemma 285; $-W_3 \leq 0$ gives the non-strict decrease. □

Lemma 287 (Strict decrease suffices for a time-invariant system)**.** *Let the state equation be $\dot{x} = f(x)$ with f continuous and $f(x_{\text{eq}}) = 0$, and let $V : E \to \mathbb{R}$ be C^1 and positive definite on a neighbourhood D of x_{eq} with $\nabla V(x) \cdot f(x) < 0$ for $x \in D \setminus \{x_{\text{eq}}\}$. Then $(t,x) \mapsto V(x)$ is a strict uniform Lyapunov function (definition 283).*

Proof. Take $W_3 = -\nabla V \cdot f$: continuous since V is C^1 and f is continuous, zero at x_{eq} since $f(x_{\text{eq}}) = 0$, and positive elsewhere on D; apply lemma 286. □

7.5 Lyapunov's direct method

For a time-varying V the derivative along the system acquires an explicit time term, and positive definiteness of $V(t,\cdot)$ on its own buys nothing uniform in t_0: V could flatten as $t \to \infty$. Sandwiching V between two positive-definite functions of the state alone is what restores uniformity. Only trajectories on segments $[t_0, t_1]$ are used, so no solution is assumed to exist for all time.

Theorem 288 (Lyapunov's uniform stability theorem)**.** *If V is a uniform Lyapunov function for x_{eq} (definition 282) on a system with state space $\mathbb{R}^n$, then x_{eq} is uniformly stable (definition 246).*

The sandwich $W_1(x) \leq V(t,x) \leq W_2(x)$ is what makes the conclusion uniform in t_0: W_1 and W_2 do not depend on t, so neither does the $\delta(\varepsilon)$ extracted from them.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.8.*

Proof. Given ε, work on a radius $r \leq \varepsilon$ whose closed ball lies in the region of the certificate. Bound W_1 below and W_2 above by class $\mathcal{K}$ functions α_1, α_2 on it (theorem 237), fix a level d below both ranges, and take $\delta = \alpha_2^{-1}(d)$: starting within δ of x_{eq} forces $V(t_0, x(t_0)) < d$. To leave the r-ball the trajectory would first cross a sphere on which $W_1 > d$; up to that time V has not increased (lemma 49), so it is still below d, a contradiction. □

Theorem 289 (Lyapunov's uniform asymptotic stability theorem)**.** *If V is a strict uniform Lyapunov function for x_{eq} (definition 283) on a system with state space $\mathbb{R}^n$, then x_{eq} is uniformly asymptotically stable (definition 251).*

A strictly negative $\dot{V}$ alone would not do: W_3 must be bounded away from zero on each annulus, which is exactly what positive definiteness of a function of x alone buys.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.9. Khalil's two "moreover" clauses are not part of this conclusion: the explicit class $\mathcal{KL}$ bound, which the proof constructs, and the global case.*

Proof. Uniform stability is theorem 288. For convergence, take class $\mathcal{K}$ bounds $\alpha_1, \alpha_2, \alpha_3$ for W_1, W_2, W_3 on a ball in the region of the certificate, and the trapping radius c of that proof. Along

a trajectory from the c-ball, $\dot V \le -W_3(x) \le -\alpha_3(\|x - x_{\text{eq}}\|) \le -\alpha_3(\alpha_2^{-1}(V))$, so $v(t) = V(t, x(t))$ satisfies $D^+v \le -\alpha(v)$ with $\alpha = \alpha_3 \circ \alpha_2^{-1}$ of class $\mathcal{K}$. theorem 241 gives a class $\mathcal{KL}$ σ with $v(t) \le \sigma(v(t_0), t - t_0)$; unwinding the sandwich,

$$\|x(t) - x_{\text{eq}}\| \le \alpha_1^{-1}(\sigma(\alpha_2(\|x(t_0) - x_{\text{eq}}\|), t - t_0)),$$

a class $\mathcal{KL}$ bound. Conclude by lemma 273. □

In the global theorem compact sublevel sets of V are the hypothesis; for a continuous V they follow from radial unboundedness.

Lemma 290 (Sublevel sets of a radially unbounded function are compact)**.** *Let E be a proper normed space. If $V : E \to \mathbb{R}$ is continuous and radially unbounded ($V(x) \to \infty$ as $\|x\| \to \infty$), then every sublevel set $\{x \mid V(x) \le c\}$ is compact.*

Proof. Closed: it is $V^{-1}((-\infty, c])$, with V continuous. Bounded: radial unboundedness gives a compact K off which $V > c$, so the sublevel set lies in K. Compact: closed and bounded in a proper space. □

Theorem 291 (Lyapunov's global asymptotic stability theorem)**.** *Let the state equation be $\dot x = f(x)$ with f continuous, on $\mathbb{R}^n$. Let $V : \mathbb{R}^n \to \mathbb{R}$ be C^1, positive definite about x_{eq} on all of $\mathbb{R}^n$ (definition 19), with compact sublevel sets $\{V \le c\}$ (equivalently, radially unbounded) and $\nabla V(x) \cdot f(x) < 0$ for $x \ne x_{\text{eq}}$. Then x_{eq} is globally uniformly asymptotically stable (definition 252).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.2.*

Proof. V does not increase along trajectories: $\nabla V \cdot f \le 0$ everywhere, since $\nabla V(x_{\text{eq}}) = 0$ at the minimum. So a trajectory starting below every value V takes outside the ε-ball never leaves it. Take $\delta(\varepsilon)$ the supremum of the radii $\rho \le \varepsilon$ whose closed ball lies strictly below those values: it is positive, since V is bounded below by a positive constant outside the ε-ball and small near x_{eq}, and it tends to ∞, since V is bounded on each ball and large far out. This is global uniform stability.

For convergence from the c-ball, trajectories stay in the compact set $\{V \le M\}$, $M = \max_{\overline{B}(x_{\text{eq}}, c)} V$, on which, outside the $\delta(\eta)$-ball, $\nabla V \cdot f \le -\gamma < 0$. So V falls at rate γ while the trajectory is outside, and since $V \ge 0$ it enters the $\delta(\eta)$-ball by time $t_0 + M/\gamma + 1$; from then on it stays in the η-ball. □

Theorem 292 (Lyapunov's exponential stability theorem)**.** *If V is an exponential Lyapunov function for x_{eq} (definition 284), then x_{eq} is locally exponentially stable (definition 253): near x_{eq},*

$$\|x(t) - x_{\text{eq}}\| \le \left(\frac{k_2}{k_1}\right)^{1/a} \|x(t_0) - x_{\text{eq}}\|\, e^{-(k_3/(ak_2))(t - t_0)}.$$

The state space is any normed space.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.10.*

Proof. The bounds give $\dot V \le -(k_3/k_2)V$, so $e^{(k_3/k_2)t}V(t, x(t))$ does not increase and $V(t, x(t)) \le V(t_0, x(t_0))\, e^{-(k_3/k_2)(t - t_0)}$ while the trajectory stays in D; the sandwich and an a-th root give the estimate. To stay in D, start within $r/(K + 1)$ of x_{eq}, where $\overline{B}(x_{\text{eq}}, r) \subseteq D$ and $K = (k_2/k_1)^{1/a}$: at the first time the trajectory reached the sphere of radius r the estimate would put it strictly inside. □

Theorem 293 (Lyapunov's global exponential stability theorem)**.** *If V is an exponential Lyapunov function for x_{eq} on the whole space (definition 284 with $D = E$), then x_{eq} is globally exponentially stable (definition 254).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.10, global case.*

Proof. The estimate of theorem 292 holds on every trajectory segment, since every segment stays in $D = E$. □

7.6 Converse Lyapunov theorems

Exponential stability has a quadratic Lyapunov function, built by integrating the squared distance along solutions over a fixed horizon. Its gradient is of the order of the distance to the equilibrium, which is what lets it absorb perturbations of the vector field.

Theorem 294 (Converse Lyapunov theorem for global exponential stability)**.** *Let x_{eq} be an equilibrium of $\dot{x} = f(t, x)$ on $\mathbb{R}^n$, with f continuously differentiable on $\mathbb{R} \times \mathbb{R}^n$ and $f(t, \cdot)$ L-Lipschitz for every $t \in \mathbb{R}$ (for C^1 f, equivalently $\|\partial f/\partial x\| \le L$). If x_{eq} is globally exponentially stable (definition 254), there is an exponential Lyapunov function V with exponent 2 on $\mathbb{R}^n$ (definition 284) and $c_4 > 0$ with $\|\partial V/\partial x\| \le c_4\|x - x_{\text{eq}}\|$ for all $t \ge 0$ and $x \in \mathbb{R}^n$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.14, global case. Differences:*

- *The Lipschitz bound replaces Khalil's equivalent Jacobian bound, and is asked for all $t \in \mathbb{R}$: differentiability of V at $t = 0$ involves solutions from negative initial times.*
- *The constants are $c_1 = \delta e^{-2L\delta}$, $c_2 = \delta k^2$ and $c_4 = 2k\delta e^{L\delta}$, cruder than Khalil's.*

Proof. Solutions $\phi(\tau; t, x)$ exist from every (t, x), since f is globally Lipschitz in x. Let $V(t, x) = \int_0^\delta \|\phi(t + \sigma; t, x) - x_{\text{eq}}\|^2 \, d\sigma$ with $\delta = \ln(2k^2 + 1)/(2\gamma)$. The exponential estimate and the lower bound $\|\phi(\tau) - x_{\text{eq}}\|^2 \ge e^{-2L(\tau - t)}\|x - x_{\text{eq}}\|^2$ sandwich V. V is differentiable: appending time to the state, solutions are differentiable in their initial data (**??**) and locally Lipschitz in it (**??**), so the integral can be differentiated under the integral sign. Along a solution, uniqueness gives $V = \int_h^{h+\delta} \|\phi(t + \sigma) - x_{\text{eq}}\|^2 d\sigma$, so $\dot{V} = \|\phi(t + \delta) - x_{\text{eq}}\|^2 - \|x - x_{\text{eq}}\|^2 \le (k^2 e^{-2\gamma\delta} - 1)\|x - x_{\text{eq}}\|^2 \le -\frac{1}{2}\|x - x_{\text{eq}}\|^2$. Finally, solutions from nearby states separate at most like $e^{L\sigma}$, which makes $V(t, \cdot)$ locally Lipschitz with constant $\approx 2k\delta e^{L\delta}\|x - x_{\text{eq}}\|$. □

Theorem 295 (Converse Lyapunov theorem for exponential stability)**.** *Let x_{eq} be an equilibrium of $\dot{x} = f(t, x)$ on $\mathbb{R}^n$, $f(t, x_{\text{eq}}) = 0$ for all t, with f continuously differentiable on $\mathbb{R} \times D$, $D = \{x : \|x - x_{\text{eq}}\| < r\}$, and $f(t, \cdot)$ L-Lipschitz on D for all t (for C^1 f, equivalently $\|\partial f/\partial x\| \le L$ there). Let $k, \gamma, r_0 > 0$ with $r_0 < r/k$, $D_0 = \{x : \|x - x_{\text{eq}}\| < r_0\}$, and suppose every trajectory on a segment $[t_0, t_1]$, $t_0 \ge 0$, with $x(t_0) \in D_0$ satisfies*

$$\|x(t) - x_{\text{eq}}\| \le k\,\|x(t_0) - x_{\text{eq}}\|\,e^{-\gamma(t - t_0)} \qquad \forall\, t \in [t_0, t_1].$$

Then there is an exponential Lyapunov function V with exponent 2 on D_0 (definition 284), that is, with

$$c_1\|x - x_{\text{eq}}\|^2 \le V(t, x) \le c_2\|x - x_{\text{eq}}\|^2, \qquad \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} f(t, x) \le -c_3\|x - x_{\text{eq}}\|^2$$

for $t \ge 0$ and $x \in D_0$, and $c_4 > 0$ with $\|\partial V/\partial x\| \le c_4\|x - x_{\text{eq}}\|$ there.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.14. Differences:*

- *f is C^1 and Lipschitz for all $t \in \mathbb{R}$, not only $t \geq 0$: differentiability of V at $t = 0$ involves solutions from negative initial times.*
- *The Lipschitz bound replaces Khalil's equivalent Jacobian bound.*
- *The estimate is assumed on trajectory segments.*

Proof. Cut the field off outside the ball: with χ a smooth bump equal to 1 on the ball of radius ρ_1, $kr_0 < \rho_1 < r$, and vanishing near the sphere of radius r, $\tilde{f} = \chi f$ is C^1 and globally Lipschitz in x, uniformly in t, and agrees with f on the ρ_1-ball. A solution of $\tilde{f}$ from D_0 never reaches the ρ_1-sphere: up to the first time it would, it is a solution of f, so Khalil's estimate keeps it within $kr_0 < \rho_1$. It is therefore a solution of f, and obeys the estimate for all time. The construction of theorem 294, run for $\tilde{f}$ from D_0, gives V; on D_0, where $\tilde{f} = f$, its derivative along $\tilde{f}$ is its derivative along f. □

7.7 Ultimate boundedness

A Lyapunov-like function whose derivative is negative only outside a ball does not certify stability, but it does bound the trajectories: they decay like a class $\mathcal{KL}$ function until they reach a neighbourhood of the ball, and stay there.

Theorem 296 (Ultimate boundedness theorem)**.** *Let the state space be $\mathbb{R}^n$, $\alpha_1 : [0, r) \to [0, b_1)$ and $\alpha_2 : [0, r) \to [0, b_2)$ class $\mathcal{K}$, and $V : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}$ differentiable at (t, x) and*

$$\alpha_1(\|x - x_c\|) \leq V(t, x) \leq \alpha_2(\|x - x_c\|), \qquad \dot{V}(t, x) \leq -W_3(x) \quad \text{for } \mu \leq \|x - x_c\|$$

whenever $t \geq 0$ and $\|x - x_c\| < r$, where W_3 is positive definite about x_c on the closed r-ball (definition 19) and $0 \leq \mu < c$, where $c < r$ and $\alpha_2(c) < b_1$. Then there is a class $\mathcal{KL}$ function β on $[0, c)$, depending only on α_1, α_2, W_3 and c — not on the system, V or μ — such that every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$, with $\|x(t_0) - x_c\| < c$ satisfies

$$\|x(t) - x_c\| \leq \max\{\beta(\|x(t_0) - x_c\|, t - t_0),\ \alpha_1^{-1}(\alpha_2(\mu))\} \qquad \forall\, t \in [t_0, t_1].$$

Khalil's estimates (4.42) and (4.43) are the two branches of the maximum.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.18. Differences:*

- *The sandwich holds on an open ball; Khalil's level $\alpha_1(r)$ becomes b_1, and the initial states form the open c-ball.*
- *$\mu = 0$ is allowed.*
- *$\beta(\rho, s) \leq \alpha_1^{-1}(\alpha_2(\rho))$ is part of the conclusion. The global case is theorem 299.*

Proof. Let $v(t) = V(t, x(t))$, $\eta = \alpha_2(\mu)$ and $d = \alpha_2(c)$. If $v > \eta$ then $\alpha_2(\|x - x_c\|) \geq v > \alpha_2(\mu)$, so $\|x - x_c\| > \mu$ and $\dot{v} \leq -W_3 \leq 0$. The trajectory therefore never reaches the sphere $\alpha_1(\|x - x_c\|) = d$: up to the first time it did, v could only fall while above η, so it would be at most $\max\{v(t_0), \eta\} < d$ (lemma 18), while on the sphere $v \geq d$.

Bound W_3 below by a class $\mathcal{K}$ α_3 (theorem 237). Above η, $\dot{v} \leq -\alpha_3(\|x - x_c\|) \leq -\alpha_3(\alpha_2^{-1}(v))$, so theorem 242 gives $v(t) \leq \max\{\sigma(\alpha_2(\|x(t_0) - x_c\|), t - t_0), \eta\}$. Apply the increasing α_1^{-1} and set $\beta(\rho, s) = \alpha_1^{-1}(\sigma(\alpha_2(\rho), s))$. □

Corollary 297 (Uniform boundedness from a Lyapunov-like function)**.** *Under the hypotheses of theorem 296, the trajectories are uniformly bounded around x_c (definition 255).*

Proof. From the c-ball, start within $a < c$: the estimate gives $\|x(t) - x_c\| \le \max\{\beta(a,0), \alpha_1^{-1}(\alpha_2(\mu))\}$, as β is increasing in its first argument and decreasing in its second. □

Corollary 298 (Uniform ultimate boundedness from a Lyapunov-like function)**.** *Under the hypotheses of theorem 296 with $\mu > 0$, if the system is forward complete from the c-ball about x_c at every $t_0 \ge 0$ (definition 69), its trajectories are uniformly ultimately bounded around x_c with ultimate bound $\alpha_1^{-1}(\alpha_2(\mu))$ (definition 257).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.18, estimate (4.43).*

Proof. The ultimate bound $b = \alpha_1^{-1}(\alpha_2(\mu))$ is positive, and $\beta(a,s) \to 0$, so $\beta(a,s) \le b$ for $s \ge T(a)$. Restrict a trajectory on $[t_0, \infty)$ to $[t_0, t]$ and apply the estimate. □

Theorem 299 (Ultimate boundedness theorem, global case)**.** *Let the state space be $\mathbb{R}^n$, V differentiable, α_1, α_2 class $\mathcal{K}_\infty$, W_3 positive definite about x_c on $\mathbb{R}^n$ (definition 19) and $\mu \ge 0$, with*

$$\alpha_1(\|x - x_c\|) \le V(t,x) \le \alpha_2(\|x - x_c\|), \qquad \dot V(t,x) \le -W_3(x) \quad \text{for } \mu \le \|x - x_c\|$$

for all $t \ge 0$ and $x \in \mathbb{R}^n$. Then there is a global class $\mathcal{KL}$ function β (definition 219), depending only on α_1, α_2 and W_3 — not on the system, V or μ — such that every trajectory on a segment $[t_0, t_1]$, $t_0 \ge 0$, satisfies

$$\|x(t) - x_c\| \le \max\{\beta(\|x(t_0) - x_c\|, t - t_0),\ \alpha_1^{-1}(\alpha_2(\mu))\} \qquad \forall\, t \in [t_0, t_1].$$

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.18, global case. Khalil's proof chooses one class $\mathcal{K}$ lower bound for W_3 on all of $\mathbb{R}^n$, which need not exist: W_3 may tend to 0 at infinity.*

Proof. For each $k \in \mathbb{N}$, apply theorem 296 with initial radius $c_k = k + 1$, on a ball large enough that $\alpha_2(c_k) < \alpha_1(R_k)$. If $\mu \ge \|x(t_0) - x_c\|$, the bound of β_k at an index with $c_k > \mu$ is at most $\alpha_1^{-1}(\alpha_2(\mu))$, so the trajectory never leaves the ultimate bound. It gives a class $\mathcal{KL}$ β_k with $\beta_k(\rho, s) \le \alpha_1^{-1}(\alpha_2(\rho))$. The sums $U(r,s) = \sum_{k \le \lceil r \rceil} \beta_k(k,s)$ are positive, decreasing to 0 in s, and increasing in r, and a trajectory from the r-ball that is outside the ultimate bound at time t is within $\beta_{\lceil r \rceil}(r, t - t_0) \le U(r+1, t - t_0)$ and within $\alpha(r) = \alpha_1^{-1}(\alpha_2(r))$ of x_c. **??** turns the class $\mathcal{K}_\infty$ bound α and the family U into a global class $\mathcal{KL}$ β above $\min\{\alpha(r), \sqrt{\alpha(r)\, U(r+1, s)}\}$. □

Corollary 300 (Global uniform boundedness from a Lyapunov-like function)**.** *Under the hypotheses of theorem 299, the trajectories are globally uniformly bounded around x_c (definition 256).*

Proof. Starting within a, the estimate gives $\|x(t) - x_c\| \le \max\{\beta(a,0), \alpha_1^{-1}(\alpha_2(\mu))\}$. □

Corollary 301 (Global uniform ultimate boundedness from a Lyapunov-like function)**.** *Under the hypotheses of theorem 299 with $\mu > 0$, if the system is forward complete from every state at every $t_0 \ge 0$ (definition 69), its trajectories are globally uniformly ultimately bounded around x_c with ultimate bound $\alpha_1^{-1}(\alpha_2(\mu))$ (definition 258).*

Proof. The ultimate bound b is positive and $\beta(a,s) \to 0$, so $\beta(a,s) \le b$ for $s \ge T(a)$; apply the estimate on $[t_0, t]$. □

7.8 Input-to-state stability

Input-to-state stability bounds the state by a decaying function of the initial condition plus a gain of the size of the input. An ISS-Lyapunov function certifies it through the global ultimate boundedness theorem, applied once per input: its class $\mathcal{KL}$ function does not depend on the input.

Definition 302 (Input-to-state stability)**.** A continuous-time system with input is *input-to-state stable* about x_{eq} when there are a global class $\mathcal{KL}$ function β (definition 219) and a class $\mathcal{K}_\infty$ function γ (definition 194) such that every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$, under any input u satisfies

$$\|x(t) - x_{\text{eq}}\| \leq \beta(\|x(t_0) - x_{\text{eq}}\|, t - t_0) + \gamma(M) \qquad \forall\, t \in [t_0, t_1]$$

whenever $\|u(\tau)\| \leq M$ for all $\tau \in [t_0, t]$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.7. Differences:

- Any bound M on the input replaces its supremum, which is equivalent.
- γ is class $\mathcal{K}_\infty$, as in Sontag, rather than class $\mathcal{K}$; a class $\mathcal{K}$ gain γ is dominated by $\gamma + \text{id}$.
- Solutions are not required to exist for all time.

Theorem 303 (Input-to-state stability implies global uniform asymptotic stability)**.** *If s is input-to-state stable about x_{eq} (definition 302), then x_{eq} is a globally uniformly asymptotically stable equilibrium of the unforced system $\dot{x} = F(t, x, 0)$ (definition 252).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Section 4.9.*

Proof. With $u \equiv 0$ and $M = 0$, $\gamma(M) = 0$ and the estimate is $\|x(t) - x_{\text{eq}}\| \leq \beta(\|x(t_0) - x_{\text{eq}}\|, t - t_0)$. The bound $\beta(r, s) + re^{-s}$ is again global class $\mathcal{KL}$, now also radially unbounded, and lemma 277 concludes. □

Theorem 304 (Input-to-state estimate from an ISS-Lyapunov function)**.** *Let the state space be $\mathbb{R}^n$, V differentiable, α_1, α_2 class $\mathcal{K}_\infty$, W_3 positive definite about x_c on $\mathbb{R}^n$ (definition 19), and $\rho : [0, \infty) \to [0, \infty)$ nondecreasing, with*

$$\alpha_1(\|x - x_c\|) \leq V(t, x) \leq \alpha_2(\|x - x_c\|), \qquad \dot{V}(t, x, u) \leq -W_3(x) \quad \text{for } \|x - x_c\| \geq \rho(\|u\|)$$

for all $t \geq 0$, x and u. Then there is a global class $\mathcal{KL}$ function β such that every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$, satisfies

$$\|x(t) - x_c\| \leq \beta(\|x(t_0) - x_c\|, t - t_0) + \alpha_1^{-1}(\alpha_2(\rho(M)))$$

for $t \in [t_0, t_1]$, whenever $\|u(\tau)\| \leq M$ for all $\tau \in [t_0, t]$.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.19, whose ρ is class $\mathcal{K}$; only its monotonicity is used.*

Proof. Fix the input and t. Freeze the input, clamped to $[t_0, t]$, into the system: $\dot{x} = F(\tau, x, u(\tau))$ is an unforced system with the same trajectories on $[t_0, t]$, and since $\|u(\tau)\| \leq M$ and ρ is nondecreasing, $\dot{V} \leq -W_3$ wherever $\|x - x_c\| \geq \rho(M)$. Apply theorem 299 with $\mu = \rho(M)$; its β does not depend on the system or on μ, so it is the same for every input. Finally $\max\{a, b\} \leq a + b$ for $a, b \geq 0$. □

Theorem 305 (ISS-Lyapunov theorem)**.** *Under the hypotheses of theorem 304 with ρ class $\mathcal{K}_\infty$, the system is input-to-state stable about x_c (definition 302) with gain $\gamma = \alpha_1^{-1} \circ \alpha_2 \circ \rho$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.19, with ρ of class $\mathcal{K}$; a class $\mathcal{K}$ ρ can be replaced by $\rho + \mathrm{id}$, which only weakens the hypothesis.*

Proof. theorem 304, with $\alpha_1^{-1} \circ \alpha_2 \circ \rho$ class $\mathcal{K}_\infty$ (definition 201, definition 202). □

7.9 LaSalle's invariance principle

When $\nabla V \cdot f \leq 0$ only, V may stall, and Lyapunov's theorems give stability but not convergence. LaSalle's principle recovers convergence for a time-invariant system: a trajectory confined to a compact set converges to its ω-limit set, on which V is constant, and forward invariance of that set turns constancy of V into $\nabla V \cdot f = 0$.

Lemma 306 (The ω-limit set is forward invariant)**.** *Let $\dot{x} = f(x)$ with f Lipschitz, and let φ be a trajectory on $[t_0, \infty)$. Then its ω-limit set $\omega(\varphi)$ (definition 52) is forward invariant (definition 80).*

Neither boundedness of φ nor compactness of an enclosing set is needed; those hypotheses of Khalil's Lemma 4.1 are what make $\omega(\varphi)$ nonempty and compact, not what make it invariant.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.1.*

Proof. Let ψ be a trajectory on $[a, b]$ with $p = \psi(a) \in \omega(\varphi)$, and fix $t \in [a, b]$. Given $\varepsilon > 0$, put $C = e^{K(b-a)}$ and pick, by the definition of the ω-limit set, arbitrarily large r with $\|\varphi(r) - p\| < \varepsilon / C$. The time-shifted curve $u(\cdot) = \varphi(\cdot + r - a)$ solves the same time-invariant equation on $[a, b]$ with $u(a) = \varphi(r)$, so Grönwall gives

$$\|\varphi(r + t - a) - \psi(t)\| = \|u(t) - \psi(t)\| \leq \|\varphi(r) - p\| \, e^{K(t-a)} < \varepsilon.$$

Since r may be taken arbitrarily large, so may $r + t - a$; hence φ is frequently within ε of $\psi(t)$, i.e. $\psi(t) \in \omega(\varphi)$.

Khalil argues instead with the flow map and the semigroup identity, which presumes a globally defined flow; comparing the two solutions directly needs only uniqueness, in its quantitative Grönwall form. □

Lemma 307 ($\dot{V}$ vanishes on the ω-limit set)**.** *Let $\dot{x} = f(x)$ with f Lipschitz, Ω compact and forward invariant, $V \in C^1$ with $\nabla V \cdot f \leq 0$ on Ω, and φ a trajectory on $[t_0, \infty)$ with $\varphi(t_0) \in \Omega$. Then $\nabla V(y) \cdot f(y) = 0$ for every $y \in \omega(\varphi)$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), proof of Theorem 4.4.*

Proof. $V(\varphi(t))$ is nonincreasing and bounded below on the compact Ω, so it converges to some L, and $V \equiv L$ on $\omega(\varphi)$ (lemma 55). Through $y \in \omega(\varphi)$ take a solution ψ on $[0, 1]$ (Picard–Lindelöf, theorem 38); it stays in $\omega(\varphi)$ by lemma 306, so $V \circ \psi \equiv L$ and its derivative at 0, which is $\nabla V(y) \cdot f(y)$, vanishes. □

Theorem 308 (LaSalle's invariance principle)**.** *Let $\dot{x} = f(x)$ with f Lipschitz, Ω compact and forward invariant (definition 80), and $V \in C^1$ with $\nabla V \cdot f \leq 0$ on Ω. Put*

$$E = \{x \in \Omega : \nabla V(x) \cdot f(x) = 0\},$$

and let M contain every forward invariant subset of E. Then every trajectory φ on $[t_0, \infty)$ with $\varphi(t_0) \in \Omega$ satisfies $\varphi(t) \to M$ as $t \to \infty$.

Khalil takes M to be the largest invariant *subset of E. Here M must absorb every* forward *invariant subset, a larger family, so the hypothesis on M is correspondingly stronger: backward invariance of $\omega(\varphi)$ would need solutions through its points to extend backwards in time.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.4.*

Proof. The ω-limit set lies in Ω, which is closed and contains the trajectory (lemma 56), and $\nabla V \cdot f$ vanishes on it (lemma 307), so $\omega(\varphi) \subseteq E$. It is forward invariant (lemma 306), so $\omega(\varphi) \subseteq M$. A trajectory confined to a compact set converges to every set containing its ω-limit set (lemma 57). □

Theorem 309 (Barbashin's theorem)**.** *Let $\dot{x} = f(x)$ on $\mathbb{R}^n$ with f Lipschitz, and $V \in C^1$ positive definite on a neighbourhood D of x_{eq} (definition 19) with $\nabla V \cdot f \leq 0$ on D. Suppose that for some $c > 0$ the sublevel set $\Omega_c = \{V \leq c\}$ is contained in D, compact and forward invariant, and that $\{x_{\text{eq}}\}$ contains every forward invariant subset of $\{x \in \Omega_c : \nabla V(x) \cdot f(x) = 0\}$. Then x_{eq} is locally asymptotically stable (definition 248).*

The hypothesis on E is the one a user of LaSalle discharges: solve $\nabla V \cdot f = 0$ and check that no solution can stay there but the equilibrium.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Corollary 4.1.*

Proof. Stability: V is a uniform Lyapunov function (lemma 285, theorem 288). Forward completeness: f is Lipschitz (theorem 73). Convergence: near x_{eq}, $V < c$, so a trajectory starting there starts in Ω_c, and theorem 308 with $M = \{x_{\text{eq}}\}$ applies. □

Theorem 310 (Krasovskii's theorem)**.** *Let $\dot{x} = f(x)$ on $\mathbb{R}^n$ with f Lipschitz, and $V \in C^1$ positive definite on $\mathbb{R}^n$, with compact sublevel sets (equivalently, radially unbounded) and $\nabla V \cdot f \leq 0$ everywhere. Suppose that for every c, $\{x_{\text{eq}}\}$ contains every forward invariant subset of $\{x : V(x) \leq c,\ \nabla V(x) \cdot f(x) = 0\}$. Then x_{eq} is globally asymptotically stable (definition 249).*

Unlike Barbashin's theorem (theorem 309) no forward invariant set is assumed: every sublevel set is compact, and $\nabla V \cdot f \leq 0$ keeps a trajectory in the one through its initial state.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Corollary 4.2.*

Proof. Stability and forward completeness as in Barbashin's theorem. The sublevel set $\{V \leq V(x(t_0))\}$ is compact and, since V does not increase along trajectories, forward invariant; apply theorem 308 with $M = \{x_{\text{eq}}\}$. □

7.10 Instability

Instability is the failure of stability, and the simplest witness is a trajectory that leaves a fixed neighbourhood from arbitrarily close initial states (lemma 259). Chetaev's theorem produces such trajectories from a function that grows along the system near the equilibrium.

The geometric criterion uses a positive certificate with positive derivative in an open region and vanishing on its boundary, with these conditions local to a closed ball.

Definition 311 (Chetaev function)**.** Let s be an input-free continuous-time system. A Chetaev certificate consists of an open region D, a point $x_{\text{eq}} \in \partial D$, a radius $\rho > 0$, and a globally C^1 function V. For $\|x - x_{\text{eq}}\| \leq \rho$, $V(x) > 0$ in D and $V(x) = 0$ on ∂D; in D the derivative $\nabla V(x) \cdot F(t, x, ())$ is strictly positive for every $t \geq 0$.

Reference: Khalil, *Nonlinear Systems*, 3rd ed., Theorem 4.3, p. 125.

Theorem 312 (Geometric Chetaev instability criterion)**.** *For an input-free system $\dot{x} = f(x)$ with f globally C^1, a Chetaev certificate at x_{eq} (definition 311) implies instability (definition 245).*

Reference: Khalil, Nonlinear Systems*, 3rd ed., Theorem 4.3, p. 125.*

Proof. Choose a positive seed near the boundary point and continue a cutoff curve. Boundary vanishing retains it in the region. The Lie derivative has a positive minimum on its compact superlevel, forcing growth beyond the ball's certificate ceiling. □

The exponential criterion is its corollary on the full positive set $D = \{V > 0\}$.

Theorem 313 (Exponential Chetaev instability criterion)**.** *Suppose the vector field f and certificate V are C^1, V is quadratically bounded on a ball, its positive set accumulates at the base point, and its Lie derivative satisfies $\dot{V} \geq 2\alpha V$ there for some $\alpha > 0$. Then the base point is unstable (definition 245) for any system whose state equation is $\dot{x} = f(x)$.*

Reference: Khalil, Nonlinear Systems*, 3rd ed. (Prentice Hall, 2002), Theorem 4.3, p. 125, for the method; this quantitative variant is not its statement.*

Proof. Take $D = \{V > 0\}$. The quadratic bound gives $V(x_{\text{eq}}) = 0$; the nearby positive seeds put x_{eq} in its closure. Continuity makes $V = 0$ on ∂D, and $\dot{V} \geq 2\alpha V > 0$ in D within the ball. Apply theorem 312. □

Chapter 8

Stability of discrete-time systems

The notions of the previous chapter for an autonomous discrete-time system $x^+ = f(k, x)$ (definition 87), under the same names. Discrete time is simpler in one respect: trajectories always exist and are unique (lemma 102, lemma 103), so every notion quantifies over trajectories on $[k_0, \infty)$ and no forward completeness is needed anywhere. Exponential estimates are geometric, ρ^{k-k_0} with $0 \leq \rho < 1$, in place of $e^{-\gamma(t-t_0)}$.

8.1 Definitions

Definition 314 (Stability of a discrete-time system)**.** The equilibrium x_{eq} of an autonomous discrete-time system $x^+ = f(k, x)$ is *stable* when for every $\varepsilon > 0$ and $k_0 \in \mathbb{N}$ there is $\delta = \delta(\varepsilon, k_0) > 0$ such that every trajectory on $[k_0, \infty)$ with $\|x(k_0) - x_{\text{eq}}\| < \delta$ satisfies $\|x(k) - x_{\text{eq}}\| < \varepsilon$ for all $k \geq k_0$.

The counterpart of definition 244. Trajectories on rays suffice: a discrete-time trajectory exists from every state for all later times (lemma 102).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4; Jungers and van de Wouw (2026), Definition 2.

Definition 315 (Instability of a discrete-time system)**.** The equilibrium is *unstable* when it is not stable (definition 314).

Definition 316 (Uniform stability of a discrete-time system)**.** The equilibrium is *uniformly stable* when δ in definition 314 can be chosen independently of k_0.

Definition 317 (Global uniform stability of a discrete-time system)**.** The equilibrium is *globally uniformly stable* when it is uniformly stable (definition 316) with a choice of $\delta(\varepsilon)$ satisfying $\delta(\varepsilon) \to \infty$ as $\varepsilon \to \infty$.

Definition 318 (Local asymptotic stability of a discrete-time system)**.** The equilibrium is *locally asymptotically stable* when it is stable (definition 314) and for every k_0 there is $c = c(k_0) > 0$ such that every trajectory on $[k_0, \infty)$ with $\|x(k_0) - x_{\text{eq}}\| < c$ satisfies $x(k) \to x_{\text{eq}}$. Unlike definition 248 no forward completeness is asked for: discrete-time trajectories always exist.

Definition 319 (Global asymptotic stability of a discrete-time system)**.** The equilibrium is *globally asymptotically stable* when it is stable (definition 314) and every trajectory on $[k_0, \infty)$ satisfies $x(k) \to x_{\text{eq}}$.

Definition 320 (Uniform convergence time, discrete time)**.** Trajectories starting within c of x_{eq} have a *uniform convergence time*: for every $\eta > 0$ there is $T = T(\eta) \in \mathbb{N}$ such that every trajectory on $[k_0, \infty)$ with $\|x(k_0) - x_{\text{eq}}\| < c$ satisfies $\|x(k) - x_{\text{eq}}\| < \eta$ for $k \geq k_0 + T$.

Definition 321 (Uniform asymptotic stability of a discrete-time system)**.** The equilibrium is *uniformly asymptotically stable* when it is uniformly stable (definition 316) and there is $c > 0$ from which convergence has a uniform convergence time (definition 320).

Definition 322 (Global uniform asymptotic stability of a discrete-time system)**.** The equilibrium is *globally uniformly asymptotically stable* when it is globally uniformly stable (definition 317) and every $c > 0$ admits a uniform convergence time (definition 320).

Definition 323 (Local exponential stability of a discrete-time system)**.** The equilibrium is *locally exponentially stable*, or *geometrically stable*, when there are $c, C > 0$ and $0 \leq \rho < 1$ such that every trajectory on $[k_0, \infty)$ with $\|x(k_0) - x_{\text{eq}}\| < c$ satisfies

$$\|x(k) - x_{\text{eq}}\| \leq C \, \|x(k_0) - x_{\text{eq}}\| \, \rho^{k-k_0} \qquad \forall \, k \geq k_0.$$

The counterpart of definition 253, with $\rho = e^{-\gamma}$.

Reference: Jungers and van de Wouw (2026), Definition 2, estimate (6).

Definition 324 (Global exponential stability of a discrete-time system)**.** The equilibrium is *globally exponentially stable* when the estimate of definition 323 holds for every initial state, with no radius c.

Boundedness is a property of the system near a point that need not be an equilibrium: trajectories stay, or end up and stay, within a prescribed distance of it.

Definition 325 (Uniform boundedness of a discrete-time system)**.** Trajectories are *uniformly bounded* around x_c when there is $c > 0$ such that for every $0 < a < c$ there is $\beta = \beta(a) > 0$ with

$$\|x(k_0) - x_c\| \leq a \;\Rightarrow\; \|x(k) - x_c\| \leq \beta \qquad \forall \, k \geq k_0$$

for every trajectory on $[k_0, \infty)$. The counterpart of definition 255.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6.

Definition 326 (Global uniform boundedness of a discrete-time system)**.** Trajectories are *globally uniformly bounded* around x_c when the bound of definition 325 holds for every $a > 0$, with no radius c.

Definition 327 (Uniform ultimate boundedness of a discrete-time system)**.** Trajectories are *uniformly ultimately bounded* with ultimate bound b around x_c when there is $c > 0$ such that for every $0 < a < c$ there is $T = T(a, b) \in \mathbb{N}$ with

$$\|x(k_0) - x_c\| \leq a \;\Rightarrow\; \|x(k) - x_c\| \leq b \qquad \forall \, k \geq k_0 + T$$

for every trajectory on $[k_0, \infty)$. The point x_c need not be an equilibrium.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.6; Jungers and van de Wouw (2026), Definition 3.

Definition 328 (Global uniform ultimate boundedness of a discrete-time system)**.** Trajectories are *globally uniformly ultimately bounded* with ultimate bound b around x_c when definition 327 holds for every $a > 0$, with no radius c.

8.2 Implications

Lemma 329 (A stable equilibrium is a fixed point)**.** *If x_{eq} is stable (definition 314), then $f(k, x_{\text{eq}}) = x_{\text{eq}}$ for every k.*

Proof. The trajectory from x_{eq} at time k stays within every $\varepsilon > 0$ of x_{eq}, so its value at $k+1$, which is $f(k, x_{\text{eq}})$, is at distance 0. □

Lemma 330 (Instability from a fixed escape radius, discrete time)**.** *Fix $\varepsilon > 0$ and k_0. If for every $\delta > 0$ some trajectory on $[k_0, \infty)$ starts within δ of x_{eq} and reaches distance at least ε from it, then x_{eq} is unstable (definition 315).*

Proof. □

Lemma 331 (Uniform stability implies stability, discrete time)**.** *A uniformly stable equilibrium (definition 316) is stable (definition 314).*

Proof. □

Lemma 332 (Global asymptotic stability implies local, discrete time)**.** *A globally asymptotically stable equilibrium (definition 319) is locally asymptotically stable (definition 318).*

Proof. □

Lemma 333 (Uniform asymptotic stability implies local, discrete time)**.** *A uniformly asymptotically stable equilibrium (definition 321) is locally asymptotically stable (definition 318). Unlike lemma 262, no forward completeness is needed.*

Proof. □

Lemma 334 (Global uniform asymptotic stability implies global, discrete time)**.** *A globally uniformly asymptotically stable equilibrium (definition 322) is globally asymptotically stable (definition 319).*

Proof. □

Lemma 335 (Exponential stability implies uniform stability, discrete time)**.** *A locally exponentially stable equilibrium (definition 323) is uniformly stable (definition 316).*

Proof. The estimate bounds the distance by C times the initial distance, so $\delta = \min(c, \varepsilon/C)$ works for every k_0. □

Lemma 336 (Exponential stability implies uniform asymptotic stability, discrete time)**.** *A locally exponentially stable equilibrium (definition 323) is uniformly asymptotically stable (definition 321).*

Proof. Uniform stability is lemma 335. From the ball of radius c the distance after T steps is below $Cc\rho^T$, which is below any $\eta > 0$ for T large, independently of k_0. □

Lemma 337 (Global exponential stability implies local, discrete time)**.** *A globally exponentially stable equilibrium (definition 324) is locally exponentially stable (definition 323).*

Proof. □

Lemma 338 (Global exponential stability implies global uniform, discrete time)**.** *A globally exponentially stable equilibrium (definition 324) is globally uniformly asymptotically stable (definition 322).*

Proof. $\delta(\varepsilon) = \varepsilon/C$ witnesses global uniform stability, and from every ball the estimate gives a uniform convergence time as in lemma 336. □

8.3 Lyapunov's direct method

For a time-invariant system $x^+ = f(x)$ the decrease of a Lyapunov function along the system is the difference $\Delta V(x) = V(f(x)) - V(x)$. Unlike continuous time, the theorems need continuity of f for a reason of their own: a trajectory can jump over a sphere in one step, so the first-exit argument of continuous time is replaced by keeping one step from a small ball inside the region where V is controlled.

Theorem 339 (Lyapunov's stability theorem, discrete time)**.** *Let E be proper, and let the system be $x^+ = f(x)$ with $f(x_{\mathrm{eq}}) = x_{\mathrm{eq}}$ and f continuous at x_{eq}. Let $\mathcal{D}$ be a neighbourhood of x_{eq} and V positive definite on $\mathcal{D}$ about x_{eq} (definition 19), with*

$$\Delta V(x) = V(f(x)) - V(x) \leq 0 \qquad \forall\, x \in \mathcal{D} \quad \{x_{\mathrm{eq}}\}.$$

Then x_{eq} is uniformly stable (definition 316).

Continuity of f at x_{eq} is not among the hypotheses of the reference, but it cannot be dropped: on $\mathbb{R}$, $V(x) = x^2/(1+x^4)$ and $f(x) = 1/x^2$ for $0 < |x| < 1$ ($f(x) = x$ otherwise) satisfy everything else, yet $x_0 = 10^{-3}$ jumps to $x_1 = 10^6$.

Reference: Jungers and van de Wouw (2026), Theorem 2, row 1.

Proof. Given ε, shrink it so that the closed ball of radius ε' lies in $\mathcal{D}$, and choose $s \leq \varepsilon'$ so that f maps the ball of radius s into the ball of radius ε'. On the compact annulus $s \leq \|x - x_{\mathrm{eq}}\| \leq \varepsilon'$, V has a positive minimum m; choose $\delta < s$ with $V < m$ on the ball of radius δ. By induction, a trajectory with $\|x(k_0) - x_{\mathrm{eq}}\| < \delta$ keeps $\|x(k) - x_{\mathrm{eq}}\| < s$ and $V(x(k)) < m$: the next state is within ε' by the choice of s, has $V(x(k+1)) \leq V(x(k)) < m$, and so cannot lie in the annulus. Nothing depends on k_0, since a trajectory from k_0 is the iterate sequence of $x(k_0)$ (lemma 104). □

Theorem 340 (Lyapunov's theorem for local asymptotic stability, discrete time)**.** *Let E be proper, and let the system be $x^+ = f(x)$ with $f(x_{\mathrm{eq}}) = x_{\mathrm{eq}}$. Let $\mathcal{D}$ be a neighbourhood of x_{eq} on which f is continuous, and V positive definite on $\mathcal{D}$ about x_{eq} (definition 19), with*

$$\Delta V(x) = V(f(x)) - V(x) < 0 \qquad \forall\, x \in \mathcal{D} \quad \{x_{\mathrm{eq}}\}.$$

Then x_{eq} is locally asymptotically stable (definition 318).

Continuity of f on $\mathcal{D}$ is not among the hypotheses of the reference, but it cannot be dropped. With $V(x) = |x|$ on $\mathbb{R}$, let f be odd with $f(0) = 0$ and $f(x) = (2^{-n-1} + x)/2$ on each interval $(2^{-n-1}, 2^{-n}]$: then $\Delta V < 0$ away from 0, but a trajectory starting in $(2^{-n-1}, 2^{-n}]$ converges to 2^{-n-1}, at every scale.

Reference: Jungers and van de Wouw (2026), Theorem 2, row 2.

Proof. Stability is theorem 339. For attractivity, choose a closed ball $K \subseteq \mathcal{D}$ around x_{eq} and, by stability, a δ whose trajectories stay in K. Along such a trajectory $V(x(k))$ is nonincreasing and bounded below, so it converges to some L. A cluster point y of the trajectory, which exists in the compact set K, has $V(y) = L$; by continuity of f at y, $x(k+1) = f(x(k))$ clusters at $f(y)$, so $V(f(y)) = L$ too. Then $\Delta V(y) = 0$, so $y = x_{\text{eq}}$. A sequence in a compact set whose only cluster point is x_{eq} converges to it. □

Theorem 341 (Lyapunov's theorem for global asymptotic stability, discrete time)**.** *Let E be proper, and let the system be $x^+ = f(x)$ with f continuous and $f(x_{\text{eq}}) = x_{\text{eq}}$. Let V be positive definite on E about x_{eq} (definition 19), with compact sublevel sets $\{V \leq c\}$ (equivalently, for continuous V, radially unbounded: lemma 290), and*

$$\Delta V(x) = V(f(x)) - V(x) < 0 \qquad \forall\, x \neq x_{\text{eq}}.$$

Then x_{eq} is globally asymptotically stable (definition 319).

Continuity of f is not among the hypotheses of the reference but cannot be dropped: with $V(x) = |x|$ on $\mathbb{R}$, the map $f(x) = x/2$ for $|x| \leq 1$, $f(x) = \operatorname{sign}(x)(1+|x|)/2$ otherwise, has $\Delta V < 0$ away from 0, yet the trajectory from 2 converges to 1.

Reference: Jungers and van de Wouw (2026), Theorem 2, row 3.

Proof. Stability is theorem 339 with $\mathcal{D} = E$. From any initial state, $V(x(k)) \leq V(x(k_0))$, so the trajectory stays in the compact sublevel set $\{V \leq V(x(k_0))\}$, and the ω-limit argument of theorem 340 applies there. □

Theorem 342 (Lyapunov's theorem for local exponential stability, discrete time)**.** *Let the system be $x^+ = f(x)$ with $f(x_{\text{eq}}) = x_{\text{eq}}$ and f continuous at x_{eq}, and let $\mathcal{D}$ be a neighbourhood of x_{eq}. Let $0 < c_1 \leq c_2$, $p > 0$ and $0 < c_3 < 1$, and let $V : E \to \mathbb{R}$ satisfy, for every $x \in \mathcal{D}$,*

$$c_1\|x - x_{\text{eq}}\|^p \leq V(x) \leq c_2\|x - x_{\text{eq}}\|^p, \qquad \Delta V(x) \leq -c_3 V(x).$$

Then x_{eq} is locally exponentially stable (definition 323).

Continuity of f at x_{eq} is not among the hypotheses of the reference, but without it a trajectory can leave $\mathcal{D}$ in a single step.

Reference: Jungers and van de Wouw (2026), Theorem 2, row 4.

Proof. Choose $r_0 > 0$ so that the ball $B(x_{\text{eq}}, r_0)$ lies in $\mathcal{D}$ and f maps it into $\mathcal{D}$, and let $C = (c_2/c_1)^{1/p} \geq 1$ and $\delta = r_0/(2C)$. By induction, a trajectory starting within δ stays in $B(x_{\text{eq}}, r_0)$ and satisfies $V(x(k)) \leq (1-c_3)^{k-k_0} V(x(k_0))$: the decrease applies at $x(k) \in \mathcal{D}$, and $x(k+1) = f(x(k)) \in \mathcal{D}$, so the sandwich gives $\|x(k+1) - x_{\text{eq}}\| \leq C\delta = r_0/2$. The geometric bound with $\rho = (1-c_3)^{1/p}$ then follows as in theorem 343. □

Theorem 343 (Lyapunov's theorem for global exponential stability, discrete time)**.** *Let the system be $x^+ = f(x)$, let $0 < c_1 \leq c_2$, $p > 0$ and $0 < c_3 < 1$, and let $V : E \to \mathbb{R}$ satisfy, for every x,*

$$c_1\|x - x_{\text{eq}}\|^p \leq V(x) \leq c_2\|x - x_{\text{eq}}\|^p, \qquad \Delta V(x) = V(f(x)) - V(x) \leq -c_3 V(x).$$

Then x_{eq} is globally exponentially stable (definition 324), with $C = (c_2/c_1)^{1/p}$ and $\rho = (1-c_3)^{1/p}$. No continuity of f is needed.

Reference: Jungers and van de Wouw (2026), Theorem 2, row 4, on the whole space. They ask for $p \geq 1$; $p > 0$ suffices.

Proof. The decrease condition is $V(f(x)) \leq (1-c_3)V(x)$, so by induction $V(x(k)) \leq (1-c_3)^{k-k_0}V(x(k_0))$. Sandwiching,

$$c_1\|x(k) - x_{\mathrm{eq}}\|^p \leq (1-c_3)^{k-k_0}c_2\|x(k_0) - x_{\mathrm{eq}}\|^p,$$

and taking p-th roots gives the geometric bound. □

Chapter 9

Stability of linear systems

The stability notions of a linear system are those of definition 244 and the definitions after it, applied to the unforced system (definition 67): there is no separate notion of stability for linear systems. What linearity adds is that each notion becomes a condition on the state transition matrix, and for constant A a condition on its eigenvalues.

9.1 Linear time-varying systems

Hespanha states Lyapunov stability for the continuous-time linear time-varying system $\dot{x} = A(t)x$ as four conditions on its homogeneous state response $x(t) = \Phi(t, t_0)x_0$. None of them is taken as a definition here: each is proved *equivalent* to the corresponding stability predicate of the unforced system, so the linear track and the nonlinear track share one notion of stability rather than two.

Two things stand between the textbook conditions and the predicates, and both are settled in this section. The predicates quantify over trajectories where the textbook quantifies over initial conditions, which the trajectory characterization below closes in both directions; and the predicates bound the response from a *ball* where the textbook bounds it from every initial state, which homogeneity closes. The second is why a linear system has no nontrivial region of attraction, recorded as lemma 350.

9.1.1 Hespanha's Definition 8.1

Lemma 344 (Stability as a bound on the state transition matrix)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\,x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 140).

The origin is a stable equilibrium (definition 244) of this system if and only if for every $t_0 \geq 0$ there is C with

$$\|\Phi(t, t_0)\,x\| \leq C\,\|x\|, \qquad \forall\, t \geq t_0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(1).*

Proof. ($\Rightarrow$) Apply stability at $\varepsilon = 1$ to get $\delta(1, t_0)$. Every response from the ball of radius δ stays inside the unit ball, and scaling turns that into the bound with $C = 2/\delta$.

($\Leftarrow$) Given ε, take $\delta = \varepsilon/(C+1)$. A trajectory is its own response (**??**), so $\|\varphi(t)\| \leq C\|\varphi(t_0)\| < \varepsilon$. □

Theorem 345 (Lyapunov stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 140).

The origin is a stable equilibrium (definition 244) of this system if and only if, for every $t_0 \geq 0$ and every initial condition x_0, the homogeneous state response $x(t) = \Phi(t, t_0)\, x_0$ is bounded on $[t_0, \infty)$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(1).*

Proof. The state-transition form is lemma 344. A bound on Φ gives the response bound $C\|x_0\|$; conversely a family of operators bounded at every point of a Banach space is norm-bounded, by the uniform boundedness principle. □

Theorem 346 (Uniform stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 140).

The origin is a uniformly stable equilibrium (definition 246) of this system if and only if there is a single C with

$$\|\Phi(t, t_0)\, x\| \leq C\, \|x\|, \qquad \forall\, t \geq t_0 \geq 0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(1), with the margin independent of t_0.*

Proof. Identical to lemma 344, with the margin δ and the bound C both carried outside the quantifier on t_0. □

Theorem 347 (Asymptotic stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 140).

The origin is an asymptotically stable equilibrium (definition 248) of this system if and only if, for every $t_0 \geq 0$ and every x_0, the homogeneous response $t \mapsto \Phi(t, t_0)x_0$ is bounded on $[t_0, \infty)$ and tends to 0 as $t \to \infty$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(2).*

Proof. The stability conjunct is theorem 345. For attractivity: an arbitrary x_0 is rescaled into the attractivity ball, and the limit scales back, since the response is linear in the initial state; conversely every trajectory is a response (**??**), so convergence of the responses is convergence of the trajectories, from any radius. □

Theorem 348 (Global exponential stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\,x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 140).

The origin is a globally exponentially stable equilibrium (definition 254) of this system if and only if there are $k, \gamma > 0$ with

$$\|\Phi(t, t_0)\,x\| \leq k\,e^{-\gamma(t-t_0)}\,\|x\|, \qquad \forall\, t \geq t_0 \geq 0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(3).*

Proof. Both directions are the trajectory characterization plus homogeneity: the exponential estimate along every trajectory is the estimate applied to the initial state, and conversely the estimate on any ball scales to the estimate everywhere. □

Theorem 349 (Exponential stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\,x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 140).

The origin is an exponentially stable equilibrium (definition 253) of this system if and only if there are $k, \gamma > 0$ with

$$\|\Phi(t, t_0)\,x\| \leq k\,e^{-\gamma(t-t_0)}\,\|x\|, \qquad \forall\, t \geq t_0 \geq 0,\ \forall\, x.$$

The condition is the one in theorem 348: the radius in the local definition carries no information here.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(3).*

Proof. As in the global case; the radius c of the local definition is absorbed by the scaling argument, and $c = 1$ serves in the converse. □

Lemma 350 (Local exponential stability is global, for a linear system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\,x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 140).

The origin is an exponentially stable equilibrium (definition 253) of this system if and only if it is globally exponentially stable (definition 254).

Proof. Both are equivalent to the same estimate, theorem 349 and theorem 348. □

For a linear system, uniform asymptotic stability already forces exponential decay of the state transition matrix: splitting time into windows of a uniform convergence time, each window contracts by a fixed factor.

Theorem 351 (Uniform asymptotic stability of a linear time-varying system is exponential)**.** *Consider the linear time-varying system $\dot{x} = A(t)\,x$ with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 140). The origin is uniformly asymptotically stable (definition 251) if and only if there are $k, \gamma > 0$ with*

$$\|\Phi(t, t_0)\,x\| \leq k\,e^{-\gamma(t-t_0)}\,\|x\|, \qquad \forall\, t \geq t_0 \geq 0,\ \forall\, x.$$

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.11.*

Proof. ($\Leftarrow$) The estimate is exponential stability (theorem 349), which is uniform asymptotic stability. ($\Rightarrow$) Uniform stability bounds $\|\Phi(t,t_0)\| \le C$ (theorem 346), and a uniform convergence time T from the c-ball to the $c/(2e)$-ball gives $\|\Phi(t,t_0)\| \le 1/e$ for $t \ge t_0 + T$, by homogeneity. Splitting $[t_0, t]$ into $N = \lfloor (t-t_0)/T \rfloor$ windows of length T and using $\Phi(t,t_0) = \Phi(t,t_0+T)\,\Phi(t_0+T,t_0)$ gives $\|\Phi(t,t_0)\| \le \max(C,1)\,e^{-N} \le e\max(C,1)\,e^{-(t-t_0)/T}$. □

Corollary 352 (Global uniform asymptotic stability of a linear time-varying system)**.** *Under the hypotheses of theorem 351, the origin is globally uniformly asymptotically stable (definition 252) if and only if $\Phi(t,t_0)$ decays exponentially, uniformly in $t_0 \ge 0$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.11.*

Proof. Global uniform asymptotic stability is uniform asymptotic stability, hence exponential decay (theorem 351); the decay is global exponential stability (theorem 348), which is global uniform asymptotic stability. □

The time-varying form of the Lyapunov equation.

Definition 353 (Differential Lyapunov equation)**.** For matrix-valued functions A, P and Q of time, P satisfies the *differential Lyapunov equation* for A with forcing Q when, for every $t \ge 0$, P is differentiable at t and

$$-\dot{P}(t) = P(t)A(t) + A(t)^{\mathsf{T}}P(t) + Q(t).$$

For constant matrices it is the continuous Lyapunov equation (definition 373).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), equation (4.28).

Lemma 354 (Constant solutions of the differential Lyapunov equation)**.** *For constant A, P and Q, the differential Lyapunov equation (definition 353) holds if and only if the continuous Lyapunov equation (definition 373) does.*

Proof. The derivative of a constant is 0, and derivatives are unique, so the differential equation says $0 = -(PA + A^{\mathsf{T}}P + Q)$. □

Theorem 355 (Converse Lyapunov theorem for linear time-varying systems)**.** *Consider $\dot{x} = A(t)x$ with A continuous and $\|A(t)\| \le L$, and let the origin be exponentially stable (definition 253). Let $Q(t)$ be continuous, symmetric, with $\|Q(t)\| \le q$ and $x^{\mathsf{T}}Q(t)x \ge q_1\|x\|^2$, $q_1 > 0$. Then there is a bounded symmetric $P(t)$ satisfying the differential Lyapunov equation (definition 353), and $V(t,x) = x^{\mathsf{T}}P(t)x$ is an exponential Lyapunov function with exponent 2 (definition 284); in particular $c_1\|x\|^2 \le x^{\mathsf{T}}P(t)x \le c_2\|x\|^2$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.12. Differences: the state space carries the sup norm, as everywhere for linear time-varying systems here.*

Proof. Let $P(t) = \int_t^\infty \Phi(\tau,t)^{\mathsf{T}}Q(\tau)\Phi(\tau,t)\,d\tau$, written as $\Phi(0,t)^{\mathsf{T}}R(t)\Phi(0,t)$ with $R(t) = \int_t^\infty \Phi(\tau,0)^{\mathsf{T}}Q(\tau)\Phi(\tau,0)\,d\tau$, which converges by exponential decay of Φ; the same decay bounds $P(t)$. Then $\dot{R}(t) = -\Phi(t,0)^{\mathsf{T}}Q(t)\Phi(t,0)$ and $\partial_t\Phi(0,t) = -\Phi(0,t)A(t)$ (**??**), and the product rule gives $-\dot{P} = PA + A^{\mathsf{T}}P + Q$. Along a solution, $x^{\mathsf{T}}P(t)x = \int_t^\infty (\Phi(\tau,t)x)^{\mathsf{T}}Q(\tau)\Phi(\tau,t)x\,d\tau$: exponential decay bounds it above, the growth bound (lemma 148) bounds it below on $[t, t+1]$, and its derivative along the solution is $-x^{\mathsf{T}}Q(t)x$. □

Theorem 356 (Instability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\,x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 140).

The origin is unstable (definition 245) for this system if and only if there are $t_0 \geq 0$ and x_0 whose homogeneous response $t \mapsto \Phi(t, t_0)x_0$ is unbounded on $[t_0, \infty)$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(4).*

Proof. The negation of theorem 345. □

9.2 Linear time-invariant systems

For constant A the state transition matrix is $e^{A(t-t_0)}$ (theorem 157), and the stability conditions of the previous section collapse to a condition on the eigenvalues of A alone. This is Hespanha's Theorem 8.1.

Clauses (1) and (4) of that theorem — marginal stability and instability — ask that the Jordan blocks at imaginary-axis eigenvalues be 1×1. They are not recorded here: they need a Jordan normal form, which is a project rather than a lemma.

9.2.1 The Hurwitz condition and its rate-indexed refinement

For a real square matrix A, Hurwitz stability is expressed through the complex eigenpairs of its entrywise complexification, over an abstract index type. The rate-indexed predicate records a strict spectral margin. Ordinary Hurwitz stability is the zero-rate case, but is stated separately rather than derived, so that it unfolds directly to $\mathrm{Re}(\mu) < 0$.

Definition 357 (Hurwitz matrix)**.** A real square matrix A is *Hurwitz* when every eigenpair (μ, v) of its complexification, with $v \neq 0$, satisfies $\mathrm{Re}(\mu) < 0$.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.1.

Definition 358 (Hurwitz with a prescribed decay rate)**.** A real square matrix A is *Hurwitz with decay rate α* when every eigenpair (μ, v) of its complexification, with $v \neq 0$, satisfies $\mathrm{Re}(\mu) < -\alpha$.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.1.

Theorem 359 (Rate-Hurwitz via a spectral shift)**.** *A real matrix A is Hurwitz with decay rate α if and only if the spectrally shifted matrix $A + \alpha I$ is Hurwitz.*

Reference: João P. Hespanha, Linear Systems Theory *(2nd ed.), continuous-time stability criterion. This spectral-shift corollary is proved directly from eigenpairs.*

Proof. A complex eigenvalue μ of A becomes $\mu + \alpha$ after the shift, and $\mathrm{Re}(\mu) < -\alpha$ is equivalent to $\mathrm{Re}(\mu + \alpha) < 0$. □

Theorem 360 (A Hurwitz matrix has a positive spectral margin)**.** *If A is Hurwitz then there is $c > 0$ such that every eigenvalue μ of A satisfies $\mathrm{Re}(\mu) < -c$.*

Proof. The spectrum is compact and nonempty, so Re attains a maximum on it at some μ_0, which is an eigenvalue and hence has $\mathrm{Re}(\mu_0) < 0$. Take $c = -\mathrm{Re}(\mu_0)/2$. □

The exponential decay estimates rest on a general Banach-algebra fact, stated with no matrices in it: an element of spectral radius below one has a contractive power.

Lemma 361 (A contractive power from a spectral radius below one)**.** *Let $\mathbb{A}$ be a nontrivial complete normed algebra over $\mathbb{C}$ and let $a \in \mathbb{A}$ have spectral radius $r(a) < 1$. Then there is an $m > 0$ with $\|a^m\| < 1$.*

Reference: Rudin, Functional Analysis *(Gelfand's spectral-radius formula).*

Proof. Gelfand's formula gives $\|a^m\|^{1/m} \to r(a)$, so $\|a^m\|^{1/m} < 1$ for all large m; pick any such m with $m \neq 0$. Were $\|a^m\| \geq 1$, raising to the positive power $1/m$ would give $\|a^m\|^{1/m} \geq 1$, a contradiction. □

The two predicates are related by a spectral shift (theorem 359), and they agree at rate zero.

Lemma 362 (Hurwitz is rate zero)**.** *A real square matrix A is Hurwitz (definition 357) if and only if it is Hurwitz with decay rate 0 (definition 358).*

Proof. Both unfold to a condition on $\operatorname{Re}\mu$, the first to $\operatorname{Re}\mu < 0$ and the second to $\operatorname{Re}\mu < -0$. □

Lemma 363 (A certified decay rate may be weakened)**.** *If A is Hurwitz with decay rate α (definition 358) and $\beta \leq \alpha$, then A is Hurwitz with decay rate β.*

Proof. Each eigenvalue satisfies $\operatorname{Re}\mu < -\alpha \leq -\beta$. □

Lemma 364 (Zero-dimensional matrices are vacuously Hurwitz)**.** *If the index type is empty then every A is Hurwitz with every decay rate α (definition 358).*

This records the convention chosen for the degenerate case: the predicate quantifies over nonzero eigenvectors, and in dimension zero there are none.

Proof. There is no nonzero vector, so the hypothesis of the predicate is never met. □

Lemma 365 (The scalar example)**.** *For $\alpha < \gamma$, the 1×1 matrix $(-\gamma)$ is Hurwitz with decay rate α (definition 358).*

Included as a concrete check on the sign convention of the definition.

Proof. The only eigenvalue is $-\gamma$, and $\operatorname{Re}(-\gamma) = -\gamma < -\alpha$. □

The Hurwitz condition bounds each eigenvalue away from the imaginary axis one at a time; because the spectrum is compact, that upgrades to a uniform margin (theorem 360). This is what turns a qualitative hypothesis into a quantitative rate, and it is used twice below.

9.2.2 Equivalent analytic conditions

Nothing in this subsection mentions a system: the Hurwitz condition, decay of e^{At}, and an explicit exponential envelope for $\|e^{At}\|$ are shown to be the same condition on the matrix. Two generic facts about exponential envelopes are recorded first.

Lemma 366 (The decay envelope vanishes)**.** *For $\gamma > 0$ and any constants K, t_0,*

$$\lim_{t \to \infty} K\, e^{-\gamma(t - t_0)} = 0.$$

Proof. The affine map $t \mapsto -\gamma(t - t_0)$ tends to $-\infty$ because $-\gamma < 0$; compose with exp and scale by K. □

Lemma 367 (Squeeze against an exponentially decaying envelope)**.** *Let E be a normed space, $f : \mathbb{R} \to E$, and $\gamma > 0$. If eventually*

$$\|f(t)\| \le k\, e^{-\gamma(t-t_0)}\, c,$$

then $f(t) \to 0$ as $t \to \infty$.

Stated over an arbitrary normed space because it is applied to both matrix- and vector-valued quantities.

Proof. The squeeze theorem: $\|f\|$ is bounded below by 0 and above by an envelope that tends to 0 (lemma 366, after reassociating $k \cdot E \cdot c = (kc) \cdot E$). □

Theorem 368 (Decay of the matrix exponential forces the Hurwitz condition)**.** *If $e^{At} \to 0$ as $t \to \infty$ then A is Hurwitz (definition 357).*

Stated from mere convergence rather than from an exponential bound, so that it serves clause (2) and clause (3) of Theorem 8.1 alike.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.1.*

Proof. Suppose $\operatorname{Re}\mu \ge 0$ for some eigenpair (μ, v) with $v \neq 0$. Complexification is continuous, so $e_{\mathbb{C}}^{At}v \to 0$ as well. But v is an eigenvector of $tA_{\mathbb{C}}$ with eigenvalue $t\mu$, hence of its exponential (lemma 164), so

$$\|e_{\mathbb{C}}^{At}v\| = |e^{t\mu}|\,\|v\| = e^{t\operatorname{Re}\mu}\|v\| \ge \|v\| > 0$$

for every $t \ge 0$, contradicting convergence to 0. □

Theorem 369 (Pointwise decay of the state forces the Hurwitz condition)**.** *If $e^{At}x \to 0$ as $t \to \infty$ for every real state x, then A is Hurwitz (definition 357).*

This is the composite that both clauses of Theorem 8.1 reach for: the stability predicates speak about states, not about the matrix.

Proof. Feeding the hypothesis the standard basis vectors returns the columns of e^{At}, and convergence in the matrix space is entrywise, so $e^{At} \to 0$; apply theorem 368. □

Theorem 370 (Hurwitz is decay of the matrix exponential)**.** *A real square matrix A is Hurwitz (definition 357) if and only if $e^{At} \to 0$ as $t \to \infty$.*

Proof. Forwards, **??** supplies an exponentially decaying envelope and lemma 367 squeezes; backwards is theorem 368. □

9.2.3 Hespanha's Theorem 8.1

Theorem 371 (Asymptotic stability of a linear time-invariant system)**.** *Consider the time-invariant system $\dot{x} = Ax$ with A a real square matrix.*

The origin is an asymptotically stable equilibrium (definition 248) if and only if A is Hurwitz (definition 357), that is, every eigenvalue of A has strictly negative real part.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.1(2).*

Proof. For constant A the state transition matrix is $e^{A(t-t_0)}$ (theorem 157), so theorem 347 turns the statement into one about $e^{At}x_0$.

If the response converges then A is Hurwitz (theorem 369). Conversely a Hurwitz A gives an exponentially decaying envelope for $\|e^{At}x_0\|$ (**??**); the envelope is bounded by $k\|x_0\|$, which supplies the stability half, and tends to zero (lemma 366), which supplies the attractivity half. □

Theorem 372 (Exponential stability of a linear time-invariant system)**.** *Consider the time-invariant system $\dot{x} = Ax$ with A a real square matrix.*

The origin is an exponentially stable equilibrium (definition 253) if and only if A is Hurwitz (definition 357).

Together with theorem 371 this says the two notions coincide for a time-invariant system — a genuine feature of constant A, not of linearity: for time-varying $A(t)$ they differ, as $\dot{x} = -x/(1+t)$ shows.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.1(3).*

Proof. As for theorem 371, but against theorem 349, which already asks for an exponential envelope. The forward direction squeezes $\|e^{At}x\|$ to zero under that envelope and applies theorem 369; the converse is **??** re-anchored from 0 to t_0. □

9.2.4 The Lyapunov equation

Definition 373 (Continuous Lyapunov equation)**.** For real square matrices A, P and Q, the matrix P satisfies the *continuous Lyapunov equation* for A with forcing Q when

$$PA + A^{\mathsf{T}}P = -Q.$$

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.2.

Definition 374 (Matrix quadratic form)**.** A real matrix P represents the quadratic form $x \mapsto x^{\mathsf{T}}Px$ on Euclidean state space.

Reference: standard quadratic Lyapunov-function construction.

Hespanha's Theorem 8.2 ties the spectral condition to solvability of $PA + A^{\mathsf{T}}P = -Q$. The direction a Lyapunov certificate is *used* in comes first: exhibiting a single positive-definite P proves the spectrum lies in the open left half-plane.

Theorem 375 (A Lyapunov certificate proves the spectrum lies in the open left half-plane)**.** *If P and Q are positive definite and $PM + M^{\mathsf{T}}P = -Q$, then every eigenvalue of M has strictly negative real part.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Complexify the equation and pair it against an eigenvector v of M with eigenvalue μ. The two terms contribute $\mu\, v^*Pv$ and $\bar{\mu}\, v^*Pv$, so the left-hand side is $2\,\mathrm{Re}(\mu)\, v^*Pv$ while the right-hand side is $-v^*Qv$. Both v^*Pv and v^*Qv are strictly positive (lemma 161), so $\mathrm{Re}(\mu) < 0$. □

Theorem 376 (The Lyapunov inequality certifies the Hurwitz condition)**.** *If P is positive definite and $-(PM + M^{\mathsf{T}}P)$ is positive definite, then M is Hurwitz (definition 357).*

This is Hespanha's Theorem 8.2, (5) ⇒ (3). It is stated through positive definiteness of the negated residual rather than through a matrix order relation, so that it needs no Loewner order.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Take the forcing to be minus the residual and apply theorem 375. □

The converse is a construction. One finite block of the integral $\int_0^m e^{tA^\mathsf{T}} Q e^{tA}\,\mathrm{d}t$ is summed into a convergent discrete Lyapunov series, with convergence and uniqueness both coming from the contraction $\|e^{Am}\| < 1$. A Hurwitz matrix supplies that contraction, which is theorem 379.

Lemma 377 (A convergent sum of positive semidefinite matrices is positive semidefinite)**.** *If (M_i) is a summable family of real positive semidefinite matrices, then $\sum_i M_i$ is positive semidefinite.*

Proof. Transposition and the quadratic form $P \mapsto x^\mathsf{T} P x$ are continuous and linear, so they commute with the sum: the sum is symmetric, and its quadratic form is a sum of nonnegative terms. □

Theorem 378 (The Lyapunov equation from a contracting exponential)**.** *Suppose $\|e^{Am}\| < 1$ for some integer $m > 0$. Then for every positive definite Q there is a positive definite P with $PA + A^\mathsf{T} P = -Q$ (definition 373), and P is the unique matrix solution.*

Reference: Khalil, Nonlinear Systems*.*

Proof. Write $L(P) = PA + A^\mathsf{T} P$ for the Lyapunov operator, $C = e^{Am}$, and $T(W) = C^\mathsf{T} W C$ for the congruence (Stein) operator.

One block. Let $R = \int_0^m e^{tA^\mathsf{T}} Q e^{tA}\,\mathrm{d}t$. Differentiating the integrand two ways and equating the derivatives by uniqueness gives the key identity $L(R) = T(Q) - Q$. R is positive definite because the integrand is, pointwise, and positive definiteness survives the integral.

Summing the blocks. $\|C\| < 1$ makes $\|T\| < 1$, so $\sum_k T^k R$ converges; call it P. It is positive definite, being a convergent sum of positive definite terms. Since A and C commute, L commutes with T, so $L(T^k R) = T^{k+1} Q - T^k Q$ and the series for $L(P)$ telescopes to $-Q$.

Uniqueness. A contraction has only the trivial fixed point, so L is injective; two solutions with the same forcing therefore coincide.

The contraction is taken in the Frobenius norm, the only place in the construction where $\|C^\mathsf{T}\| = \|C\|$ is needed. □

Theorem 379 (Lyapunov equation for a Hurwitz matrix)**.** *If a real matrix A is Hurwitz, then for every positive-definite Q there is a positive-definite matrix P satisfying $PA + A^\mathsf{T} P = -Q$, and this P is the unique matrix solution.*

omit [DecidableEq X] in Reference: Khalil, Nonlinear Systems*.*

Proof. A contracting integer-time matrix exponential is obtained from the Hurwitz spectrum. Integrating over one time block and summing the resulting discrete Lyapunov series constructs P; contraction proves convergence and uniqueness. □

Lemma 380 (From every forcing to one certificate)**.** *If every positive definite Q admits a positive definite solution of $PM + M^\mathsf{T} P = -Q$ (definition 373), then some positive definite P makes $-(PM + M^\mathsf{T} P)$ positive definite.*

This is Hespanha's Theorem 8.2, (4) ⇒ (5)*.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Instantiate the forcing at $Q = I$; the residual is then $-I$, whose negation is the identity. The edge carries no mathematics beyond the instantiation. □

9.2.5 The rate form

The Lyapunov inequality $PA + A^\mathsf{T}P \preceq -2cP$ carries an explicit decay rate: along a trajectory it gives $\dot{V} \leq -2cV$ for $V = x^\mathsf{T}Px$. It is not one of Hespanha's clauses, but it is the shape every contraction and guaranteed-decay-rate condition is written in, so it is recorded as a sixth equivalent.

Theorem 381 (The rate form certifies the Hurwitz condition)**.** *If P is positive definite, $c > 0$, and $PA + A^\mathsf{T}P \preceq -2cP$ — that is, $-(PA + A^\mathsf{T}P) - 2cP$ is positive semidefinite — then A is Hurwitz (definition 357).*

Along a trajectory this inequality gives $\dot{V} \leq -2cV$ for $V = x^\mathsf{T}Px$, so it certifies not merely stability but a decay rate.

Proof. Split $-(PA + A^\mathsf{T}P)$ as $(-(PA + A^\mathsf{T}P) - 2cP) + 2cP$: positive semidefinite plus positive definite is positive definite, so theorem 376 applies. □

Theorem 382 (A Hurwitz matrix admits a rate certificate)**.** *If A is Hurwitz (definition 357) then there are a positive definite P and a rate $c > 0$ with $PA + A^\mathsf{T}P \preceq -2cP$.*

Proof. Solve the Lyapunov equation for the *shifted* matrix. A Hurwitz A has a positive spectral margin c (theorem 360), so $A+cI$ is still Hurwitz (theorem 359). Solving $P(A+cI)+(A+cI)^\mathsf{T}P = -I$ (theorem 379) and expanding gives $PA+A^\mathsf{T}P = -I-2cP$, so the residual $-(PA+A^\mathsf{T}P)-2cP$ is *exactly* the identity.

No domination bound between two positive definite matrices is needed: the residual is not merely bounded below, it is I. □

9.2.6 Theorem 8.2

Theorem 383 (Hurwitz is solvability of the Lyapunov equation for every forcing)**.** *A real square matrix A is Hurwitz (definition 357) if and only if for every positive definite Q there is a positive definite P with $PA + A^\mathsf{T}P = -Q$ (definition 373), unique among all matrix solutions.*

This is Hespanha's Theorem 8.2, (3) ⇔ (4)*.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Forwards is theorem 379. Backwards, instantiate the forcing at $Q = I$ and apply theorem 375. □

Theorem 384 (Hurwitz is the existence of one Lyapunov certificate)**.** *A real square matrix A is Hurwitz (definition 357) if and only if some positive definite P makes $-(PA + A^\mathsf{T}P)$ positive definite.*

This is Hespanha's Theorem 8.2, (3) ⇔ (5)*, and the form in which a Lyapunov certificate is actually supplied: one matrix, not a family.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Forwards, theorem 379 produces a solution for each forcing and lemma 380 extracts one certificate from them; backwards is theorem 376. □

Theorem 385 (Hurwitz is the existence of a rate certificate)**.** *A real square matrix A is Hurwitz (definition 357) if and only if there are a positive definite P and a rate $c > 0$ with $PA+A^\mathsf{T}P \preceq -2cP$.*

This is clause 6 of theorem 386, which is not in Hespanha's list. It is the rate-carrying refinement of theorem 384, and the shape every contraction and guaranteed-decay-rate condition is written in.

Proof. theorem 382 and theorem 381. □

Theorem 386 (Hespanha's Theorem 8.2)**.** *For the time-invariant system $\dot{x} = Ax$ the following are equivalent.*

1. *The origin is asymptotically stable (definition 248).*
2. *The origin is exponentially stable (definition 253).*
3. *A is Hurwitz (definition 357).*
4. *For every positive definite Q there is a positive definite P with $PA + A^{\mathsf{T}}P = -Q$ (definition 373), and this P is the unique matrix solution.*
5. *There is a positive definite P with $-(PA + A^{\mathsf{T}}P)$ positive definite.*
6. *There are a positive definite P and a rate $c > 0$ with $PA + A^{\mathsf{T}}P \preceq -2cP$.*

Clause 6 is not in Hespanha's list. It is the rate-carrying form of clause 5: along a trajectory it gives $\dot{V} \leq -2cV$ for $V = x^{\mathsf{T}}Px$, hence $V(t) \leq V(0)e^{-2ct}$, and it is the shape every contraction and guaranteed-decay-rate condition is written in.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Every clause is tied to clause 3 and the cycle is closed by transitivity: $1 \Leftrightarrow 3$ is theorem 371, $2 \Leftrightarrow 3$ is theorem 372, $3 \Leftrightarrow 4$ is theorem 383, $3 \Leftrightarrow 5$ is theorem 384, and $3 \Leftrightarrow 6$ is theorem 385. □

9.2.7 Quadratic forms along a linear vector field

What makes a solution of the Lyapunov equation a Lyapunov *function*: the forcing matrix is exactly the dissipation rate of the associated quadratic form.

Theorem 387 (A Lyapunov solution makes its quadratic form decrease)**.** *Let P solve $PA+A^{\mathsf{T}}P = -Q$ (definition 373). Then the derivative of the centered quadratic form $x \mapsto (x - x_{\mathrm{eq}})^{\mathsf{T}}P(x - x_{\mathrm{eq}})$ (definition 374) along the linear vector field $y \mapsto Ay$ is*

$$-(x - x_{\mathrm{eq}})^{\mathsf{T}}Q\,(x - x_{\mathrm{eq}}).$$

This is what makes a Lyapunov solution a Lyapunov function*: the forcing Q is exactly the dissipation rate.*

Proof. The derivative of the quadratic form in the direction Ay is $\langle y, Py'\rangle + \langle y', Py\rangle$ with $y' = Ay$, which is the form represented by $PA + A^{\mathsf{T}}P$. Transporting the equation through `toEuclideanCLM` — under which transpose becomes adjoint — rewrites that as $-Q$. □

Lemma 388 (Identity forcing gives decrease at the squared norm)**.** *For P solving $PA+A^{\mathsf{T}}P = -I$ (definition 373), the derivative of the centered quadratic form (definition 374) along $y \mapsto Ay$ is $-\|x - x_{\mathrm{eq}}\|^2$.*

Proof. theorem 387 at $Q = I$, where the quadratic form is the squared Euclidean norm. □

9.2.8 A quadratic certificate for an unstable eigenmode

The instability counterpart (theorem 395): an eigenvalue with positive real part yields a real quadratic Chetaev certificate. The construction solves a shifted Lyapunov–Sylvester equation, whose injectivity rests on a nonresonance condition on the spectrum — which is what the lemmas below supply.

Lemma 389 (A nonresonant bilinear form vanishes on a pair of generalized eigenspaces)**.** *Let T be an endomorphism of a complex vector space and B a bilinear form satisfying the Sylvester-type identity*

$$B(x, Ty) + B(Tx, y) = c\, B(x, y) \qquad \forall\, x, y.$$

If $\xi + \nu \neq c$, then B vanishes on $\ker(T - \xi)^k \times \ker(T - \nu)^l$ for all k, l.

Proof. Strong induction on $k + l$. The base cases are the zero generalized eigenspaces, where the vector is 0. Otherwise write $Tx = N_x + \xi x$ and $Ty = N_y + \nu y$ with N_x, N_y one order lower; the induction hypothesis kills $B(N_x, y)$ and $B(x, N_y)$, leaving $(\xi + \nu - c)B(x, y) = 0$, and the coefficient is nonzero by nonresonance. □

Lemma 390 (A nonzero generalized eigenvector witnesses its eigenvalue)**.** *If $x \neq 0$ lies in the maximal generalized eigenspace of T at ξ, then ξ is an eigenvalue of T.*

Proof. Membership gives $x \in \ker(T - \xi)^k$ for some k, necessarily $k \neq 0$ since $x \neq 0$; so that generalized eigenspace is nontrivial, and a nontrivial generalized eigenspace forces an eigenvalue. □

Lemma 391 (A nonresonant bilinear form vanishes identically)**.** *Let V be a finite-dimensional complex vector space, T an endomorphism, and B a bilinear form with $B(x, Ty) + B(Tx, y) = c\, B(x, y)$. If no pair of eigenvalues of T sums to c, then $B = 0$.*

This is the uniqueness half of solvability for the Sylvester-type equation: the associated operator has trivial kernel exactly when the spectrum is nonresonant.

Proof. Over $\mathbb{C}$ the generalized eigenspaces of T span V, so it suffices to check $B(x, y) = 0$ for x, y in generalized eigenspaces. A nonzero such vector witnesses its eigenvalue (lemma 390), so nonresonance applies and lemma 389 gives $B(x, y) = 0$. □

Lemma 392 (Right multiplication acts on the right argument)**.** *For square matrices H, A over a commutative semiring, $(HA)(x, y) = H(x, Ay)$ as bilinear forms.*

Proof. Both sides unfold to $x \cdot (HAy)$ by associativity of matrix-vector multiplication. □

Lemma 393 (Left multiplication by the transpose acts on the left argument)**.** *For square matrices H, A over a commutative semiring, $(A^{\mathsf{T}}H)(x, y) = H(Ax, y)$ as bilinear forms.*

Together with lemma 392 this is what turns the matrix equation $HA + A^{\mathsf{T}}H = cH$ into the Sylvester-type identity on bilinear forms that lemma 391 consumes.

Proof. Move the transpose across the dot product: $x \cdot (A^{\mathsf{T}}Hy) = (Ax) \cdot (Hy)$. □

Lemma 394 (A symmetric matrix represents a symmetric form)**.** *If H is a symmetric real matrix then $H(x, y) = H(y, x)$ as a bilinear form.*

Proof. Move the transpose across the dot product and apply $H^{\mathsf{T}} = H$. □

Theorem 395 (Quadratic instability certificate from an unstable eigenpair)**.** *A complex eigenpair of a real matrix A with $\operatorname{Re}\mu > 0$ yields a Hermitian real matrix H, a shift $\alpha > 0$, and a direction w such that $w^{\mathsf{T}} H w > 0$ and*

$$HA + A^{\mathsf{T}} H - 2\alpha H$$

is positive definite.

Reference: Hahn, Stability of Motion*; Khalil,* Nonlinear Systems*.*

Proof. Choose a shift avoiding the finitely many pairwise spectral resonances, solve the resulting Lyapunov–Sylvester equation by finite-dimensional injectivity, and extract w from the real or imaginary part of the supplied eigenvector. □

9.3 Discrete-time linear systems

For $x(t+1) = A(t)x(t)$ Hespanha's conditions are again theorems about the stability predicates of the unforced system, now those of definition 314 and the definitions after it. The state transition matrix is a finite product, so no regularity of A is needed, and trajectories always exist, so no forward completeness enters. For constant A the spectral condition is the *Schur* condition: every eigenvalue in the open unit disc.

9.3.1 Hespanha's Definition 8.2

Lemma 396 (Discrete-time stability as a bound on the state transition matrix)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$ with state transition matrix Φ (definition 172), the origin is stable (definition 314) if and only if for every t_0 there is $C \geq 0$ with*

$$\|\Phi(t, t_0)\, x\| \leq C\, \|x\|, \qquad \forall\, t \geq t_0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(1).*

Proof. ($\Rightarrow$) Stability at $\varepsilon = 1$ gives a margin δ; every response from the δ-ball stays in the unit ball, and homogeneity scales that to the bound with $C = 2/\delta$. ($\Leftarrow$) Given ε, take $\delta = \varepsilon/(C+1)$: a trajectory is its own homogeneous response (theorem 192). □

Theorem 397 (Lyapunov stability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is stable (definition 314) if and only if, for every t_0 and every initial condition x_0, the homogeneous response $x(t) = \Phi(t, t_0)\, x_0$ (definition 183) is bounded on $t \geq t_0$.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(1).*

Proof. The state-transition form is lemma 396. A bound on Φ bounds each response by $C\|x_0\|$; conversely a family of operators bounded at every point of a Banach space is norm-bounded, by the uniform boundedness principle. □

Theorem 398 (Uniform stability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is uniformly stable (definition 316) if and only if there is a single C with $\|\Phi(t, t_0)\, x\| \leq C\, \|x\|$ for all $t \geq t_0$ and all x.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(1), with the margin independent of t_0.*

Proof. As lemma 396, with the margin and the bound both carried outside the quantifier on t_0. □

Theorem 399 (Asymptotic stability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is locally asymptotically stable (definition 318) if and only if, for every t_0 and every x_0, the homogeneous response $t \mapsto \Phi(t, t_0)x_0$ is bounded on $t \geq t_0$ and tends to 0.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(2).*

Proof. The stability conjunct is theorem 397. For attractivity an arbitrary x_0 is rescaled into the attractivity ball and the limit scales back; conversely every trajectory is a response (theorem 192). □

Theorem 400 (Global exponential stability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is globally exponentially stable (definition 324) if and only if there are $c > 0$ and $0 \leq \rho < 1$ with*

$$\|\Phi(t, t_0)\, x\| \leq c\, \rho^{t-t_0}\, \|x\|, \qquad \forall\, t \geq t_0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(3).*

Proof. The estimate along every trajectory is the estimate applied to its initial state, since a trajectory is its own homogeneous response; conversely the response from x is a trajectory. □

Theorem 401 (Exponential stability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is locally exponentially stable (definition 323) if and only if there are $c > 0$ and $0 \leq \rho < 1$ with $\|\Phi(t, t_0)\, x\| \leq c\, \rho^{t-t_0}\, \|x\|$ for all $t \geq t_0$ and all x: the condition of theorem 400.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(3).*

Proof. As in the global case; the radius of the local definition is absorbed by the scaling argument, and radius 1 serves in the converse. □

Lemma 402 (Local exponential stability is global, for a discrete-time linear system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is locally exponentially stable (definition 323) if and only if it is globally exponentially stable (definition 324).*

Proof. Both are equivalent to the same estimate: theorem 401 and theorem 400. □

Theorem 403 (Instability of a discrete-time linear time-varying system)**.** *For the discrete-time linear system $x(t+1) = A(t)x(t)$, the origin is unstable (definition 315) if and only if there are t_0 and x_0 whose homogeneous response $t \mapsto \Phi(t, t_0)x_0$ is unbounded on $t \geq t_0$.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.2(4).*

Proof. The negation of theorem 397. □

9.3.2 The Schur condition

Nothing in this subsection mentions a system: the Schur condition, decay of A^k, and a geometric envelope for $\|A^k\|$ are the same condition on the matrix. The proof runs through Gelfand's formula, as for the Hurwitz condition, with powers in place of the exponential.

Definition 404 (Schur matrix)**.** A real square matrix A is *Schur* when every eigenpair (μ, v) of its complexification, with $v \neq 0$, satisfies $|\mu| < 1$. The discrete-time counterpart of definition 357.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.3.

Theorem 405 (A Schur matrix has geometrically decaying powers)**.** *If A is Schur (definition 404) then there are $C > 0$ and $0 \leq \rho < 1$ with*

$$\|A^k\| \leq C\,\rho^k, \qquad \forall\, k \in \mathbb{N}.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.3.*

Proof. The spectral radius of $A_{\mathbb{C}}$ is below 1, so by Gelfand's formula (lemma 361) some power has $\|A^m\| < 1$; fix $\|A^m\| \leq c < 1$ with $c > 0$, and let M bound $\|A^r\|$ for $r < m$. Writing $k = r + mq$ with $r < m$, submultiplicativity gives $\|A^k\| \leq Mc^q$, and with $\rho = c^{1/m}$, $c^q = \rho^{mq} \leq \rho^k/c$. □

Theorem 406 (Geometric decay of the state under a Schur matrix)**.** *If A is Schur (definition 404) then there are $C > 0$ and $0 \leq \rho < 1$ with $\|A^k x\| \leq C\,\rho^k\,\|x\|$ for every k and x.*

Proof. theorem 405 and $\|Ax\|_\infty \leq \|A\|_\infty \|x\|_\infty$. □

Theorem 407 (Decay of the matrix powers forces the Schur condition)**.** *If $A^k \to 0$ as $k \to \infty$ then A is Schur (definition 404).*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.3.*

Proof. Suppose $|\mu| \geq 1$ for an eigenpair (μ, v) of $A_{\mathbb{C}}$ with $v \neq 0$. Complexification is continuous, so $A_{\mathbb{C}}^k v \to 0$; but $A_{\mathbb{C}}^k v = \mu^k v$ (lemma 166) has norm $|\mu|^k \|v\| \geq \|v\| > 0$. □

Theorem 408 (Pointwise decay of the state forces the Schur condition)**.** *If $A^k x \to 0$ as $k \to \infty$ for every real state x, then A is Schur (definition 404).*

Proof. The standard basis vectors return the columns of A^k, so $A^k \to 0$; apply theorem 407. □

Theorem 409 (Schur is decay of the matrix powers)**.** *A real square matrix A is Schur (definition 404) if and only if $A^k \to 0$ as $k \to \infty$.*

Proof. Forwards, theorem 405 supplies a geometric envelope; backwards is theorem 407. □

Theorem 410 (Schur is a geometric envelope for the matrix powers)**.** *A real square matrix A is Schur (definition 404) if and only if there are $C > 0$ and $0 \leq \rho < 1$ with $\|A^k\| \leq C\rho^k$ for all k.*

Proof. Forwards is theorem 405; backwards, the envelope forces $A^k \to 0$ and theorem 409 applies. □

9.3.3 Hespanha's Theorem 8.3

Clauses (1) and (3) of the theorem, marginal stability and instability, ask that the Jordan blocks at eigenvalues of magnitude one be 1×1; as in continuous time they need a Jordan normal form and are not recorded here.

Theorem 411 (Asymptotic stability of a discrete-time linear time-invariant system)**.** *For the time-invariant system $x^+ = Ax$, the origin is locally asymptotically stable (definition 318) if and only if A is Schur (definition 404): every eigenvalue of A has magnitude strictly smaller than* 1*.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.3(2).*

Proof. For constant A the state transition matrix is A^{t-t_0} (theorem 188), so theorem 399 turns the statement into one about $A^k x_0$. If the responses converge then A is Schur (theorem 408); conversely a Schur A has a geometric envelope $\|A^k x\| \le C\rho^k \|x\|$ (theorem 406), which bounds the responses and makes them vanish. □

Theorem 412 (Exponential stability of a discrete-time linear time-invariant system)**.** *For the time-invariant system $x^+ = Ax$, the origin is locally exponentially stable (definition 323) if and only if A is Schur (definition 404).*

Together with theorem 411 this says the two notions coincide for a time-invariant system.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.3(2).*

Proof. Against theorem 401, which already asks for a geometric envelope $\|A^{t-t_0}x\| \le c\rho^{t-t_0}\|x\|$. Forwards, at $t_0 = 0$ the envelope makes $A^k x \to 0$, so A is Schur (theorem 408); backwards is theorem 406. □

9.3.4 The discrete Lyapunov equation

The discrete-time counterpart of $PA + A^\mathsf{T}P = -Q$ is $A^\mathsf{T}PA - P = -Q$. Unlike continuous time, the solution needs no integral: it is the convergent series $\sum_k (A^\mathsf{T})^k Q A^k$.

Definition 413 (Discrete Lyapunov equation)**.** For real square matrices A, P and Q, the matrix P satisfies the *discrete-time Lyapunov equation* for A with forcing Q when

$$A^\mathsf{T}PA - P = -Q.$$

Along $x(t+1) = Ax(t)$ the quadratic form $V(x) = x^\mathsf{T}Px$ then changes by $V(Ax) - V(x) = -x^\mathsf{T}Qx$.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.4.

Lemma 414 (Decrease of the quadratic form along a discrete LTI system)**.** *If $A^\mathsf{T}PA - P = -Q$ (definition 413), then for every x,*

$$(Ax)^\mathsf{T}P(Ax) - x^\mathsf{T}Px = -x^\mathsf{T}Qx.$$

Proof. $(Ax)^\mathsf{T}P(Ax) = x^\mathsf{T}(A^\mathsf{T}PA)x$, and $A^\mathsf{T}PA = P - Q$. □

Lemma 415 (The transpose of a Schur matrix is Schur)**.** *If A is Schur (definition 404) then so is A^T.*

Proof. A and A^T have the same characteristic polynomial, hence the same spectrum over $\mathbb{C}$, and the spectrum is the set of eigenvalues (lemma 162). □

Theorem 416 (Discrete Lyapunov equation for a Schur matrix)**.** *If A is Schur (definition 404), then for every positive definite Q there is a positive definite P with $A^\mathsf{T}PA - P = -Q$ (definition 413), and P is the unique matrix solution.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Take $P = \sum_k (A^\mathsf{T})^k Q A^k$. It converges because both A and A^T are Schur (lemma 415), so their powers decay geometrically (theorem 405). Shifting the index, $A^\mathsf{T}PA = P - Q$. The first term is Q and the others are positive semidefinite, so P is positive definite (lemma 377). For uniqueness, a difference W of two solutions satisfies $W = A^\mathsf{T}WA$, hence $W = (A^\mathsf{T})^k W A^k \to 0$. □

Theorem 417 (A discrete Lyapunov certificate proves the spectrum lies in the unit disc)**.** *If P and Q are positive definite and $A^\mathsf{T}PA - P = -Q$, then A is Schur (definition 404).*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Complexify the equation and pair it against an eigenvector v with eigenvalue μ: the left-hand side is $(|\mu|^2 - 1)\, v^* P v$ and the right-hand side is $-v^* Q v$. Both quadratic forms are strictly positive (lemma 161), so $|\mu|^2 < 1$. □

Theorem 418 (The discrete Lyapunov inequality certifies the Schur condition)**.** *If P is positive definite and $-(A^\mathsf{T}PA - P)$ is positive definite, then A is Schur (definition 404).*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Take the forcing to be minus the residual and apply theorem 417. □

Theorem 419 (Schur is solvability of the discrete Lyapunov equation for every forcing)**.** *A real square matrix A is Schur (definition 404) if and only if for every positive definite Q there is a positive definite P with $A^\mathsf{T}PA - P = -Q$ (definition 413), unique among all matrix solutions.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Forwards is theorem 416. Backwards, instantiate the forcing at $Q = I$ and apply theorem 417. □

Theorem 420 (Schur is the existence of one discrete Lyapunov certificate)**.** *A real square matrix A is Schur (definition 404) if and only if some positive definite P makes $-(A^\mathsf{T}PA - P)$ positive definite.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Forwards, solve the equation with $Q = I$ (theorem 416); the residual is then $-I$. Backwards is theorem 418. □

9.3.5 Hespanha's Theorem 8.4

Theorem 421 (Hespanha's Theorem 8.4)**.** *For the time-invariant system $x^+ = Ax$ the following are equivalent.*

1. *The origin is asymptotically stable (definition 318).*
2. *The origin is exponentially stable (definition 323).*
3. *A is Schur (definition 404).*

4. *For every positive definite Q there is a positive definite P with $A^\mathsf{T}PA - P = -Q$ (definition 413), and this P is the unique matrix solution.*

5. *There is a positive definite P with $-(A^\mathsf{T}PA - P)$ positive definite.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.4.*

Proof. Every clause is tied to clause 3: $1 \Leftrightarrow 3$ is theorem 411, $2 \Leftrightarrow 3$ is theorem 412, $3 \Leftrightarrow 4$ is theorem 419, and $3 \Leftrightarrow 5$ is theorem 420. □

Chapter 10

Lyapunov's indirect method

Near an equilibrium a nonlinear system behaves like its linearization: if the Jacobian is Hurwitz the equilibrium is locally exponentially stable, and if the Jacobian has an eigenvalue with positive real part it is unstable. Both branches run the same way. A quadratic certificate for the linear system (theorem 379, theorem 395) is a certificate for the nonlinear one as well, once the linearization error is absorbed near the equilibrium.

10.1 The linearization error

Lemma 422 (Absorbing the linearization remainder)**.** *Let f be C^1 with $Df(x_{\mathrm{eq}}) = A$, let M be a matrix, and let $c > 0$. Then there is $r > 0$ such that for every x with $\|x - x_{\mathrm{eq}}\| < r$,*

$$\left| D\, q_M(x)\, [\, f(x) - f(x_{\mathrm{eq}}) - A(x - x_{\mathrm{eq}})\,] \right| \le c\, \|x - x_{\mathrm{eq}}\|^2,$$

where q_M is the centred quadratic form $q_M(x) = \langle M(x - x_{\mathrm{eq}}),\, x - x_{\mathrm{eq}} \rangle$.

The bracket is the first-order remainder of f at x_{eq}. The point is that c is arbitrary*: near enough to the equilibrium the remainder perturbs $\dot{q}_M$ by less than any prescribed multiple of $\|x - x_{\mathrm{eq}}\|^2$, which is the order of $\dot{q}_M$ itself. This is why the linearization decides stability.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.7 (the quadratic-Lyapunov proof of Lyapunov's indirect method).*

Proof. Write $y = x - x_{\mathrm{eq}}$ and e for the remainder. Differentiability of f at x_{eq} gives, for any $\eta > 0$, a radius within which $\|e\| \le \eta \|y\|$. The derivative of a centred quadratic form is bounded by $|Dq_M(x)[e]| \le 2\|M\|\, \|y\|\, \|e\|$, so the product is at most $2\|M\| \eta\, \|y\|^2$. Choosing $\eta = c/(4(\|M\| + 1))$ makes $2\|M\|\eta \le c$. □

10.2 The stable branch

The stable-branch theorem gives uniform local exponential decay on every finite forward solution segment, without silently excluding solutions that are not globally defined.

Theorem 423 (Lyapunov's indirect method, stable branch)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be C^1, and let x_{eq} be an equilibrium. If its Jacobian A at the equilibrium is Hurwitz, then x_{eq} is locally exponentially stable (definition 253) for any system whose state equation is $\dot{x} = f(x)$.*

Reference: Khalil, Nonlinear Systems*.*

Proof. Solve the identity-forced Lyapunov equation for a positive-definite P; after absorbing the $o(\|x - x_{\text{eq}}\|)$ linearization remainder on a small ball, $V(x) = (x - x_{\text{eq}})^{\mathsf{T}} P (x - x_{\text{eq}})$ is an exponential Lyapunov function with exponent 2 there, so theorem 292 applies. □

10.3 Time-varying systems

For a time-varying system the linearization is itself time-varying, and exponential stability transfers between the two in both directions. The converse Lyapunov function of the nonlinear system serves for its linearization; conversely, the solution of the differential Lyapunov equation of the linearization serves for the nonlinear system.

Theorem 424 (Exponential stability passes to the linearization)**.** *Let x_{eq} be an equilibrium of $\dot{x} = f(t, x)$ on $\mathbb{R}^n$, with f continuously differentiable on $\mathbb{R} \times D$, $D = \{x : \|x - x_{\text{eq}}\| < r\}$, and $f(t, \cdot)$ and $\partial f / \partial x(t, \cdot)$ Lipschitz on D, uniformly in t. If x_{eq} is locally exponentially stable (definition 253), then it is a locally exponentially stable equilibrium of the linearization*

$$\dot{x} = A(t)(x - x_{\text{eq}}), \qquad A(t) = \frac{\partial f}{\partial x}(t, x_{\text{eq}}).$$

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.15, "only if" direction. Khalil's bounded Jacobian is the Lipschitz bound on $f(t, \cdot)$.*

Proof. Let V be the converse Lyapunov function of the nonlinear system (theorem 295). Along the linearization,

$$\dot{V}_{\text{lin}} = \dot{V} + \frac{\partial V}{\partial x}(A(t)(x - x_{\text{eq}}) - f(t, x)) \leq -c_3 \|x - x_{\text{eq}}\|^2 + c_4 L_1 \|x - x_{\text{eq}}\|^3,$$

by **??**. On a small ball this is at most $-\frac{c_3}{2}\|x - x_{\text{eq}}\|^2$, so V is an exponential Lyapunov function for the linearization (theorem 292). □

Theorem 425 (Lyapunov's indirect method for time-varying systems)**.** *Under the hypotheses of theorem 424, if x_{eq} is a locally exponentially stable equilibrium of the linearization $\dot{x} = A(t)(x - x_{\text{eq}})$, $A(t) = \frac{\partial f}{\partial x}(t, x_{\text{eq}})$, then it is locally exponentially stable for $\dot{x} = f(t, x)$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.13. Khalil's bounded Jacobian is the Lipschitz bound on $f(t, \cdot)$.*

Proof. In coordinates the linearization is $\dot{v} = A(t)v$ with A continuous and bounded, and exponentially stable (lemma 268). Theorem 355 with $Q = I$ gives a bounded $P(t)$ with $c_1\|x\|^2 \leq x^{\mathsf{T}} P(t) x \leq c_2 \|x\|^2$ and derivative $-x^{\mathsf{T}} x$ along the linearization. Along f, $V = (x - x_{\text{eq}})^{\mathsf{T}} P(t)(x - x_{\text{eq}})$ has derivative

$$\dot{V} \leq -c_3 \|x - x_{\text{eq}}\|^2 + 2\|P\| L_1 \|x - x_{\text{eq}}\|^3,$$

by **??**, which is at most $-\frac{c_3}{2}\|x - x_{\text{eq}}\|^2$ on a small ball; conclude by theorem 292. □

Theorem 426 (Exponential stability and the linearization)**.** *Under the hypotheses of theorem 424, x_{eq} is locally exponentially stable for $\dot{x} = f(t, x)$ if and only if it is for the linearization.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.15.*

Proof. Theorem 424 and theorem 425. □

10.4 The unstable branch

The quadratic certificate is fed to Chetaev's theorem (theorem 313).

Theorem 427 (Lyapunov's indirect method, unstable branch)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be C^1 with $f(x_{\text{eq}}) = 0$. If the Jacobian at x_{eq} has a complex eigenvalue with positive real part, then x_{eq} is unstable (definition 245) for any system whose state equation is $\dot{x} = f(x)$.*

Reference: Khalil, Nonlinear Systems *(Lyapunov's indirect method).*

Proof. Build the real quadratic certificate supplied by theorem 395, absorb the first-order nonlinear remainder on a small ball, and apply the exponential Chetaev criterion. □

Theorem 428 (Instability of a linear system)**.** *If $A \in \mathbb{R}^{n \times n}$ has a complex eigenvalue μ with $\operatorname{Re} \mu > 0$, then the origin is unstable (definition 245) for any system on $\mathbb{R}^n$ whose state equation is $\dot{x} = Ax$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.7.*

Proof. The field $x \mapsto Ax$ is C^1, vanishes at 0 and is its own Jacobian, so this is theorem 427 at $x_{\text{eq}} = 0$. □

Chapter 11

Controllability and observability

11.1 Observability and controllability matrices

For a finite-dimensional linear system

$$\dot{x} = Ax + Bu, \qquad y = Cx,$$

with state matrix A, input matrix B, and output matrix C, this section collects the (finite-horizon) controllability and observability matrices and the bridge theorem connecting the textbook observability predicate to the kernel of the observability matrix.

11.1.1 Observability matrix

Definition 429 (Observability matrix)**.** The *observability matrix* of a pair (A, C) with $A \in \mathbb{F}^{n\times n}$ and $C \in \mathbb{F}^{p\times n}$ is the block-row matrix

$$\mathcal{O}(A, C) = \begin{bmatrix} C \\ C\,A \\ C\,A^2 \\ \vdots \\ C\,A^{n-1} \end{bmatrix} \in \mathbb{F}^{(np)\times n}.$$

Rows are indexed by $\mathrm{Fin}\, n \times \mathrm{Fin}\, p$, so that A^k is available without casting $k : \mathrm{Fin}\, n$ through Fin.val.

Definition 430 (Observability)**.** A linear system (A, C) is *observable* when the only state $x \in \mathbb{F}^n$ for which

$$C\,A^k\,x = 0 \qquad \text{for every } k = 0, 1, \ldots, n-1$$

is the zero state. This phrasing does not mention $\mathcal{O}(A, C)$, so the bridge theorem 436 has real content.

Lemma 431 (Entries of the observability matrix)**.** *Block-row entry shape: at row (k, i) and column j, the observability matrix coincides with the (i, j) entry of $C\,A^k$:*

$$\mathcal{O}(A, C)_{(k,i),\,j} = (C\,A^k)_{i,\,j}.$$

Proof. Holds definitionally by the encoding chosen for definition 429. □

Lemma 432 (Action of the observability matrix)**.** *For every state $x \in \mathbb{F}^n$ and every $(k, i) \in \operatorname{Fin} n \times \operatorname{Fin} p$,*

$$(\mathcal{O}(A, C) \cdot x)_{(k,i)} = (C\, A^k \cdot x)_i.$$

This is the definitional bridge that powers theorem 436.

Proof. Holds by definitional unfolding of $\mathcal{O}(A, C)$ (see definition 429). □

11.1.2 Controllability matrix

Definition 433 (Controllability matrix)**.** The *controllability matrix* of a pair (A, B) with $A \in \mathbb{F}^{n\times n}$ and $B \in \mathbb{F}^{n\times m}$ is the block-column matrix

$$\mathcal{C}(A, B) = \begin{bmatrix} B & A\,B & A^2\,B & \cdots & A^{n-1}\,B \end{bmatrix} \in \mathbb{F}^{n\times(nm)}.$$

Columns are indexed by $\operatorname{Fin} n \times \operatorname{Fin} m$, so that A^k is available without casting $k : \operatorname{Fin} n$ through Fin.val.

Lemma 434 (Entries of the controllability matrix)**.** *Block-column entry shape: at row i and column (k, j), the controllability matrix coincides with the (i, j) entry of $A^k\,B$:*

$$\mathcal{C}(A, B)_{i,\,(k,j)} = (A^k\,B)_{i,\,j}.$$

Proof. Holds definitionally by the encoding chosen for definition 433. □

Definition 435 (Controllability)**.** A linear system (A, B) is *controllable* when every target state $x \in \mathbb{F}^n$ is reachable from the origin in n steps: there exist input vectors $u_0, u_1, \ldots, u_{n-1} \in \mathbb{F}^m$ such that

$$x = \sum_{k=0}^{n-1} A^k\,B\,u_k.$$

This phrasing does not name the controllability matrix, so the bridge theorem 438 has real content.

11.1.3 Milestone: observability iff trivial kernel of $\mathcal{O}$

Theorem 436 (Observability via a trivial kernel)**.** *A finite-dimensional linear system (A, C) is observable in the sense of definition 430 if and only if the observability matrix $\mathcal{O}(A, C)$ has trivial kernel under matrix-vector multiplication:*

$$\operatorname{IsObservable}(A, C) \iff (\forall x,\ \mathcal{O}(A, C) \cdot x = 0 \Rightarrow x = 0).$$

Proof. Both directions follow from the bridge lemma 432, which identifies the (k, i)-coordinate of $\mathcal{O}(A, C) \cdot x$ with the i-coordinate of $(C\,A^k) \cdot x$. The forward direction reads the per-power kernel out of the assembled kernel coordinatewise; the reverse direction packs them back. □

11.1.4 Rank characterizations

Theorem 437 (Observability rank test)**.** *A finite-dimensional system (A, C) is observable if and only if the observability matrix $\mathcal{O}(A, C)$ has full column rank, i.e.*

$$\operatorname{rank} \mathcal{O}(A, C) = n,$$

where n is the state dimension.

Proof. Chain the kernel-form milestone theorem 436 with the matrix-level bridge $(\forall x,\ M \cdot x = 0 \Rightarrow x = 0) \iff \operatorname{rank} M = n$ from 'MatrixAlgebra.Rank', applied with $M = \mathcal{O}(A, C)$. □

Theorem 438 (Controllability rank test)**.** *A linear system (A, B) is controllable if and only if the controllability matrix has full row rank,*

$$\operatorname{rank} \mathcal{C}(A, B) = n,$$

where n is the state dimension.

Proof. Combine the reindexing identity $\mathcal{C}(A, B) \cdot u = \sum_k A^k B \cdot u_k$ (with $u_k(j) = u(k, j)$) with the matrix-level bridge $(\forall y,\ \exists x,\ M\,x = y) \iff \operatorname{rank} M = n$ from 'MatrixAlgebra.Rank', applied with $M = \mathcal{C}(A, B)$. □

11.1.5 Hautus observability

Definition 439 (Unobservable subspace)**.** The *unobservable subspace* of (A, C) is the A-invariant subspace

$$\mathcal{N}(A, C) \;=\; \{\, v \in \mathbb{C}^n \;:\; C\,A^k\,v = 0 \text{ for every } k = 0, 1, \dots, n-1 \,\}.$$

Equivalently, $\mathcal{N}(A, C) = \ker \mathcal{O}(A, C)$, but here it is phrased without naming $\mathcal{O}$ so the bridge theorem 440 reads as content.

Theorem 440 (Observability via the unobservable subspace)**.** *A finite-dimensional system (A, C) is observable in the sense of definition 430 if and only if its unobservable subspace is trivial:*

$$\mathcal{N}(A, C) = \{0\} \iff \mathrm{IsObservable}(A, C).$$

Proof. Both directions are membership unfoldings of definition 439 against definition 430. □

Lemma 441 (A-invariance of the unobservable subspace)**.** *The unobservable subspace is closed under the action of A: for every $v \in \mathcal{N}(A, C)$, also $A\,v \in \mathcal{N}(A, C)$.*

Proof. For $k = 0, \dots, n-2$ this is direct from the definition. For $k = n-1$ we land at $C\,A^n\,v$, which Cayley-Hamilton rewrites as a $\mathbb{C}$-linear combination of $C\,A^i\,v$ for $i < n$. Each of those is zero, so the combination is zero. □

Definition 442 (Hautus observability matrix)**.** The *Hautus observability matrix* of (A, C) at a complex number μ is the block-row matrix

$$H_{A,C}(\mu) \;=\; \begin{bmatrix} \mu I - A \\ C \end{bmatrix} \in \mathbb{C}^{(n+p)\times n}.$$

Lemma 443 (Kernel of the Hautus matrix)**.** *For every $\mu \in \mathbb{C}$ and $v \in \mathbb{C}^n$,*

$$H_{A,C}(\mu) \cdot v = 0 \iff (A\,v = \mu\,v \ \wedge\ C\,v = 0).$$

Proof. □

Theorem 444 (Unobservability gives a Hautus rank drop)**.** *If (A, C) is not observable, then there exists $\mu \in \mathbb{C}$ and a nonzero vector $v \in \mathbb{C}^n$ such that*

$$H_{A,C}(\mu) \cdot v = 0,$$

i.e. μ is an eigenvalue of A with an eigenvector in $\ker C$*.*

Proof. The unobservable subspace definition 439 is nontrivial by theorem 440. It is A-invariant by lemma 441, hence finite-dimensional and invariant; over $\mathbb{C}$ this guarantees a nonzero eigenvector inside it. That vector is automatically in $\ker C$ (membership at $k = 0$), so it lies in the kernel of $H_{A,C}(\mu)$ via lemma 443. □

Theorem 445 (A Hautus rank drop gives unobservability)**.** *If $\mu \in \mathbb{C}$ and $v \neq 0$ satisfy $H_{A,C}(\mu) \cdot v = 0$, then (A, C) is not observable.*

Proof. The witness gives $Av = \mu v$ and $Cv = 0$. By induction on k, $A^k v = \mu^k v$, so $C\,A^k\,v = \mu^k\,Cv = 0$ for every k. The vector v is therefore in the unobservable subspace and is nonzero, so observability fails. □

Theorem 446 (Hautus (PBH) test for observability)**.** *A finite-dimensional system (A, C) over $\mathbb{C}$ is observable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H_{A,C}(\mu)$ has trivial kernel:*

$$\mathrm{IsObservable}(A, C) \iff \forall \mu \in \mathbb{C},\ \ker H_{A,C}(\mu) = \{0\}.$$

Proof. Combine theorem 444 (failure direction) and theorem 445 (converse). □

11.1.6 Hautus controllability

Definition 447 (Hautus controllability matrix)**.** The *Hautus controllability matrix* of (A, B) at a complex number μ is the block-column matrix

$$H^{\mathrm{ctrl}}_{A,B}(\mu) \;=\; \begin{bmatrix} \mu I - A & B \end{bmatrix} \in \mathbb{C}^{n \times (n+m)}.$$

Theorem 448 (Hautus (PBH) test for controllability)**.** *A finite-dimensional system (A, B) over $\mathbb{C}$ is controllable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H^{\mathrm{ctrl}}_{A,B}(\mu)$ has full row rank:*

$$\mathrm{IsControllable}(A, B) \iff \forall \mu \in \mathbb{C},\ \operatorname{rank} H^{\mathrm{ctrl}}_{A,B}(\mu) = n.$$

Proof. Combine the duality bridge `isControllable_iff_isObservable_transpose` with the observability Hautus iff theorem 446, then convert the kernel-form RHS to the rank form via the matrix bridge `mulVec_kernel_trivial_iff_rank_eq_card_cols` from 'MatrixAlgebra.Rank', and finally identify the transposed Hautus matrices. □

11.2 Reachable subspace and Kalman decomposition

Definition 449 (Reachable subspace)**.** The reachable subspace of a pair (A, B) is the column span of its controllability matrix:

$$\mathcal{R}(A, B) = \operatorname{range} \mathcal{C}(A, B).$$

Thus it is exactly the span of $B, AB, \dots, A^{n-1}B$.

Reference: Hespanha, *Linear Systems Theory.*

Lemma 450 (Membership in the reachable subspace)**.** *Let $A \in \mathbb{F}^{n\times n}$ and $B \in \mathbb{F}^{n\times m}$, and let $\mathcal{C}(A, B) \in \mathbb{F}^{n\times(nm)}$ be their controllability matrix. For $x \in \mathbb{F}^n$,*

$$x \in \mathcal{R}(A, B) \quad \iff \quad \exists u : \operatorname{Fin}(n) \times \operatorname{Fin}(m) \to \mathbb{F}, \qquad \mathcal{C}(A, B)u = x.$$

Here $\mathcal{R}(A, B) = \operatorname{range} \mathcal{C}(A, B)$ is the reachable subspace.

Proof. □

Theorem 451 (Controllability via the reachable subspace)**.** *A pair (A, B) is controllable if and only if its reachable subspace is the whole state space.*

Reference: Hespanha, Linear Systems Theory*.*

Proof. □

Lemma 452 (A-invariance of the reachable subspace)**.** *The reachable subspace is A-invariant: $A\mathcal{R}(A, B) \subseteq \mathcal{R}(A, B)$.*

Reference: Hespanha, Linear Systems Theory*.*

Proof. □

Definition 453 (Kalman decomposition)**.** A Kalman decomposition of (A, B, C) records four coordinate sectors adapted to the reachable and unobservable subspaces. Their direct sums recover those subspaces and the full state space. The chosen complements are noncanonical and are not individually asserted to be invariant under A.

References: Hespanha, *Linear Systems Theory*; Kailath, *Linear Systems.*

Theorem 454 (Existence of the Kalman decomposition)**.** *Every finite-dimensional complex state-space system has four coordinate sectors adapted to its reachable and unobservable subspaces. The complements are noncanonical vector-space complements and are not individually asserted to be invariant under A.*

References: Hespanha, Linear Systems Theory*; Kailath,* Linear Systems*.*

Proof. □

Definition 455 (Adapted coordinate equivalence)**.** Addition of the four Kalman coordinate sectors defines a linear equivalence $X_{c\bar{o}} \times X_{co} \times X_{\bar{c}\bar{o}} \times X_{\bar{c}o} \simeq \mathbb{C}^n$. Choosing sector bases therefore gives an adapted basis of the original state space.

Theorem 456 (Block-zero pattern in adapted coordinates)**.** *In a basis adapted to the four Kalman coordinate sectors, ordered as controllable-unobservable, controllable-observable, uncontrollable-unobservable, uncontrollable-observable, the system has the forced-zero pattern*

$$A' = \begin{bmatrix} * & * & * & * \\ 0 & * & 0 & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{bmatrix}, \qquad B' = \begin{bmatrix} * \\ \\ 0 \\ 0 \end{bmatrix}, \qquad C' = \begin{bmatrix} 0 & * & 0 & * \end{bmatrix}.$$

Every displayed zero is asserted entrywise; no vanishing claim is made for the starred blocks. The chosen complement sectors are not individually asserted to be invariant under A.

References: Hespanha, Linear Systems Theory*; Kalman, "Mathematical Description of Linear Dynamical Systems" (1963); Kailath,* Linear Systems*.*

Proof. Reachable-subspace invariance forces the lower-left state and input zeros, unobservable-subspace invariance forces the remaining state zeros, and the zeroth observability condition forces the output zeros. Coordinate-basis representation lemmas translate these component statements to individual matrix entries. □